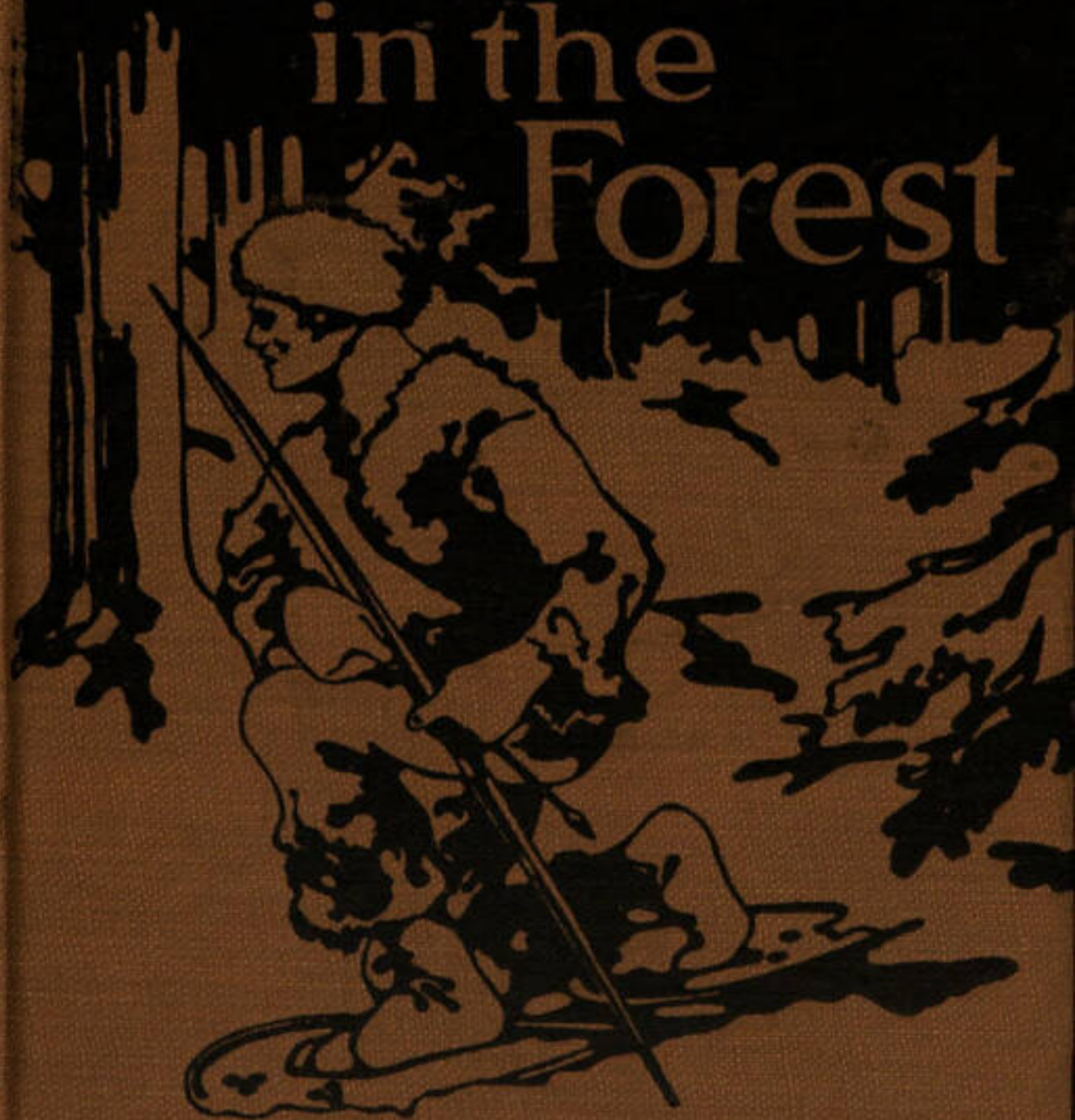
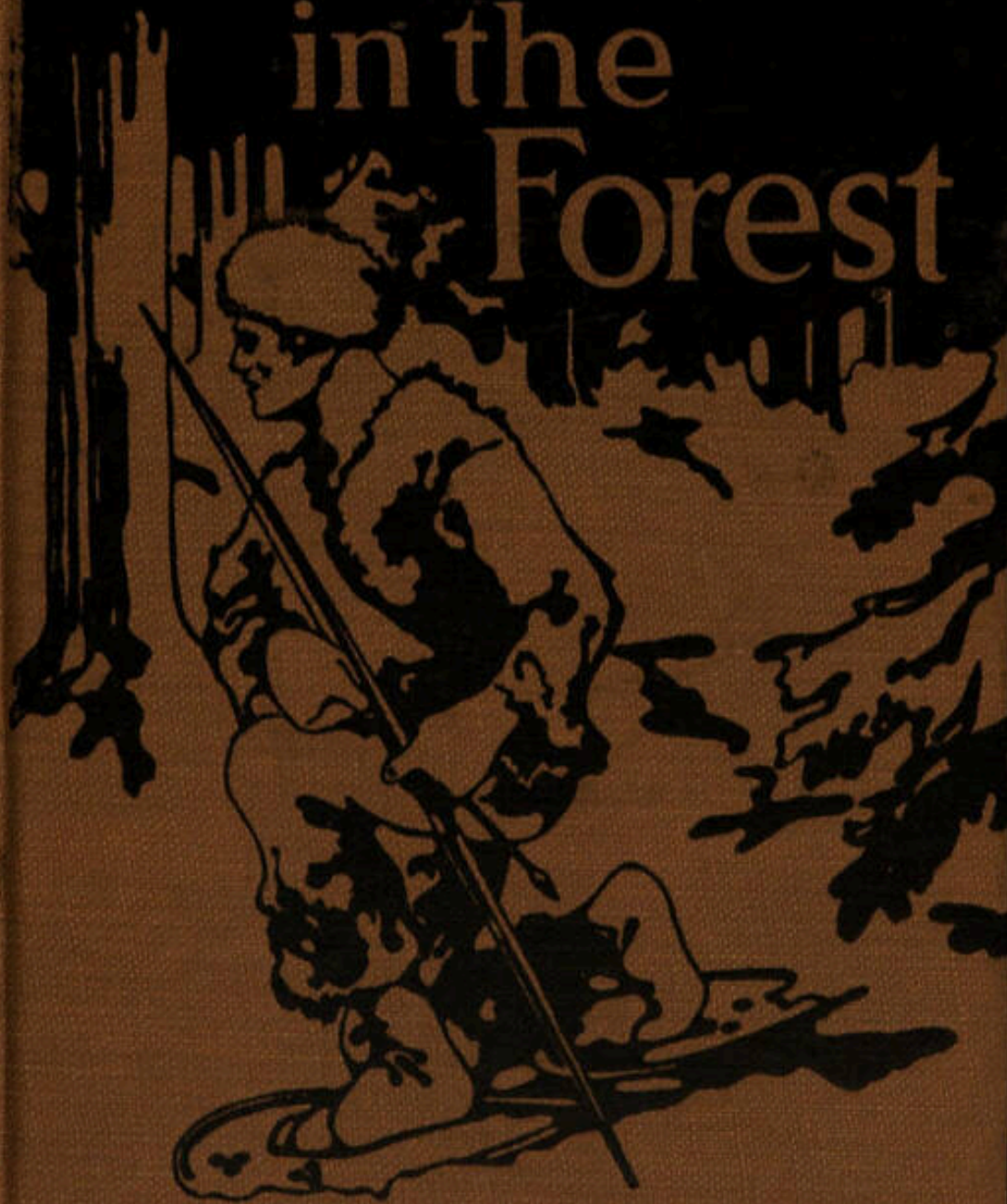


Marooned in the Forest



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THE FOREST: THE STORY OF A PRIMITIVE FIGHT FOR LIFE ***



A CRUSOE OF THE WILDERNESS

MAROONED IN THE FOREST
THE STORY OF A PRIMITIVE FIGHT FOR LIFE

BY
A. HYATT VERRILL

AUTHOR OF
"HARPER'S BOOK FOR YOUNG NATURALISTS" ETC.

ILLUSTRATED

HARPER & BROTHERS PUBLISHERS
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BOOKS BY
ALPHEUS HYATT VERRILL

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“WITH STEADY STROKES I WHIRLED THE DRILL AROUND AND AROUND”

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“I LOOKED UPON A BROAD, FAIR VALLEY”

PREFACE

If a man or a well-grown boy is lost in the wilderness, what can he do? Shall he whimper and give up? Never, if he has real blood in his veins. He faces a primitive struggle for life. It is a question of reinventing primitive means of living. How to make a fire, how to obtain food, how to clothe and shelter himself—these are the immediate problems to be met. He is a Robinson Crusoe of the wilderness.

This story of a modern Crusoe in the far Northern forests embodies many actual experiences, and it is an epitome of the basic facts of outdoor life. In books like Harper's *Camping and Scouting*, *Outdoor Book*, *Young Naturalists*, and others, the appliances of civilization are always at hand. It is a very different situation when one is lost in the depths of the forest without food, fire, weapons, or compass. But the problem of working out means of existence is one that will interest every lover of outdoor life, whether his interest is in camping, canoeing, fishing, or hunting, whether he is a member of the Boy Scouts or the Woodcraft Indians or simply an individual who knows the call of the wild. The adventures of Mr. Verrill's hero forth a story of thrilling interest and constant suspense. And it is also full of suggestions which will stimulate many readers to work out some of the hero's problems for themselves.

MAROONED IN THE FOREST

CHAPTER I

ALONE IN THE WILDERNESS

It all happened in the twinkling of an eye. I turned quickly at a sudden cry from Joe—my half-breed guide—in time to see him cast the handle of his broken paddle aside and leap forward for the extra paddle. Before he could reach it the canoe swerved, swung broadside to the rushing current, crashed sickeningly against the jagged rocks, and the next instant I was floundering about in the icy, swirling water. Bumping against rocks, struggling for breath, battling frantically with the torrent, I was swept down the river. Time and again my feet touched bottom, but each time, ere I could gain a foothold, I was drawn under, and each second I realized that my strength was growing less, that my lungs were bursting for air, and that in a few more moments all would be over. Down, down, I sank; above me the green water closed in and from my mouth and nostrils tiny bubbles of escaping air rose upward despite my every effort to withhold the scanty breath within my lungs. I was drowning I knew, and vaguely I wondered what had become of Joe, and how my friends would take the news of my loss here in this river of the great wilderness. Suddenly my foot touched a hard object. I threw all of my last remaining strength into a spasmodic kick and lost consciousness.

Slowly I opened my eyes and with wonder looked upon a strip of deep-blue sky against which the dark-green boughs of evergreens were sharply outlined. For a space I marveled, for so firmly convinced had I been that I was drowned I could scarce realize that I was not looking with spiritual eyes at a scene in another world. Then it dawned upon me that through some miracle I had been saved, and with a mighty effort I sat up.

I found myself upon the very brink of a little precipice—a natural dam over which the river fell in a miniature cataract, although the greater portion of the current swept to the left and poured like a mill-race through a narrow channel in the rocks. In a moment I realized how I had escaped. My final kick had driven me beyond the sweep of the current, I had been washed upon the edge of the waterfall, and my position had allowed the water to drain from my lungs. I was still terribly weak, I was choking with the water I had swallowed, my head swam, and with the utmost difficulty I half

crawled, half waded to the shore and threw myself upon the moss-covered bank where rays of sunshine penetrated the foliage overhead.

Although I was saved from death in the river by the merest chance, still my plight was desperate, for I was alone in the heart of the great woods, miles from civilization or settlements and without food, weapons, shelter, or anything save the clothes upon my back and the few trifles my pockets might contain. Possibly, I thought, the canoe might be washed ashore with its contents, or Joe might be safe and in the vicinity; and with these ideas strong in my mind I rose and slowly walked along the river's bank. I was now rapidly regaining strength, and, with the aid of a stout pole of dead wood which I picked up, I had little trouble in making my way up the stream. Presently I called out Joe's name, but only the soft echo of the woods replied. Again I trudged on, frequently calling and ever searching the edges of the stream and the eddies for the wreckage of the canoe, but not a sign of my guide or of my outfit could I find. At last, firmly convinced that Joe had been lost and that the canoe and its contents were gone forever, I seated myself upon a log and strove to collect myself and look squarely at the future. It would have been bad enough to be cast away in a country which I knew, but here I was completely at a loss. I had trusted entirely to Joe, and I knew nothing of this wilderness nor of the direction or route to the settlements; while, to make matters still worse, my compass had been lost in the river.

The last was really the least of my troubles, for I had little doubt that I could readily determine which direction was east and which west by the sun, and I had also heard that the moss grew thickest on one side of the trees; but as to whether that side was north or south I could not remember, cudgel my brains as I might. I also knew, in a general way, that the settlements were southward from the camp we had left, and I knew that Joe had expected to reach them by running down with the current, paddling across a lake, and tramping through the woods, and that he had stated the entire trip would consume about five days. However, I could not even I guess how many miles we had traveled before the canoe upset, and I had taken no notice of the turns and twists in the river. For all I knew, the stream might flow east, or even north, at the spot where I had crawled ashore, and if I attempted to travel in any direction—using the flow of the current as my guide—I might easily travel directly *away* from my fellow-men.



I CALLED OUT JOE'S NAME

My sole hope of reaching civilization would be in following the banks of the river, and this I realized would mean many weary days of tramping alone and unguided through the great forest.

Vainly I regretted having trusted so completely to Joe that I had paid no attention to the surroundings as we swept down the stream, and for that matter had not even asked for information which would have proved so valuable to me now. But it was wasting valuable time to spend the few remaining hours of daylight in regrets, and I was thankful for the few odds and ends of woodcraft and forest lore I had picked up during my life in the woods.

My clothing had partly dried, but with the passing of the bit of sunlight from the opening between the trees the air had become chilly and I was shivering with cold, the strain of my recent experience and my forebodings for the future. Rising from my seat, I strode back and forth, swinging my arms and striving by exercise to regain in some measure the circulation of my blood and a feeling of warmth. Activity, even of this forced sort, did me a world of good, and I began to plan for my immediate wants. Shelter I must have, and warmth, before night fell, and while I was not at the moment hungry, I realized that food of some sort would become a most pressing need by the following morning. Shelter without warmth would be of little value, and I thought with longing of the roaring fires which Joe had built before our camps each night and about which we had lounged while telling tales of past adventures.

Fire I *must* obtain, and in a mad hope that at least one good match might still remain in my pockets, I sought feverishly and emptied every one of my pockets upon a smooth rock. My total possessions thus displayed consisted of a small bunch of keys, a few small coins, a cambric handkerchief, a heavy jackknife, and the headless sticks of some matches from which the phosphorus had been completely soaked off. I gazed at these few articles with the bitterest disappointment, for of them all the knife was, as far as I could see, the only thing of any value to me in my present plight. With it I thought I might be able to fashion a bow-drill and spindle and thus obtain fire, for in my youth I had accomplished this feat when "playing Indian," but I well knew the difficulty in obtaining just the proper kinds of wood and I realized that a search for them would consume much valuable time, whereas but an hour or two of daylight now remained. Then flint and steel occurred to me. I had the steel in my knife, but I did not know whether flint

was to be obtained in the vicinity. However, I rose, made my way to the stony edge of the river, and sought diligently for some bit of rock which resembled flint. Each piece that struck my fancy I tried with my knife, and several gave off faint, bright sparks. All these I pocketed and, having obtained quite an assortment, I retraced my way to the rock whereon I had left my other possessions and prepared to try my hand at obtaining fire by means of my knife and the pebbles.

I realized that the tiny sparks which I could obtain in this way would never ignite a twig, or even a bit of bark, and that some inflammable tinder, which would catch the spark and which could then be fanned to a flame, must be secured before I could hope to succeed. As I was thinking of this my gaze fell upon a black-edged hole in my handkerchief. It had been burned, a couple of days before, by a spark from Joe's pipe blown back by the wind. The incident was too trivial to have filled my thoughts for an instant at another time, but now all its details came back to me with a rush and I gave a shout of joy as I suddenly realized that this burnt hole and the events which had caused it had actually solved my puzzle. Seizing the square of cotton cloth, which was now quite dry, I weighted it down with bits of stone—for the apparently useless handkerchief had now become of the utmost value to me—and hurried into the woods in search of dry twigs and other inflammable material. I had not long to hunt, for dead and dried trees were all about; several white birches furnished sheets of paper-like bark, and with a great armful of fire-wood I returned to my rock. Gathering the handkerchief into a loosely crumpled mass, I placed it on the rock, held the most promising of my pebbles close to it, and struck the stone sharply with the back of my knife-blade. A little shower of sparks flew forth at the blow, but none fell upon the handkerchief. Again and again I tried, each time holding the stone in a different position and trying my best to cause the sparks to fall upon the handkerchief. Finally I gathered the cloth in my hand, held the pebble in the midst of its folds, and struck it.

Sparks gleamed against the handkerchief, but no sign of charring cloth or wisp of smoke rewarded me. Surely, I thought, these sparks must be as hot as the tiny, glowing ember from Joe's pipe, and I unfolded and examined the handkerchief about the burned spot. Perhaps, I thought, this particular part of the cloth was more inflammable than the rest, and again gathering up the handkerchief, with the old burn close to the pebble, I again struck it with my knife.

Carefully I examined the cloth and the next instant dropped knife and pebble and cried aloud in triumph, for at one edge of the charred hole a tiny speck of red glowed in the dusk of coming evening, and spread rapidly in size. Carefully I blew upon it, folded another corner of the cloth against it, and waved it back and forth. Brighter and brighter it gleamed; a tiny thread of pungent smoke arose from it and an instant later a little tongue of flame sprang from the cambric, and I knew that fire, warmth, and comfort were mine. It was but an instant's work to ignite a piece of birch bark and push it among the pile of wood and twigs, and then, carefully extinguishing the handkerchief—for it had now grown very precious in my eyes—I squatted before the blazing fire and reveled in the comforting warmth from its glow. Although it was too late to consider ways and means of shelter that night, I knew that I could keep warm, and as soon as the chill and stiffness had been driven from my bones and muscles I set diligently at work gathering great piles of fuel to feed the flames during the night. Several large logs were close by, and these, with much labor, I dragged to the fire and placed near at hand to use later on when I went to sleep. By the time I had accumulated a supply which I judged would last through the night, I discovered that I was very hungry. I had not eaten since the forenoon, and I had worked strenuously, to say nothing of the utter exhaustion occasioned by my semi-drowning. My efforts to obtain fire and the extent to which I had concentrated my mind on this problem had kept me up and doing until now, but, once the fire was blazing merrily and an ample supply of fuel was at hand, I felt weary beyond words, famished, and absolutely worn out.



"I EMPTIED THE CONTENTS OF MY POCKETS ON A ROCK"



“I STRUCK THE STONE WITH MY KNIFE-BLADE”

The sun had set and the forest was black as midnight, but the sky was still faintly bright with the afterglow and the river shone silvery as it swirled and eddied between its shadowy banks. There was no hope of finding berries, roots, or other edibles in the woods after dusk. I had no means of catching game or fish, which, I knew, were abundant, and I commenced to think that I would die a miserable death of starvation before morning, when I suddenly recollected having seen a number of fresh-water mussels in some shallow backwaters of the river while hunting for my flinty pebbles. I had never eaten these shell-fish, but I felt sure they were edible, and, seizing a blazing pine knot from the fire, I made my way to the shore and soon found the pools where I had noticed the mollusks. There were not many—a bare dozen were all I could find that night—but these I felt would be far better than nothing, and in a few moments I had them baking on a bed of hot coals. Hardly waiting for them to cook, I raked them forth and devoured them ravenously, and never did choicest food taste so delicate, so delicious, and so welcome to my lips as did those half-baked, slimy, unseasoned mussels eaten beside my fire in the midst of the wilderness. Few as they

were, they served to refresh me greatly and to drive away the most pressing pangs of hunger, and, much as I desired more, I had not the strength or ambition to trudge up and down the river-bank searching for the shells. Piling several huge logs on the fire, I formed a rude bed of fir twigs and, casting myself upon this, fell instantly into a deep, dreamless sleep.

CHAPTER II

FOOD AND SHELTER

I was awakened by a shaft of sunlight striking my face, and opened my eyes to find the day well advanced. My first thought was of the fire, which had burned completely out. A thread of bluish smoke rose from the heap of ashes, however, and by raking these aside and thrusting bits of birch bark amid the embers I soon had a new blaze started, which I piled high with dry wood. I was wonderfully strengthened and refreshed by my long sleep, but I was all but famished, and as soon as the fire was going well I hurried to the river for more mussels. I found a few here and a few there, and with a dozen or two went back to the fire and presently was breakfasting off the shell-fish. I realized that while these would serve to prevent me from dying of hunger and they were wonderfully welcome in my present starved condition, I would be forced to search for something else to eat very soon. In the first place, the supply of the bivalves was limited. They would, I felt, prove far from palatable save when I was very hungry, and I doubted how much nourishment was contained in their flabby meat.

Had I possessed firearms or even fishing-tackle my plight would not have been bad, for birds and animals could, I knew, be readily found in the woods, while trout and other fish were abundant everywhere in the wilderness streams. As I ate my mussels I sought to devise some method of securing game, but every plan that occurred to me was spoiled by some unsurmountable obstacle which arose. I had often snared game and had even caught partridges with a slender noose on the end of a pole—for in the north woods these birds sit stupidly upon the low fir-trees and allow the hunter to pull them from their perches without taking flight. But a snare required a fine line, a slender wire, or a horsehair, and I had none. Fishing with a line was cast aside as out of the question for the same reason, with the added lack of a hook. Then a bow and arrow occurred to me, but I soon realized that arrows without feathers or sharp, heavy points would be impossible, and that neither heads nor feathers were within reach. Then I

thought of spears, for I knew that many savage tribes used spears both in fishing and in hunting, and I decided to try my skill at harpooning some unsuspecting fish or some unusually stupid partridge. It was a long time before I could find a straight, light stick for a haft, but at last I found a slender pole of weathered, dried spruce cast up by the river, and, by dint of whittling and trimming, this was worked into a very straight, well-balanced shaft which I judged would fulfil my requirements. I tried throwing it several times and found it easy to handle, but that it could not be depended upon, for one end was nearly as heavy as the other and it would fly sideways and strike a glancing blow as frequently as it would strike end on.

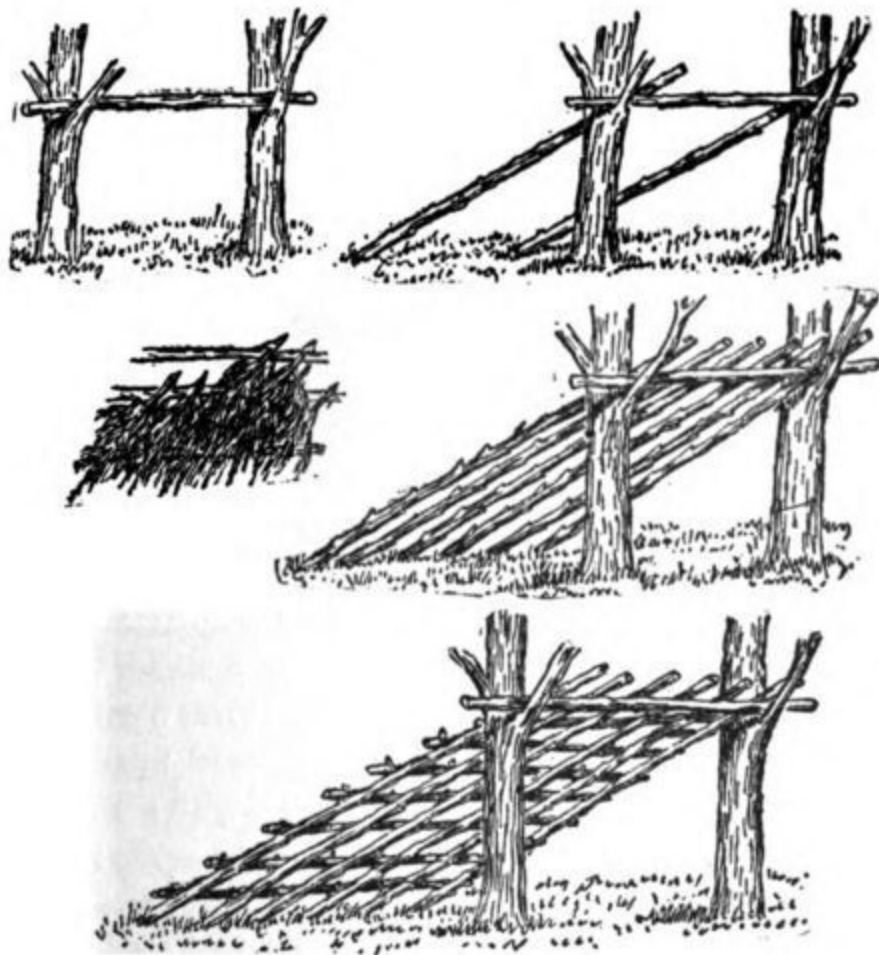
I realized that a head of some sort was required, but this I could not furnish, and rather than lose all the time I had spent on it I determined to try my hand at spearing a fish before throwing my weapon aside. Whittling the end to a sharp point and cutting numerous barbs, or notches, in it, I walked to the river and looked carefully into each pool and backwater. I saw several fish, but each flitted out of view as the spear was plunged downward, and I was about to abandon my attempts when luck favored me. Approaching one small pool, I gave a little start as a great bullfrog leaped almost from beneath my feet with a loud croak. A moment later he appeared on the farther side of the pool, his goggly eyes just showing above the water, and, approaching him carefully, I drove my sharpened stick at his big, green body. It was a lucky stroke, for the frog was fairly impaled upon the stick, and I drew my first victim from his watery home with a wonderful feeling of elation to think that unaided and alone I had actually succeeded in hunting and capturing a live, wild creature to serve my needs.

Strangely enough, frogs had not occurred to me hitherto, but, now that I had obtained one, I bestirred myself to capture a number. I realized that with my crude spear I could not expect to kill many frogs, and that my first success was pure luck more than anything else. Many a time when a boy I had speared frogs when spending my summers on a farm, and now that frogs were in my mind I remembered the two- or three-pronged spears which the farmers' boys used. I was still hungry, and while my frog was broiling I busied myself in making a real frog-spear. It was not a difficult task. I had only to attach two slender, barbed pieces of hard wood to the sides of my spear. I had some trouble in binding them on, but I sacrificed strips of my clothing for the purpose, and although the completed spear was very crude, I felt sure it would serve its purpose. I knew, however, that it

would soon be blunted and broken among the rocks of the river and I also knew that in such spots frogs would be scarce and that in muddy or stagnant pools I would stand a much better chance of finding them. No swamps or pools were in the immediate vicinity, but I had little doubt that I could find some by a short tramp. I was very anxious to try my spear, but I also realized that I must give time and thought to constructing a shelter to protect me in case of rain, and, reluctantly abandoning my frog-hunt for the time being, I gave my whole attention to the problem of house-building. I had seen many a shack or "lean-to" built and had helped at the work myself, but without an ax I knew that to build even the smallest and simplest shelter would necessitate a tremendous amount of hard labor and would present almost insurmountable difficulties. With only a pocket-knife to cut the necessary trees, poles, and branches, I would be obliged to make the shack of small stuff, and I trembled to think what fate might have in store for me if I should break my knife in an attempt to cut tough branches from the trees.

However, if I was to have a shelter at all it behooved me to begin at once, and I started forth to select a site for my home. I found a sheltered, dry knoll with good drainage a short distance from the river and with plenty of building material in the form of balsam firs, pines, and birches near at hand. I first selected two young trees, about five feet apart, and from these I cut the lower branches, leaving the stubs projecting a few inches. Across two of these I placed a light spruce pole and from the ends of this I laid other poles extending back at an angle to the ground.

This all sounds very simple and easy, now that I come to write it down, but as a matter of fact it required hours of hard, back-breaking, hand-blistering work, and by the time this much was accomplished I was faint with hunger. I succeeded in finding and eating a few mussels, but I had no time to devote to frog-hunting, and hurried back to my house-building. Across the two slanting poles other lighter poles were placed, and over these the broad "fans" of fir were spread like shingles, the lowest layer being placed first with each succeeding layer overlapping the last. This was comparatively easy work, for the twigs were small and easy to cut, and by late afternoon I had a shack which, though not by any means complete, was far better than nothing but the blue sky for a shelter.



HOW THE "LEAN-TO" WAS BUILT

I had an hour or two of daylight left, and determined to look for a likely spot for frogs. I dared not walk far into the forest for fear of losing my way in the fading afternoon light, but even a tramp of a few hundred yards away from the river was enough to convince me that there were no swamps or ponds in the vicinity, for the ground was quite hilly and rocky. Deciding that my only chance lay in finding stray frogs in the pools of the river, I walked down-stream for some distance, searching carefully wherever there was a backwater or a puddle of water along the shore. I found a number of mussels, which I pocketed, but no sign of frogs until I had traveled perhaps half a mile from my fire. At this point a small brook fell in a tiny cascade over the bank into the river, and, clambering up, I found that the little stream ran through an open vale or glade luxuriant with ferns, brush, and

rank-growing plants. The stones over which it flowed were dark with a coating of moss, and in the deep, still pools between the boulders I caught glimpses of great speckled trout lurking in the shadows. It was an ideal trout-brook and I tried my best to spear one of the beautiful fish, but without success. However, I was rather pleased at my discovery, for even without fishing-tackle I felt confident that I could dam up one of the pools, bail out the water, and catch the trout with my hands. But there was no time for this just then. In the hope of finding a frog I went on up the brook. I had all but given up in despair when I reached a second miniature waterfall, and above this cascade I came upon a little pond surrounded by alders and birches. It was a cool, shady spot and the dark, black water flecked with patches of green weeds and lily-pads gave promise of frogs. Hardly had I reached the edge of the pool when I spied a fine bullfrog squatting among the weeds, and a moment later he had been successfully speared. I was delighted with the success of my crude weapon and crept cautiously around the pond, seeking more victims. Frogs were plentiful and were very tame, for probably man had never disturbed them, and before the growing dusk warned me that it was time to return to my camp I had obtained seven fine, big hoppers. As I was making my way toward the brook and the cascade I was startled by some good-sized creature which sprang from the grass at the border of the pond and plunged into the water. A moment later I saw a furry, brown head followed by a silvery, rippling wake, cleaving the placid surface of the pond, and realized that the animal which had caused my momentary fright was merely a harmless muskrat. I stopped and watched the creature for several moments and longed to be able to secure him, for I well knew that muskrats are edible and are even esteemed a delicacy. More than once I had eaten their tender, white meat when cooked by Joe. It was useless to give the matter any consideration, however, for without a gun the muskrat was far beyond my reach, and reluctantly I proceeded on my way.

Presently I noticed a path-like trail winding through the grass and weeds, and, looking closely, discovered the imprint of little feet upon the soft and muddy ground. I recognized the muskrat's runway, and with the realization came the thought that I might trap the rats. To be sure, I had no traps at hand, but I had seen deadfalls set in the woods by the fur trappers and, while my memory was hazy as to just how they were arranged, I felt quite confident that my ingenuity would find a way to rig up some sort of snare or deadfall which would serve my purpose. With my mind filled with such

thoughts I made my way back to my fire, which I reached just as darkness fell upon the wilderness. I dined well that night on frogs, and placed my mussels in a pool beside the river as a reserve for another day.

Much of the evening I spent experimenting with bits of twigs and sticks of wood, endeavoring to devise a deadfall, and by dint of racking my memory for details of traps I had seen, and by trying various methods, I finally discovered several different triggers which I felt would work, and, well satisfied with my day's labors and success, I fell asleep upon a bed of soft fir branches in the lean-to.

A couple of the frogs, which I had kept over, with a few mussels, served for my breakfast the next morning, and I then set diligently at work to complete my shelter, for a light shower had fallen during the night and my clothes were soaking wet when I awoke. To make the roof water-tight was my first consideration and to accomplish this I peeled sheets of birch bark from the trees, laid them like shingles on the roof, and secured them in place by rocks from the river-bed. At first I had trouble in preventing the stones from sliding and rolling off the slanting roof, but I soon devised a means of holding them in position by placing light branches across the roof and catching their ends on the projecting stubs of the roof timbers. In many ways I was greatly handicapped for want of string or rope. It occurred to me that strips of birch bark might serve, but I soon found that this had no strength to speak of, and I determined to try other materials. The Indians, I well knew, used bark, roots, and withes for rope, but I had no knowledge of the particular barks, roots, or withes which they employed, and I set myself to experimenting with everything that grew in the neighborhood. I soon eliminated many as useless, although certain roots appeared tough and fibrous, but these were all too gnarled and knobby or too short to serve as string. It was then that I began to realize how little I really knew of woodcraft or forest lore, although I had spent so many vacations in the woods. No doubt Joe or any other woodsman would have found life easy and simple if cast, as I was, upon his resources in the forest, but I had depended so completely upon others' knowledge that I was obliged to seek blindly for the simplest things and only occasionally remembered some trifling bit of woodcraft which I had seen when in Joe's company in the forest.

While thinking of this I was sitting beside my hut. When I attempted to rise, my hand came in contact with a sharp stub projecting from the earth. It

was a small thing—merely a twig which I had cut off while clearing the open space before my shelter—and to avoid further trouble with it, I grasped it and strove to pull it up. Much to my surprise, it resisted my efforts. Seizing it with both hands, I jerked at it with all my strength. Slowly it gave, and then, with a ripping sound, broke from the loose, thin earth, and I tumbled backward and sprawled upon the ground. I was curious to learn how such a small thing could be so strongly embedded in the soil and I examined it carefully. Attached to the bit of stem was a mass of long, fibrous roots. Seizing one of these, I attempted to break it. I twisted and pulled, but the root remained intact, and suddenly it dawned upon me that here was the very material I desired—that these roots were as strong and tough as hempen rope, and that by merest accident I had stumbled upon the very thing for which I had been searching. Unfortunately, I did not know what plant the roots belonged to, for only an inch or two of stem remained, and while the supply of roots it bore would serve my present needs, I was very anxious to learn the identity of the useful growth in case I should require more roots in the future. With this end in view I set about comparing the bark and wood with other young sprouts in the vicinity, and whenever one resembled it I pulled it up and examined the roots. I searched for some time before I was rewarded, and discovered that my lucky find was a young hemlock. Pine fir, spruce, and other trees I had tried in vain, but hemlocks were not abundant, and those about were mostly large and had been passed by in my former search. Now that I had discovered a source of supply of binding materials, many problems which had confronted me were simplified and I was greatly encouraged.

It must not be supposed that during these first days of my life in the wilderness I had given no thought to making my way to the settlements. In fact, this matter was ever present in my mind, but the very first day I had decided that before I attempted to make my way out of the woods I must be equipped to secure food, provide shelter, and make fires. Anxious as I was to reach civilization, yet I knew how foolhardy it would be to start blindly forth, trusting to luck for food or shelter, and with my limited knowledge of woodcraft. Here, where I had been cast ashore, I was safe, at any rate, provided I could secure enough to eat, and I determined to make my headquarters at this spot until I could learn by experience something of the resources of the forest and how to make use of them. Already I had acquired much useful knowledge, and I felt that if I could only succeed in trapping

animals or snaring birds I could start forth on my weary tramp in comparative safety as far as starvation was concerned.

I should have felt far more confident if I could have carried food with me, and I wondered if it would be possible to dry or cure frogs, mussels, or other meat. I knew that the Indians dried venison and made pemmican, which I had frequently eaten, and I had heard of certain tribes who subsisted upon dried salmon, but venison was unattainable with my present resources, and I was not at all sure that trout, even if I succeeded in obtaining them, would dry like salmon. Finally I decided to experiment, and, lacking all else, to carry a supply of live mussels along when I set forth. These shells, I knew, would live for several hours without water, and, as I intended to follow the river, I could easily keep them alive by frequent immersions in the water. Such thoughts brought up the question of vegetables, and I wondered if in these woods there were edible roots or tubers of any kind.

I remembered many boyhood books and stories telling of men lost in the woods and subsisting upon roots and berries, but, try as I might, I could not remember a single one which told just *what* roots and berries provided sustenance for the fictitious heroes.

Berries, I felt sure, existed somewhere in the woods, but, aside from blueberries or blackberries and the tiny scarlet partridge berries, I knew of none which were edible, and I smiled to think how hungry I would be if I depended upon the meager and uncertain supply of such things for a livelihood. Once, when a youngster, I had dug up and eaten ground-nuts, but they were gritty, tasteless things, and moreover I could only tell where they grew by the delicate white flowers which bloomed only in the spring. Nuts did not exist in this forest, or, if they did, they were not ripe at this season, and I therefore cast aside all ideas of securing a supply of vegetable food. Determined to try my hand at trapping and also to attempt to capture some trout, I started again for the brook, carrying a supply of hemlock roots and my spear. It occurred to me that by braiding fine roots together I could devise a fishing-line, but the question of a hook then confronted me and I decided to try my plan of bailing the water from a pool before experimenting with hookmaking.

I soon found a pool containing several fine fish, and cautiously, for fear the trout might slip out among the stones, I piled gravel and small rocks in all the visible crevices which connected the pool with the running waters of

the brook. This accomplished, I piled rocks across the little channel where the brook ran into the pool, and by chinking all the crevices with grass, twigs, and mud I at last had the satisfaction of seeing the water diverted to one side. The pool, with its fish, now remained cut off from the surrounding water, and all I had to do was to scoop out the contents, leave the trout floundering about on the bottom, and pick them up with my hands. This all sounds very simple and easy, but I had no scoop with which to bail out the water, and until I attempted the work I did not dream what a task I had set myself. I first tried bailing out the water with my hands, but as fast as I threw it out more oozed in through tiny crevices and I soon gave this up as impossible. Then it occurred to me that one of my shoes might serve as a dipper and, removing it from my foot, I tried to throw out the water by this means. I *did* succeed in making some progress, but very little, and I commenced to think that all my work had gone for naught when a bit of birch bark caught my eye and I had an inspiration. Many a time I had used birch-bark dippers and cups for drinking, when in camp with Joe, and I had seen boxes, packs, and other utensils made of the material. In fact, Joe had once proved to me that water could be boiled in a birch-bark dish, and I laughed to think how I had so far overlooked the manifold uses to which the bark could be put. It took but a few moments to strip a large sheet of bark from a convenient tree, and but a few moments more to bend this into a deep, boxlike form. The ends were easily secured by means of the hemlock roots, and with the bark dipper, which would easily hold a gallon of water, I proceeded to empty the pool. In a very short time the water was reduced to an inch or two at the bottom and the flashing, bright-colored fish were flopping about among the stones.

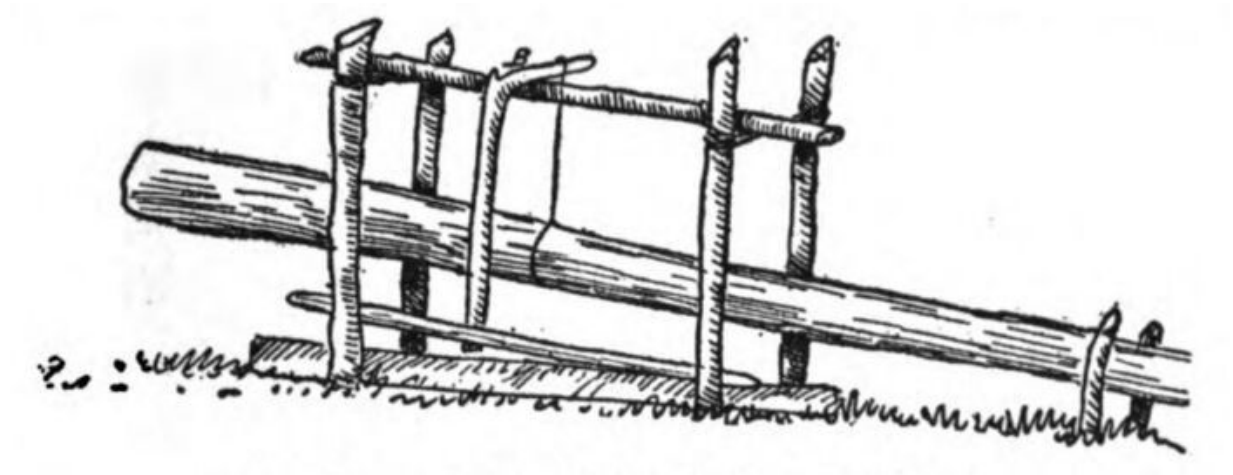
Four fine trout were the reward of my labors, and, placing them in my birch-bark dipper and covering them with cool leaves, I set them among the bushes beside the brook to await my return and then made my way toward the muskrat runway to set the trap.



SETTING THE DEADFALL

I did not know what sort of food the muskrat ate and I therefore decided to arrange a trap which would be sprung by the rat passing along the trail. First I placed a smooth stick of wood across the run, and on either side among the grass I drove two stakes with a space of a few inches between them. In this space I slipped a fairly heavy log which I found beside the pond, and I then lashed the tops of the stakes together so the log could slide readily up and down between the stakes which served as guides, and across the lashing of roots I laid a light stick. For a trigger I selected an "L"-shaped twig, and from one end of this I tied a strong root, with the other end of the fastening looped about the heavy log. This was adjusted until, when the trigger was placed across the light stick between the uprights, the heavy log was raised a few inches above the log set in the pathway. Next a very light stick was placed just above the lower log and the end of the trigger was placed resting against this, so that the pressure of the drop-log forced the trigger against the stakes. It was a very simple arrangement, but I knew that if any creature attempted to pass over the log upon the ground he would of necessity move the trigger-stick and allow the log to drop upon his back. The trap being set, I spent some time in securing a supply of frogs about the pond, and then started toward camp. I soon reached the brook and turned

aside for the trout in their birch-bark receptacle, thinking with pleasurable anticipation of the fine meal in store for me.



HOW THE DEADFALL WAS ARRANGED

I passed the pool, which was now rapidly filling up again, pushed aside the bushes, and gave a gasp of astonishment—the birch-bark dish was lying on its side, absolutely empty.

CHAPTER III

IMPORTANT DISCOVERIES

For a moment I was almost stunned by the discovery, but presently I realized that some prowling creature had robbed me of the fish which I had taken such pains to capture, and that I had only myself to blame for leaving the trout so carelessly within reach of any four-footed thief that might pass by.

It was a keen disappointment to be deprived of my expected feast, but there was nothing to be done save to drain another pool and capture more fish, if I wanted to eat trout that day.

I was anxious, however, to discover what manner of beast had stolen my fish, and I sought carefully in the soft earth and among the vegetation for signs of footprints. I had not long to search and soon discovered a number of tracks which I recognized as those of a fisher-cat, a large, marten-like animal which every woodsman knows for an inveterate thief. My first thought was to set a trap to capture the fisher, but, knowing the flesh to be unfit to eat, I abandoned the idea as a waste of time and trouble and set about my work of draining another pool. This time I selected a rocky basin worn by the water of the brook in the ledge itself—a sort of pot-hole—with solid walls which obviated the necessity of chinking up the openings and crevices as I was obliged to do in the pool I had drained before. With my birch-bark dipper the work of bailing out the pot-hole was simple and I soon secured a couple of good-sized trout.

With these and my frogs I dined well and decided to set forth on my tramp as soon as possible, for, now that I could obtain fish so readily, I had little fear of starving, for I knew that every brook and river in the forest swarmed with trout. I deemed it wise, however, first to wait until I could be sure of determining the exact points of the compass, and I also wished to determine the success or failure of my deadfall. Although the sun shone through the cleft in the forest formed by the stream, yet it gave me only a vague idea of direction, and while I knew by the sun that the river flowed in

a more or less southerly course at this spot, yet I wished to familiarize myself with the various compass points and to discover some other means of distinguishing north from south and east from west, for I had little doubt that there would be many days on which the sun would not shine. Accordingly, on the following morning I started into the woods while the sun was yet low, to study and reason out any signs which would aid me in maintaining a straight course through the forest.

As soon as I was well into the woods I looked about with minute care for any details which would be of use and also examined the trees very carefully for moss and lichens, for, as I have already mentioned, I had heard that moss grew more abundantly on one side of trees than the other, but I had forgotten which side it was.

Nearly every tree was well covered with lichens and moss and I could not see that these growths were any thicker on one side than the other. I was about to abandon this scheme for determining direction when I made a discovery. Glancing up and down the trunks in search of the moss growths, I noticed that one side of every tree was dark-colored and damp, whereas the other side was grayish and drier, and the damp side I soon found corresponded to the north as determined by my glimpse of the sun above the river. I was quite elated by this and I now noticed that the mosses *did* appear heavier and more luxuriant on the damp side of the trees than on the dry side. A further scrutiny and comparison of the various trees also convinced me that the branches, twigs, and leaves were thicker and more regular on the south side of the trees than on the north, and that more dried and dead branches and stubs projected from the north side of the trees than from the south side. Fixing these facts in my mind, I determined to test my discoveries by actual experiment, and without looking at the tree trunks I wandered aimlessly ahead for several hundred yards. Then, closing my eyes, I walked slowly about for some time, bumping into numerous trees and tripping over fallen branches several times—until I felt that I had lost all sense of direction. Then, opening my eyes, I looked about. I was out of sight or sound of the river, the only signs of sunshine were faint, bright patches amid the lofty foliage of the trees, and nothing was in view which seemed familiar. For a moment my heart thumped and I shuddered to think what might happen if my signs failed and I could not find my way back to the river. It was a dangerous experiment, the peril of which I did not fully realize even then, but, pulling myself together, I focused my attention on the

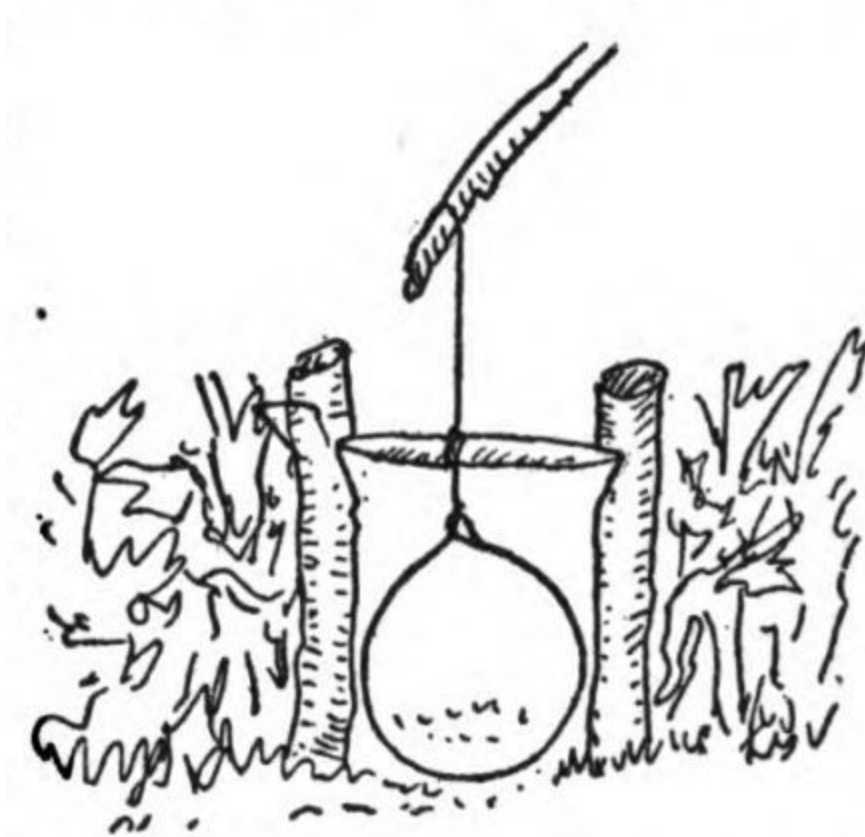
trees about me. There was no question about it, scarcely a glance was needed to show me which side of the trees faced the north and which the south, and, knowing that the river flowed to the east of the woods wherein I stood, I turned and started to retrace my steps. Even as I did so I realized how important was my newly acquired knowledge of this feature of woodcraft, for the direction which I had felt sure would lead me toward the river was *exactly opposite* to that which was shown to be right by the trees.

I was greatly pleased, for now I knew that in case rapids, cascades, or cliffs prevented me from following the river I could make detours through the forest, and, moreover, where the river turned and swung from its southerly course I could save miles of weary tramping by cutting across through the woods.

Thinking of such matters and only glancing now and then at the trees to assure myself of my direction, I was suddenly aroused by a large hare or rabbit which leaped from beside a dead stump almost at my feet and scampered off among the shadows. For a moment I stood still, watching the creature as he flashed across the open spaces and thinking regretfully what a fine supply of food was flitting beyond my reach. Then glancing down, I caught sight of a great mass of fungous growth upon the base of the stump from which the hare had jumped. The fungus was dull orange or yellow and grew in a form resembling sponge or coral. I had often seen the same thing before and had never given it more than a momentary glance, but this mass instantly riveted my attention, for one side of it had been eaten away and bits of the nibbled fungus were strewn upon the earth. This, then, was what the hare had been eating and I realized that by setting a snare or trap beside it I might be able to capture the rabbit. There was no time like the present for attempting the feat, and I at once set about preparing a trap. It was merely a simple "twitch-up," such as every farmer's boy uses for catching rabbits, partridge, and other small creatures, and while a few days before it would have been beyond me, it was now simple, with my knowledge of hemlock roots and the self-reliance which I was so rapidly acquiring.

Cutting a number of short sticks, I pushed them into the earth about the fungus, thus inclosing it on all sides but one. On either side of the opening thus left I drove two stout stakes with notches near their upper ends. From a bit of dead wood I then whittled out a spindle-shaped piece just long enough to reach from one of these stakes to the other. Then with a fine hemlock root I formed a noose, tied the spindle to the fiber just above it,

and fastened the end of the root to the tip of a small sapling close by. Bending down the latter, I slipped the spindle into the notches in the stake, spread the noose across the opening, and my snare was completed. I was very proud of my work, simple as it was, and was quite confident that when the hare returned to finish his meal he would push his head through the noose, dislodge the spindle, and would be jerked into the air and killed by the spring of the sapling. I stood for a moment looking at the snare and the fungus and suddenly roared with laughter at my own stupidity. Here I had been working for nearly an hour to set a trap which might or might not catch the rabbit, and within a few inches was a supply of food of far more value and to be had without the least effort. Surely if a rabbit could eat the fungus, so could I, and I plucked a bit of the queer growth and tasted it.



HOW I SET THE "TWITCH-UP" FOR THE HARES

It had a rather musty but not unpleasant taste with a slight nutty flavor, and I judged that, cooked, it might be very palatable. The question of eating mushrooms had occurred to me before this, but I knew nothing as to the edible qualities of fungus except that certain species were deadly and some nutritious, and I had not dared attempt eating them. Now, by the merest chance, I had discovered an edible species, and with a feeling of intense gratitude to the hare, I determined that his life should not be forfeited to my appetite and that he should be rewarded by being spared. Without more ado I removed the snare which I had taken so much trouble to prepare, and pocketed a large section of the fungus. That there was an abundant supply of this growth in the forest I was confident, and as I walked toward the river I searched on every log and stump for more. Several large masses were found, and, as many of these had been partly devoured by small animals, I felt reassured as to the edible and nutritious qualities of the sponge-like material.



“I PLUCKED A BIT OF THE QUEER GROWTH AND TASTED IT”

I reached my shelter without further adventure and at once prepared to cook and sample the fungus. I was not at all sure as to the best method of cooking it, and decided to try a small quantity in various ways. I therefore placed a lump among the hot coals to roast like a potato, while another lump was hung on a green stick before the fire to broil.

Hitherto broiling and roasting had been my sole means of cooking food, but now, having remembered that Joe had once showed me how to boil water in birch bark, I made a rude pot of this material, placed water and fungus within, and set the whole over a bed of hot coals covered with ashes. The bit of fungus to be broiled soon shriveled up and was transformed into a leathery-like material, tasteless and useless, while the piece roasting in the coals sputtered and sizzled, and might as well have been a bit of pine bark at the end of a few minutes. Both of these methods were undoubtedly failures, and I watched with some anxiety the piece boiling in the birch-bark pot. When it had boiled for some minutes I fished a bit out and, as soon as it had cooled, proceeded to taste it. Much to my joy, it had quite lost its musty, woody flavor and was as sweet, nutty, and palatable as a boiled chestnut, and I at once drew forth all that remained in the pot and dumped in all I had left. Words cannot express the satisfaction I felt at thus having discovered a source of vegetable food which I could gather as I traveled along and which would assure me a supply of provisions without the trouble and labor of trapping animals, catching fish, or hunting frogs and mussels.

As soon as my meal of fungus was finished I arose and, taking my frog-spear, made my way to the brook and my muskrat-trap. It was with quite a little excitement that I pushed my way through the thick growth toward the runway where the deadfall was placed, for even with my newly acquired knowledge of edible fungus I felt that meat would be necessary, or at least welcome, during my tramp, and the success or failure of my first trap meant much to me. But I had no cause to worry, the deadfall had been sprung and had served its purpose well, for projecting from beneath the log was a furry head. Even before I reached the trap I thought it the largest muskrat I had ever seen, and as I stooped down to lift the log I uttered an involuntary cry of amazement. The creature I had caught was no muskrat, but a great, fat beaver. Truly, my first attempt at trapping had been a huge success.

CHAPTER IV

I SET FORTH ON MY JOURNEY

Although I felt a hunter's elation at having captured the beaver, he was really of less value to me than a muskrat. His flesh, especially his tail, was edible, I knew, but I doubted if I would care to devour his meat unless very hungry, for the scent and taste of castor would be too strong. His fur, although thick, was by no means in good condition, and even if it had been "prime" it would have been of little value to me in the forest, but, nevertheless, I foresaw that I might find use for skins, and very wisely, as it turned, decided to skin the creature and dry and preserve the hide.

While skinning the beaver I was attracted by the strong, white tendons of his legs and tail, and, knowing how useful such tough, thread-like material might prove, I carefully removed and washed the tendons and placed them in a safe spot to dry.

The beaver's meat looked white, clean, and tender and I decided to cook and taste some of it. The tail I also decided to cook, for I knew the Indians and trappers considered beaver tails a great delicacy. The meat was placed to broil above the hot coals, and the tail, which seemed tough, was placed to boil in the birch-bark pot—or rather, in a fresh receptacle—for I found that after once using the bark for boiling it was worthless and that a new dish must be provided each time I wished to boil anything.

While the meat and tail were cooking I spread the skin of my beaver to dry and it then occurred to me that perhaps beaver flesh might be jerked or dried as well as venison. Accordingly, I cut strips from the carcass and hung them up. By the time this was done the meat was thoroughly broiled and ready to taste. Much to my surprise, there was but a very slight musky taste to the flesh, and while it was far from delicious without salt or seasoning, yet it was much better than mussels, and I greatly relished the flavor of real meat once more. The tail proved too gristly and tough to suit me and I doubt if I could have devoured it unless I were actually starving. It reminded me of pig's feet, and I wondered how any human beings could like it. No doubt

if properly prepared it might be far more palatable, but I then and there decided that beavers' tails would be eliminated from my menu unless I was face to face with starvation.

I was not sorry to discover that beaver flesh was edible, for I knew that where there was one beaver there were doubtless more, and that I might reasonably expect to catch others, but unless the meat could be dried and preserved it would be of little value for my purposes. I determined to try to dry trout. While thinking of this my mind turned to the matter of tackle with which to capture fish without the time and labor of bailing out the pools—a slow method at best and only possible where there were deep pools or basins. With the hemlock roots I could braid lines which I felt sure would serve my purpose, but I could not conceive of any way by which I could form a hook. I happened to notice the carcass of the beaver and picked it up to throw it into the river, when I noticed the sharp, chisel-like teeth and strong bones. For a moment I stood regarding them, turning over in my mind my various wants and striving to think of some purpose for which I could use either teeth or bones, for it seemed a pity to waste anything that might serve any useful purpose. I thought of fish-hooks, for I had heard of certain savage races using bone hooks, but I could not imagine a way of transforming either teeth or bones into trout-hooks, and I was on the point of throwing the body into the stream when bows and arrows again came to my mind, and instantly it occurred to me that the bones of my beaver might be sharpened and used for arrow-heads.

At any rate, it was a scheme worth trying, and I promptly began to dissect out the leg bones from the remaining meat. Lest I should want other material at some future time, I also removed and set aside the huge front teeth. This occupied a long time, and I had barely time to walk out to the trout-brook, catch two fish, after a deal of labor, and return to camp ere night fell. One of the trout served for my supper and the other was split, cleaned, and hung up to dry with the beaver meat.

The following morning I awoke to find the woods dripping and the world gray with a cold, drizzling rain. From my fire a thin, blue wisp of smoke arose, and I hurried to replenish the fuel and save the little life there was left in the embers. Before I could fan the coals into flame the lowering, gray sky poured forth a torrent of rain, and with a faint hiss the last hot coals grew black and dead.

Soaked through, chilled, miserable, and disgusted, I crept into my hut and, seeking a sheltered spot, sought to secure another fire with my knife, pebble, and handkerchief. What was my disappointment to find the handkerchief damp and soggy with moisture, and while one or two spots appeared quite dry, my utmost endeavors failed to ignite the cotton cloth. For an hour or more I labored, until my hands were cut and bleeding and the back of my knife-blade was worn rough and battered, and then, thoroughly disheartened, I gave up in despair. Hungry as I was, I had nothing save uncooked fungus to eat, for I had not yet reached the point where raw mussels, raw frogs, or raw fish could be considered.

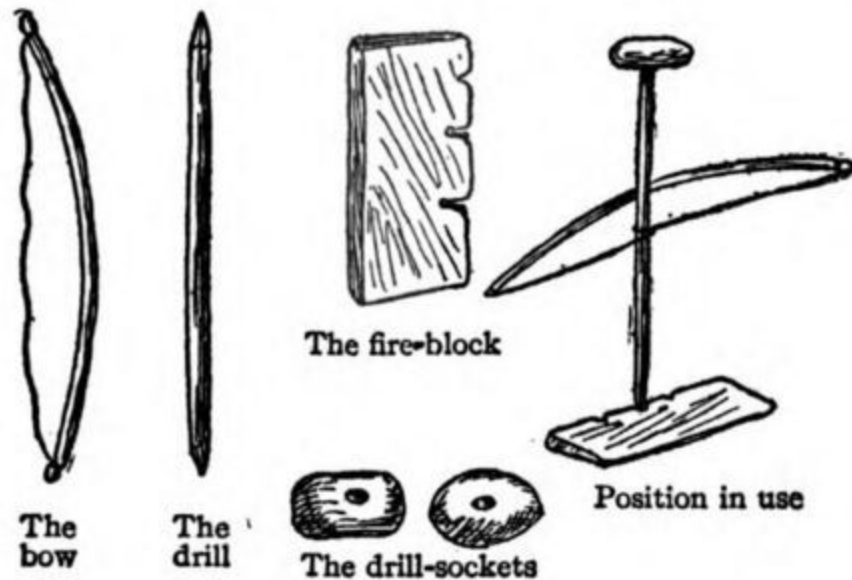
Sitting in the partial shelter of my lean-to, I spent a dreary and forlorn morning, for while the roof was fairly tight the rain drove in at front and sides, and only in the very center of the hut could I remain fairly dry. My wet clothes clung to my skin, chilling me to the bone each time the cold wind whistled down the river, and my reflections were far from cheering, for I knew that this was but a sample of what I might expect. Summer was over and the autumn rains had begun, and in a few weeks more icy winds and snow-squalls would succeed them. With a roaring fire all might have been well, and I could have laughed at the elements, but without fire I realized how helpless I was and ever uppermost in my mind was how to safeguard myself against the loss of my fire in the future, provided I again succeeded in starting a blaze—something which I considered very doubtful.

Toward noon, however, the rain ceased, the sky cleared, and by mid-afternoon the sun was shining brightly. I lost no time in finding a sunny stone whereon to spread my handkerchief, and as soon as the bit of cloth was dry I again essayed to ignite it with a spark from my flint. This time I met with more success, and after several trials I obtained a blaze and soon had a roaring fire. As soon as the fire was burning well I cooked food and while this was being done busied myself in making a neat, tight box or case of birch bark in which to carry my handkerchief. I was fearful lest the cotton cloth should give out long before I reached the end of my journey, for only a small portion remained intact. To provide against such a loss I tore bits of cloth from my shirt, charred spots on the strips with coals from the fire, and packed these carefully in additional birch-bark receptacles. To make doubly sure that these were water-tight, I smeared the edges of the packages with pitch, as I had seen Joe repair rents in the canoe, and having thus provided against future showers as far as was possible, I sat down to

my meager meal and the world and my future took on a more roseate hue. While I was fireless during the forenoon I had determined to try a bow-drill and spindle for making fire, for I felt that if I could obtain the proper materials this would be a far easier and quicker way of making fire than by flint and steel, which could be reserved for emergencies.

With this object in view I entered the woods and searched diligently for materials for my fire-making apparatus. As I have already mentioned, I had made fire by this crude, savage method when a boy and I knew by experience the materials best suited to the purpose.

The bow was an easy matter, for spruce was as good as anything, and this tree was abundant everywhere. While cutting the bow for fire-making I remembered my determination to attempt the manufacture of bow and arrows and I selected several likely-looking spruce boughs for this purpose. I next looked about for a suitable stick for the drill and selected some straight, old, dry fir roots from a tree which had been torn up and blown over by some winter's storm years before. A piece of the dry, weathered wood from the same tree served as material from which to make a fire-block, and from beneath the bark of a dead pine I secured a good supply of "punk." A hard pine knot was selected for a drill-socket, but despite every endeavor I could find nothing which I was sure would serve as tinder. Shredded cedar bark I knew was as good as anything, but not a cedar could I find, and finally I decided to try the thin, papery, dried birch bark which flaked in little curling rolls from the trees. Armed with these various things, I returned to my lean-to and was soon busily preparing the materials for use. The flexible, springy spruce limb was whittled down to a rude bow, and not until then did I remember that all my youthful attempts at thus making fire had proved failures until I used a leather bowstring. For a moment I was nonplussed, for leather was out of the question, until I thought of my shoe-laces. One of these was sacrificed and replaced by hemlock roots, and I then whittled down a fir root into a double-pointed, octagonal spindle about fifteen inches long. With the tip of my knife-blade I dug out a recess in the pine knot and whittled the outside to fit easily in my hand and then turned my attention to the fire-block. A piece of the dry, seasoned fir was split into a little slab about three-fourths of an inch thick with notches cut along one edge, and I was ready for my experiment at fire-making.



MY FIRE-MAKING APPARATUS

Upon a smooth, dry stone I placed a piece of the dry pine punk, with another piece close at hand. Next I set the fire-block upon the piece of pine and with the bowstring took a turn about the center of the drill. Setting one end of the drill in a notch of the fire-block, I placed the drill-socket, made from the knot, upon the other end of the drill, and steadying the fire-block with my foot I pressed firmly down upon the socket with my left hand and drew the bow back and forth with my right hand. With even, steady strokes I whirled the drill around and around, and presently a little mound of brownish, powdery sawdust began to accumulate on the punk beneath the fire-block. Gradually the pile increased, the hole made by the drill in the block grew larger and larger, and a faint smell of scorching wood greeted my nostrils. Harder and harder I pressed down on the socket, faster and faster I twirled the drill, and an instant later the sawdust turned black and a slender column of smoke rose from it. Dropping drill and bow, I stooped and blew gently on the smoldering powder, and as the smoke increased I lifted the fire-block from the punk beneath, slipped a few bits of the papery birch bark into the powder, clapped the second pine punk on top of all, and, seizing the whole in my hand, waved it swiftly back and forth. Hardly had I swept it through the air when the bark burst into flame, and, knowing success was mine, I danced and capered about, as pleased as the first time I had accomplished the feat, years before. The tinder, punk, fire-block, and

socket were inclosed in birch-bark packages, the drill and bow were laid carefully in the roof of my hut, and I felt sure that I would be able at any time to secure a fire in dry weather and, unless soaked with rain, that I could be reasonably sure of kindling a flame even in wet weather—for I now had two distinct methods of obtaining fire.



“WITH STEADY STROKES I WHIRLED THE DRILL AROUND AND AROUND”

My fire-making apparatus was such a success that I was anxious to go ahead with my bow and arrows, and I spent a long time scraping and whittling down the best of my spruce branches to form a bow. The ones I had selected were dead, seasoned limbs, for I well knew that green wood would warp and would have a very limited spring. At last one of the boughs was fashioned to suit me and I looked about for a bowstring. Hemlock roots seemed the only available material and a long time was spent in braiding enough of the fine roots together to form a string for the bow. Eager to try the new weapon, I cut a notch in one end of a fairly straight stick, placed it on the string, and drew the bow. As I released the string the bow sprang straight with a delightful “twang,” and the stick went humming through the air, but with a loud snap the string parted. I was so greatly pleased at the

strength and elasticity of my bow that the mere matter of the parted string troubled me very little, for I felt confident I could make some sort of a cord which would be strong enough for the purpose and I dropped my bow and hurried into the woods to search for suitable sticks from which to make arrows. Sticks there were in plenty, but, although I sought everywhere, I was unable to find one which was really straight and smooth. Cutting the best I could find, in the hope that I might be able to whittle them into presentable shape, I made my way back to camp.

I was exceedingly hungry, and with my mind on food I examined the beaver meat and fish which I had hung up.

It was an ill-smelling mess, and without more ado I cast it into the river and dined on mussels and fungus, for I was too tired to attempt a trip to my frog-pond or the brook. The next morning, however, I visited the brook and my deadfall, but the latter was empty, although sprung, and I failed to secure a single trout. The reason was simple. The brook had been so swollen by the recent rains that it was impossible to dam up any of the pools, while the pond was filled to overflowing and only one small frog could be found by dint of the most careful search. Despite my ill luck, however, I returned to camp quite elated, for while making my way about the little pond in search of frogs I had discovered some thick bushes with reddish stems so straight, smooth, and polished that they at once struck me as being perfectly adapted for arrows. Not until long afterward did I learn that this bush was known as "arrow wood" and that the Indians formerly used it for their arrows.

With a supply of this useful bush I busied myself at arrow-making, for although I had no feathers I thought that I might be able to make arrows which would serve to kill the tame and unsuspicious birds and animals, and I had but to kill one large bird in order to obtain feathers to make better arrows. Several times I had seen partridges or grouse, and on one or two occasions I had attempted to snare them by means of a hemlock-root noose on the end of a light pole, but the material was too coarse for the purpose and the birds invariably avoided the snare. Once or twice I had attempted to kill them with stones or clubs, and once I had even thrown my spear at them, but in every instance they had escaped. Perhaps it was the season, perhaps the birds were suspicious of the first man they had seen, but whatever the reason, the fact remains that they were far wiser and more wary than the grouse I had often seen when hunting in Joe's company.

It was a simple matter to cut notches in one end of each arrow, but it was a far more difficult job to fit heads. The beaver's bones were the only material I had for this purpose and I found it hard work indeed to cut and sharpen these into any semblance of an arrow-head. Indeed, I found it so difficult that I even sought to chip arrow-heads from the pebbles of the river, but I had not the remotest idea how stone arrow-heads were made and my efforts in this direction resulted only in bruised fingers and irregular, broken stones of no earthly use for my purpose.

By dint of hard work and the expenditure of many hours I finally cut and ground down some bones until they had sharp points at one end and a recess at the other, and to these I bound my arrow sticks with the sinews taken from the beaver. I still had a bowstring to make, and as I worked away at the bones I busied my mind trying to invent some sort of cord which would stand the strain of the bow. I thought of the tendons of the beaver, but these were neither long enough to serve the purpose nor were there enough to braid together to form a string, and I was at last compelled to fall back upon hemlock roots. An examination of the broken bowstring revealed the fact that it had parted at the knot at one end, and to avoid this I decided to braid or lash a loop in the new string. I made this new cord much heavier than the old, selected the fibers with greater care, and smeared the whole with pitch. The loops at the ends were twisted in and lashed in place with tendons, and when all was done I drew the bow with some trepidation for fear all my hard labor would be wasted. Much to my satisfaction, the string withstood the strain and I practised until dark with straight sticks which had bits of stone gummed on with pitch for heads, and I found that up to twenty feet I could frequently hit a mark the size of a partridge.

Anxious to test my weapons on real game, I arose early the following morning and entered the woods in search of partridge. I soon flushed a flock of grouse from among the young fir-trees, and as they perched upon the branches and craned their heads to view the intruder I approached closely, placed an arrow on the string, drew the bow, and let drive. I doubt if I was a dozen feet distant from the birds and they were packed so closely together on the branch that I could scarcely have missed them, but when the bone-tipped stick struck one of the grouse in the breast and with a great flapping he came tumbling to earth, I felt as if I was the most marvelous archer in the world. As the partridge fell the others took wing and whirled out of sight, but I paid little attention to them and hurried to pick up my first

feathered game. The arrow was still sticking in the bird's flesh, although the stick had been broken in his fall, but the head was the only valuable portion, and I hurried back to my fire, happy in the thought that I now had a weapon with which I could actually kill game.

The wing-feathers of the grouse were carefully saved, and after I had dined from the delicate meat and had picked every bone clean I devoted all the rest of the day to feathering and pointing my arrows. How to carry them was the next question, and here the beaver skin came into mind. I was learning rapidly to think out and to find ways and means, and was acquiring a store of useful knowledge, and I smiled to myself as I thought how far better equipped I was to make my journey out of the woods now than I would have been when first I scrambled out of the river not so many days before.

The beaver's skin furnished an excellent quiver, or case, for my bow and arrows, with plenty of room for a supply of mussels and fungus, and my fire-bow and drill in addition, and as there was nothing more to detain me here I decided to start on my tramp the next morning.

I ate a plentiful breakfast of fungus and mussels, and then, with the skin filled with my possessions on my back, with pockets bulging with hemlock roots, tendons, bones, and flint, and with a number of mussels and some fungi tied in a bundle in one hand, and my frog-spear in the other, I set out on my long tramp. As I reached a bend of the river and glanced back for a last look at the little lean-to beside the river, I felt as if I was leaving home. The wilderness had been kind to me and I had fared far better than I had dared hope in this spot. As I turned again toward the south and picked my way along the river-bank, little did I dream what fate had in store for me or how many dreary months would pass ere I reached my goal.

CHAPTER V

I GO A-FISHING

All day long I tramped onward, following the course of the river, but frequently entering the woods and trudging through the forest for several miles to avoid impassable portions of the river-bank. Quite frequently the shores rose in steep, rocky bluffs, between which the torrent roared and foamed, while at other times fallen trees, driftwood, and logs made progress along the shore impossible. Many a time that first day of my journey did I have cause for thankfulness that had taken the precaution to learn means for determining the points of the compass, for the knowledge saved me many a weary mile.

Late in the afternoon I made camp at a little cove where the river cut into the woods and where a crystal brook babbled through a fern-grown ravine and gave promise of trout and frogs. My first work was to build a tiny lean-to, and in doing this I saved myself a deal of labor by using dead and fallen branches for the timbers of my shelter instead of cutting them from living trees. I soon started a fire, and then walked up the brook in a search of game. I had expected to find a few frogs or perhaps to obtain some trout, but presently a flock of grouse whirled up from the ferns and alighted on a low spruce a few yards away. It took me but an instant to fit an arrow to my bow and to let it fly at one of the birds. I made a clean miss and, rather chagrined, I tried again. Once more I missed, but the stupid birds remained motionless and not at all frightened by the passing arrows. As I watched them and wondered if it would be possible to approach more closely I remembered the beaver sinews and determined to attempt snaring the grouse. Rapidly forming a noose with one of the fine tendons, I attached it to the butt of my frog-spear and cautiously crept toward the unsuspecting birds. When within reach I slowly pushed the pole forward, and although the grouse craned their necks, moved about a little, and showed some nervousness, they remained upon their perches, and an instant later the noose was slipped over the head of the lowest and with a quick jerk I

brought him fluttering to the ground. Even then the other birds did not take flight, and three fine grouse were mine ere the others realized their danger and winged their way to safer quarters. I was greatly elated at my success and dined royally on partridge, and had enough left over for my food for the next day.

As I sat by my fire that evening I thought over my life since the day when I was cast into the river, and, much to my surprise, I found it difficult to fix the days and the sequence of events in my mind. Then for the first time I realized that if I was to keep account of time I must devise some means of recording the days.

My first idea was to cut notches in a stick, one for each day, but I at once gave this up as impracticable, for I foresaw that the numerous notches representing the days I had already passed in the woods would prove confusing, and that this method would merely enable me to tell how many days had passed and would fail to give me an idea of the day of the week or the month. Moreover, to carry sticks for this purpose would be a nuisance, and after some time I decided to make a rude calendar by means of beaver tendons. My scheme was very simple and consisted of using two tendons, one for the week-days and the other for the months. Each day I would tie a knot in the week string and when the seventh day was reached I would tie a large knot. Then when the days made up a month I would tie a knot in the month string. To think was to act, and selecting a smooth, long tendon I tied knots to represent the seven days I had already been in the forest, with the last knot double the size of the others, and as the canoe had been wrecked on Wednesday, the 2d of August, I tied nine knots in my month string, which gave Wednesday as the day of the week and the ninth day of the month as the correct date. I could easily remember the month itself and I had not the least expectation of being in the wilderness long enough to require a means of keeping track of future months, but as it turned out, many a month string was tied into knots ere I came to my journey's end.

For several days I tramped onward without adventure or incident, save that I fared ill for meat and was obliged to depend almost entirely upon the mussels in the river and the fungus in the woods. Over and over again I gave thanks to the rabbit which had first led me to this supply of food, for without the fungus I would have gone to bed hungry on many a night. Several times I saw hares and once or twice I flushed partridges. I repeatedly tried to kill these creatures with my bow and arrows, but I failed

each time. Moreover, the grouse seemed wild and suspicious and I could not approach closely enough to snare them, while the brooks I passed, although alive with trout, had no deep pools or isolated basins which I could bail out to secure the fish. Anxious as I was to get out of the woods without delay, my longing for better food finally overcame my impatience and I decided to make a halt for a day and endeavor to trap or snare some sort of game.

Accordingly, I made camp at midday and spent the afternoon preparing twitch-ups and deadfalls. It was while setting one of the latter that an accident gave me an idea which proved of the utmost value and made my lot far easier. Bending over and endeavoring to lift a log, my belt parted, and to my chagrin I discovered that the stitches which held the buckle had ripped out. Holding it in my hand and thinking of a way of lashing it to the leather again, it suddenly occurred to me that in this bit of metal I had the means of fashioning a fish-hook. The buckle was a fairly large one, with a strong, sharp tongue, and one end of this was already formed in an eye. All that was necessary was to detach the tongue from the buckle-frame and bend it into a hooked form. Instantly the deadfall was forgotten and I set to work hammering the buckle with stones and bending it back and forth in order to remove the tongue.

The amount of labor which was required to accomplish this simple matter is almost incredible. My fingers were bruised and torn, my hands were cramped, and my arms ached ere the buckle-frame finally parted and the tongue was free. Even more difficult was the task of bending this, for my only tools were the water-worn pebbles. Time and again the bit of metal slipped and I yelled with pain as my crude stone hammer struck my fingertips. At last I hit upon the plan of heating it and placed the tongue amid the hot coals of my fire. While it was heating I cudgeled my brains to devise some method of holding the hot metal while bending it, and at last hit upon what I considered a very clever scheme indeed. With my knife I cut a notch in a piece of green wood and in the center of this dug out, with great labor, a little depression at right angles to the notch. The metal was now red-hot, and, carefully lifting it with a green stick, I laid it in the notch with its center above the hollow. Then a chip of stone, which I had already selected, was placed against the steel and with a rock as a hammer I drove down upon the metal until it bent into the recess below and took its shape. The steel bent so readily when hot and the whole operation was so simple

that I mentally called myself a fool for spending so much time and mashing my fingers trying to bend the metal when cold by means of stones. As soon as the buckle-tongue was bent I threw water upon it to cool it and examined the result of my ingenuity with the greatest pride. Undoubtedly it was a hook, but doubts assailed me as to its value for catching fish, for it was large, coarse, and clumsy and was scarcely better than an enormous bent pin. However, I had seen trout caught with bent pins and the only way to prove whether or not my trouble had been for naught was to test the hook. It required but a short time to gather a quantity of fine roots and tie these together in a line, but I found it a hard matter to secure a root fine and strong enough to pass through the eye of my hook, which had been squeezed out of shape while bending it. Again the useful beaver tendons came into mind and with one of these I readily attached the crude hook to my line. Equipped with this primitive tackle and with worms, which I found beneath stones, for bait, I hurried to the neighboring stream and dropped the baited hook into a shaded, deep pool among the rocks. Hardly had the hook touched water before a silvery body flashed, and by the sharp, hard tug on the line I knew I had hooked a trout. Fearful lest my captive should slip from the barbless hook, and tingling with excitement, I pulled in the line; but I was doomed to disappointment. Scarcely had his flashing body reached the surface of the water when, with a flap of his tail, the trout leaped into the air, shook himself free, and dropped back into the pool while I stood gazing stupidly at the empty hook dangling at the end of the line. I was filled with bitter disappointment, for I had been confident that the fish was firmly hooked, and for several moments I could not overcome my chagrin at the loss of the fine trout. However, I had learned patience and perseverance, and, again baiting the hook, I tried my luck once more. Again came the sharp tug at the line as a fish took the bait, and this time, instead of pulling slowly on the line, I gave a quick, hard jerk and to my intense joy saw the flashing trout flung from the pool and landed safely upon the mossy bank. Now that I had learned the trick, I found little difficulty in securing several more trout, and while one out of every two or three managed to escape, yet I had plenty of fish for my meal by the time I had stopped. With a supply of trout thus assured, I had no need to worry over my food in the future, but I was terribly afraid of breaking my line and losing the precious hook, and to avoid all chance of this I spent the remaining hours of daylight in carefully braiding a better line from fine, selected hemlock roots.

My thoughts were so fully occupied with fishing that I almost forgot about the traps and snares I had set, and I was on the point of again setting out on my tramp when I remembered them. One deadfall had been sprung and contained a tiny ground-squirrel, but a skunk or fisher-cat had visited it before me and only a few bones and a little fur of the chipmunk remained in the trap. My twitch-ups were far more successful and in one I found a fine, fat hare and in another a red squirrel.

These I decided to use for my midday meal, and, well pleased with my prowess as a trapper and fisherman, I continued on my way.

For some time I made good progress, but soon the country grew rough and more mountainous, while the river flowed for long distances between rocky, precipitous banks which compelled me to make my way through the forest. Here it was also difficult traveling, for fallen trees were scattered everywhere, the ground was rocky and full of holes and clefts, and I was compelled to go far out of my way in order to avoid such obstructions. So hard was the way that I longed again to be able to follow the shores of the river and made frequent trips to the edge of the bank, hoping each time to find the bluffs were passed and that I could again travel in the open beside the stream. On one such occasion I was standing at the verge of a high, steep bank with the river tumbling and roaring in masses of foam among the jagged rocks far below. Suddenly I felt the earth give way beneath me and with a cry of terror I clutched frantically at the bushes about me. With a roar and rumble a great slice of the bank fell crashing down to the river. The branch which I had seized snapped, I felt myself slipping to certain death, and the next instant found myself poised in midair above the precipice. Although the branch had snapped off, it still supported me, and, sick with fear and with pounding heart, I drew myself, hand over hand, up the edge of the bank and fell panting on the earth.

Terribly shaken and helplessly weak from the shock, I lay trembling upon the ground, for I had missed death, or terrible injuries, by a hair's breadth. I had been saved as by a miracle, and I breathed a prayer of thanks that Providence had guided my hand to grasp a branch which was strong enough to withstand the terrific strain of my falling body. Then, having in a measure regained my breath and self-control, and curious to see the sapling which had saved my life, I rose unsteadily to my feet and cautiously approached the precipitous bank. Lying prone on my stomach, I peered over the edge and a wave of faintness swept over me as I gazed down at the

tumbling rapids and jagged black rock at the foot of the sheer decline. Close at hand was the slender growth which had proved my salvation; bent, bruised, and drooping from my struggles, but still intact. It was scarcely as large as my finger. Filled with amazement that such a tiny bush could have supported me, I examined it with minute care. The wood was cracked and broken in a dozen spots, the bark was split and separated from the wood, but it was still as tough as a leathern thong, flexible as whalebone, and fibrous as a rope. Carefully cutting the branch, I withdrew to a point of safety to investigate its marvelous strength. Twist it, bend it, or pull it as I would, I could make no impression upon it, and it came at once to my mind that here, indeed, was a natural rope of immense strength and which would be of inestimable value to me. I also discovered that the strength was all in the bark, and by stripping off slender pieces I found that with them I could form cords, threads, and lines equal to hempen strands. Once again an accident had led to a valuable discovery, and as the shrub was abundant everywhere along the river's bank I knew that I now possessed an inexhaustible supply of lines and ropes which I could use for numberless purposes. Not till long afterward did I learn that this was "moosewood"; that its properties were known to every woodsman, hunter, and trapper of the wilderness, and that to the Indians it served every purpose of string and rope and was in constant use a thousand and one ways. I was still so upset by my terrible experience that I abandoned any idea of proceeding farther that day, and made camp a short distance away beside a little mountain stream. Fascinated by the tough and flexible character of the moosewood bark, I spent hours braiding and twisting it into cords of various sizes, and it was so far superior to the hemlock root that I made a new and better bowstring and a new fish-line from the material.

Then having secured a good rest and having quite recovered from my fright and shock I tried my hand at fishing. It took me but a short time to catch a number of trout, as they were abundant in the brook and I had now learned the knack of jerking them from the water so quickly that they could not flap loose from the hook. The next morning I resumed my tramp and for five days walk steadily onward without any incident worth of record.

With every mile the river increased in width, sand-bars and rocky islets rose in its midst, the current became less swift, and by the sixth day the stream stretched in a broad, sluggish expanse of silver a quarter of a mile wide.

Late in the forenoon of the seventh day I toiled up the slope of a low ridge and amid the tree trunks on its summit I saw the bright sky glimmering through the forest ahead. My heart leaped with joy at the sight, for I knew that the wilderness must be nearly at an end, that open country must be just beyond, and that my journey must be nearly over. Tired as I was, I hurried onward, thrilled to think that my tramp would soon be finished, that but a few hundred yards more and I would break from the woods and look upon open fields, a clearing, or some similar scene of the outposts of civilization, and that ere nightfall I would be talking with my fellow-men. Forgetting my weary, blistered feet, forgetting that my clothes were torn and ragged, forgetting the hardships I had suffered, I pushed rapidly forward, my eyes fixed upon the sunlit sky among the trees to the south, and all unmindful of the fallen branches, the thick underbrush, and the brambles that beset my path. Already the character of the woods had changed. Among the somber evergreens deciduous trees grew thickly; open, brush-filled glades were here and there; patches of blueberry-bushes grew in the hollows beneath the trees, and on every hand were indications that I was approaching the verge of the forest. A few moments more and between the tree trunks I caught a glimpse of light, broken by breeze-swayed golden leaves, and with a glad cry I broke into a run and dashed forward. I crashed through the last small growth, burst forth into glorious sun-filled, open air, and the next instant sank, exhausted and bitterly disappointed, to the earth.

CHAPTER VI

I MEET WITH AN ACCIDENT

I had expected to see tilled fields, shingled houses, perhaps a distant village or at least a clearing, with blackened stumps, a tiny shack, or rude buildings marking the outposts of the settlements. Instead, I found myself upon the banks of a great lake, the wooded shores unbroken by clearing, field, or house as far as eye could reach. A lake of the wilderness, silent, bathed in a vast solitude, and with naught but the forest-clad hills reflected on its placid blue bosom. Far out upon its surface a solitary loon swam slowly, a tiny ripple marking his wake, and from his throat came a peal of weird, maniacal laughter.

Echoing from the hills, multiplied a thousandfold, from every direction the sounds poured forth, as though the woods were filled with demons, mocking my disappointment and my plight.

A half-hour previously I had been filled with happiness and elation at the thought that my journey was over. Now I was cast down, discouraged, and ready to give up, for I realized that to cross the lake was impossible and that for miles and miles I must push through the forest around its shores ere I could hope to proceed on my southerly course. Dimly I remembered that Joe had spoken of a lake and no doubt this was the very one he had in mind, but whether the settlements were to the south, east, or west I knew not. With wistful, weary eyes I gazed across the broad, blue expanse of waters, and the forests beyond seemed vast indeed, and to penetrate them and find my way to civilization through their labyrinths seemed a hopeless, endless task.

Gradually, however, a measure of common sense returned to me, and I realized that I was terribly hungry and that here was as good a place as any in which to camp. No doubt the lake teemed with fish, and presently I had baited my hook and had cast the line into deep water from a jutting ledge. Scarcely had the hook descended when it was seized and a moment later a large salmon-trout was floundering about among the ferns behind me. Three of the great fellows were landed, and, well satisfied with my success, I soon

had one broiling over my fire. After my meal I felt far better and gradually my first disappointment wore away and I began to plan for the future. If I was compelled to walk about the lake I would not want for food, at any rate, and, moreover, I thought, traveling along the shore would be far easier than tramping through the woods or clambering among the boulders and up and down the high banks of the river. By the time the setting sun painted the lake with crimson and gold and the distant forests turned purple and blue in the fading light I was again cheerful and went to sleep upon a bed of fragrant fir in a lean-to, fully expecting to resume my way upon the morrow.

Traveling along the shore of the lake was by no means as easy as I had anticipated, for dense brush and a tangle of brambles grew close to the water's edge in many places and here and there deep alder swamps compelled me to make long detours. But in the swamps I found frogs, which gave me a change of diet, and I managed to capture a fine, big turtle that was sunning himself on a log on the lake's shore. Two days of weary tramping brought me to the western end of the lake and once more I became utterly discouraged as I saw that my efforts had been in vain. The outlet of the lake was a broad, swiftly flowing river which barred me from the farther shore as effectually as the lake itself. My case seemed hopeless indeed, for the river, where it emptied into the lake, was, I knew, fully as wide and impassable as the outlet, and I could see no way of reaching the country to the south save by retracing my steps for miles, crossing the river where it was narrow, and then following down its other bank. The very thought of this made me sick at heart, for it meant over a week's steady walking through the forest. Moreover, my shoes—light, rubber-soled affairs which I had donned only to wear in the canoe—were almost worn out, my clothes were in rags and tatters, and my flesh was scratched, bruised, and torn by brambles, brush, and stones. My coat, flannel shirt, hat, corduroy trousers, and heavy boots had all gone down with the canoe and I had been compelled to journey through the forest and endure all my hardships in light tennis-shoes and with only a white cotton shirt and a pair of thin worsted trousers for my costume.

Although I had satisfied my most pressing needs, which were food, fire, and shelter, yet I had suffered a great deal from exposure, the bites of black flies and mosquitoes, and the scratches of brush and brambles. Hitherto my mind had been so filled with more important matters that I had paid little

heed to these discomforts, yet now my sufferings came back to me and I could scarce bring myself to think of again facing the long journey necessary to carry me back to a spot where I could ford the river.

To add to my misery it commenced to rain and I was forced to build a lean-to and a fire and to camp where I was by the outlet of the lake. It was no drizzle, such as I had experienced hitherto, but a torrential downpour, the water falling in perfect sheets and at times completely blotting the farther shores from my view. Within a few hours the river was a foaming torrent and the waters of the lake had risen perceptibly. Huddling in the shelter of my rude hut and striving to keep warm in the glow from my fire, I gazed upon the gray, wind-lashed lake and the dripping, sodden shores, feeling unutterably miserable. Near my camp a large log was stranded on the beach and idly I watched it as the waters crept higher and higher, and I began to wonder if the lake might not rise to the spot whereon I had made my lean-to. I was pondering on this and had almost decided to brave the elements and make another shelter on higher ground when a sudden gust of wind whipped the water into foam-capped waves and with a final lurch the log slipped from the shore and floated, bobbing up and down, on the surface of the lake. A moment later it had felt the suction of the current and, lurching and rolling, was drawn rapidly toward the river. It was a small matter, but it was of the greatest importance to me, as it pointed a way by which I might escape from the wilderness.

For the first time it now occurred to me that by means of a few logs I could build a raft and cross the lake as soon as the storm passed and the waters were again calm and peaceful.

My first idea was to gather together logs which had been cast on the beach, float them in shallow water to a spot near my camp, and then bind them together with moosewood bark, vines, and roots until I had formed a raft which would support my weight. Accordingly, I set forth on the following morning to gather my logs together, but I soon found that my scheme was worthless, for the high water and wind had carried off any stranded logs and trees which were buoyant enough to float, and during the entire forenoon I did not succeed in finding a single log which would serve my purpose. At one spot, however, I discovered a number of dead trees lying on the bank well away from the shore, and I decided that if these could be rolled or pulled to the lake they would serve excellently. It seemed an easy matter to accomplish, but the very first log which I selected proved

how futile were my efforts. The fallen tree was some twenty feet in length and over eight inches in diameter at the largest part, and while its weight was not so tremendous, yet its length and the numerous stubs of branches which projected from it prevented me from making the least headway. After hours of back-breaking toil I succeeded in prying it up and in placing small limbs beneath it, but despite every effort I was utterly unable to force the log sideways or endways or to roll it along. Finally exhausted, hungry, and discouraged, I gave up in despair and made my way to my fire to cook my meal. The fire had been built near a good-sized tree and as I approached my camp I discovered that the flames had spread to the tree, had eaten their way through the trunk, and that it had fallen upon my lean-to, which was a broken, flattened mass of twigs and branches. It was a small loss, but I was thankful that I had not been within the shelter when the tree came crashing down upon it. The next instant I gave a shout of joy, for the fire had shown me a way to solve my difficulties with the logs. Dolt that I was not to have thought of it before. I could cut the logs into easily handled lengths by means of fire, and for that matter could fell standing trees by the same means, more easily, if more slowly, than by an ax.

The tree which had been accidentally cut down was scarcely six inches through, but it was a green, live tree and I realized that much larger dead trees could be cut in less time, and as soon as I had eaten I gathered up a firebrand and started toward the logs on which I had labored so strenuously and with such poor success. Placing a pile of dry twigs and branches under a log, I soon had them ablaze and threw more dry sticks around and over the trunk. Very soon the dry wood of the tree had caught fire and was burning merrily, but, much to my chagrin, I found that the flames were spreading along the surface of the log, fanned by the breeze from the lake. For a moment I was in a quandary to prevent the entire log from being consumed before it was cut through. Then, gathering up handfuls of the soft, damp earth and moss, I plastered it over the surface of the log on either side of the fire. Had I possessed a birch-bark dipper I could easily have thrown water upon it, but I had not used a birch-bark dish for days and had no time in which to fashion one at the moment. The wet earth checked the spread of the flames, and as fast as the heat dried it off I threw on more, and within an hour the log cracked, sagged, and a moment later broke in two. My experiment was a success, and, quickly raking away the fire, I started it in another place. All through the afternoon I worked and before nightfall

had six lengths of smooth, dry logs cut and ready to roll to the lake. They were all free from projecting branches and stubs, which I had also burned off. My eyes were smarting with smoke, my skin was scorched, and my hands black with muck and grime, and I was inexpressibly tired, but I was supremely happy, for I felt that the worst of my labors were over and that within a couple of days I would be paddling across the lake toward the southern shore.

A bath in the lake did much to refresh me, for although the water was icy cold, it served to relieve my numerous burns and the irritation of my eyes, and the next morning I felt well able to resume my work of getting the logs to the water's edge.

It was not difficult to pry up the short, eight-foot lengths and to place branches beneath them, and at the end of an hour I saw the first log slip down the bank to the shore and splash into the water. I fairly cheered as the piece of trunk floated within a few feet of the beach and I felt as much elated and as proud as though I had launched a ship. Hastily dragging it to the shore, I secured it to the bushes and climbed up the bank to attack a second log.

Experience had taught me much and I found far less work in prying up and sliding this log than I had expended on the first. In a short time I had it beside its fellow on the beach, and when I at last ceased my labors to prepare my midday meal only two logs remained upon the bank.

I was very tired, for I had worked ceaselessly since early morning, but I had accomplished a great deal and felt that I had earned a few hours' rest. It was a pleasant afternoon, warm and balmy, with a soft, blue haze half-veiling the distant woods and hills, and had I possessed the necessities of life and a canoe I would have been glad to spend many days camping on the borders of the lovely forest lake. As I rested, lying upon the mossy bank and gazing off across the sparkling water, I was attracted by a slight rustle near at hand, and, turning my head, saw a graceful deer step forth from the ferns. For a moment the splendid creature stood, suspiciously sniffing the air, and then, scenting no danger, trotted to the water's edge and plunged into the lake.

Straight toward the opposite side he swam, his antlers, ears, and nose alone showing above the rippling surface of the water, and a long, silvery wake trailing behind him. How I envied his powers of swimming and how I longed for a means to slay him, for my admiration of his beauty and grace

was as nothing compared with my longing for broiled venison and for some tough buckskin with which to cover my poor, blistered, bruised feet. Perhaps, I thought, I might devise some trap with which to capture a deer, or by finding a runway I might noose one, and I sighed to think how I was compelled to suffer and to win a bare existence although surrounded by game which would furnish me with innumerable comforts.

But if I was to complete my raft it was high time I resumed my labors. Arising regretfully, for my rest upon the moss was delightful, I tramped through the woods to the two remaining logs.

Presently I had pried one of the logs from behind some projecting stumps and, having placed branches beneath it to hold it up, I dropped my lever and started back to secure some pieces of wood I had used as rollers. In doing this I stepped upon the other log, my foot slipped, I fell crashing backward, my arm knocked the props from under the log, and with a thud it dropped, pinning my foot beneath it.

A terrific pain shot through my leg, I writhed and twisted, shrieking in agony as I strove to free my foot, and sweat poured from my face with the torture, but still my leg remained fast wedged between the logs, and I saw before me a lingering, terrible death.

Of what use had been all my trials my hard, weary tramping, and my strivings to reach my fellow-men? Of what avail my brave conquering of nature and my struggle to live? Caught like a rat in a trap, I would die by inches here by the borders of the lake and years later some wandering hunter would find my whitened bones and in them read the story of my awful fate. Better by far if I had shared Joe's death in the swirling waters of the river. Half-crazed by pain and maddened to think of my helplessness, I gathered all my strength, gritted my teeth, and with a last despairing effort threw myself sideways and shoved with my free foot upon the log. My foot seemed torn from the ankle, and my whole spine seemed seared with red-hot iron, but despite the agony I knew that I was free; I rolled forward on my face and mercifully lost consciousness.

CHAPTER VII

CRIPPLED

I opened my eyes to find that night had fallen. My first sensations were of unutterable pain; I was chilled through, racked with agony, and weak and faint from my injuries, and my first thought was to strive to reach my camp. Groaning at each motion, I dragged myself forward a few inches, but could scarce move a yard before I was compelled to sink again to the earth, for the torture of dragging my wounded foot through the brush and over the rough earth was more than I could bear.

Thinking to allay the pain somewhat, I sat up, tore off the rags of my shirt, and started to bind this about my foot, but with every motion and with each touch I cried aloud, and only by gritting my teeth and by frequently stopping to rest was I at last able to wrap the cloth about my ankle. I judged that it was crushed and broken, but in the darkness I could not tell the extent of the injury, and merely knew that the pain was excruciating and that the leg and ankle were terribly swollen and caked with blood.

With the wrapping protecting the foot, I found it pained me less as I dragged myself along, but even then the suffering it caused was more than flesh and blood could stand, and before I had covered half the distance to my camp I fainted.

When I once more regained consciousness the sun was rising above the dark forest beyond the lake and a dull numbness had taken the place of the pain in my leg and foot. I was burning with fever. I rolled and dragged myself to the edge of the lake, where I drank and bathed hands and face, and then lay there with my wounded leg soaking in the cold water.

The shock of the cold revived me wonderfully, and as there was now no sensation whatever in my foot or leg I again started toward the camp. I had crawled but a few feet when it occurred to me that a crutch might enable me to walk, and I soon found a stout stick with a fork near one end, which I thought would serve my purpose. Pulling myself up beside a sapling, I placed the stick beneath my arm and, much to my delight, found that I

could hobble along far more rapidly and with less exertion than by crawling on all-fours.

In this manner I reached the camp, but the fire had long since burned out and it was some time before I could gather sufficient strength to start a new one.

When at last the fire was blazing and I had cooked and eaten some food, I felt much better, but I realized that something must be done for my foot at once. To neglect it might result in blood-poisoning and death, and even if this did not occur I would be disabled and prevented from escaping from the forest before winter set in unless a rapid recovery was assured.

Even now I thought it might be too late, for, after lying on the damp ground all night with the foot left to itself, I feared that cold might have settled in the wound or that dangerous inflammation might have set in.

My first care, then, must be to make an attempt to reduce the swelling and bind up the wound with some pain-allaying and healing substance, and here again my lack of knowledge of woodcraft and forest resources made me pitifully hopeless. I tried to remember any scraps of conversation or any incidents which I had heard or seen in the woods and which might help me, and at last I *did* succeed in bringing to mind two instances in which my guides had made use of nature's remedies in curing wounds. On one occasion Joe had applied bear's grease to a cut upon his hand, and at another time he had gathered some herbs as we passed through a little glade and to my questions had replied that it was arnica and was used in curing bruises and sprains.

I had no bear's grease and I did not know if arnica was found in the neighborhood, but I decided that the wound must be cleansed and bandaged, at any rate, and that the sooner it was done the better, for the numbness was now passing off and the leg was commencing to pain again.

Carefully I unrolled the shirt, and as I exposed the limb I grew sick at the sight, for leg and ankle were black with congested blood, terribly swollen and misshapen, and with the flesh and skin deeply torn and cut. I could not tell whether it was dislocated or broken; although when I felt of it and moved it carefully with my hand it seemed so loose and caused me such agony that I feared the worst.

It was a long, tedious, and torturing operation to bathe the foot and ankle, wash out the cuts, and bind up the limb anew, but at last it was

accomplished, and with sticks bound about it as splints to protect it, I determined to hobble about and try to discover some arnica plants before I became unable to move.

By means of a crutch I could travel fairly easily over the more open ground near the lake, and I made my way slowly and painfully toward a little open spot I had noticed a few days before. Here rank weeds and plants grew in abundance, and after searching for some time I discovered a clump of herbs which I felt sure were the same as those I had seen Joe gather.

While searching for the arnica I had wandered to the farther side of the glade, and as I glanced about to get my bearings my eyes fell upon a well-marked opening or trail leading into the woods close at hand. Filled with wonder at this, for I knew that the trail must have been made by human hands, I examined it intently. Yes, there could be no doubt of it. There were lichen-covered stumps showing the ax-marks and I realized that by chance I had stumbled upon an old wood road. Thrilled with excitement and almost forgetting the pain in my injured leg, I hobbled forward along the old road, for I knew that it must lead to some definite goal.

I was too overcome with my discovery to think that the road might lead for miles through the woods and that in my injured and weakened condition I was running a grave risk in following it away from my camp, and it was fortunate for me that the end came so quickly.

Hardly had I hobbled forward for a dozen rods along the road when a turn ahead disclosed a broad, sunlit clearing and in its center a house!

For an instant I was so utterly dumfounded that I could scarce believe my eyes, for to find a human dwelling here in the forest and within a few rods of my own camp seemed absolutely incredible. But there it stood, a tiny log cabin basking in the sunshine of the clearing, and with a glad shout, and quite forgetting my injured leg, I hurried forward as fast as my condition enabled me.

No voice replied to my cry, and as I reached the open doorway I realized that the hut had been long deserted and my heart sank. Across the floor dead leaves had drifted ankle-deep; moss grew upon the threshold; lichens covered the walls; sunlight streamed through many a hole and crevice in the roof, and a squirrel scampered into the fireplace and up the chimney as I stepped within the door. Along one side of the cabin extended a rude bunk, a table of slabs stood near the open fireplace, some moldy, cobwebbed fragments of clothing hung from pegs driven in the walls, a rusty iron pot

stood upside down beside the hearth, and a small grindstone was propped against one wall.

It was a curious sensation to stand once more within the four walls of a building, small and rude as it was, and I realized instantly how far superior it was to the best lean-to which I could build.

With my injured leg I would need rest and shelter, and here was a building provided as if by a miracle for my needs.

Throwing down my bundle of arnica, I made my way back to camp, gathered up my few simple belongings, and carried them to the cabin. Then, despite my disabled condition, I made another trip and brought over some blazing brands to kindle a fire in my new home, for the trip of a few score yards was less arduous than the labor of starting a new fire with my crude appliances.

My first care was to make use of the arnica, for my leg was again torturing me, and while I knew nothing of the method of using the herb, I assumed that it should be steeped or boiled and used as a compress, or poultice, and I soon had the freshly plucked plants simmering over the fire in the old iron pot.

Meanwhile I examined the interior of the hut with more care and found that none of the openings in the roof were directly over the bunk, so that I could be reasonably sure of protection from rain until I was able to repair the leaks. The leaves which had sifted within the door were soft, clean, and dry, and I piled a great mass of these upon the bunk. Upon them I placed soft balsam-fir twigs and thus secured a bed which was luxurious and comfortable indeed compared with the crude makeshifts to which I had become accustomed.

By the time the bed was prepared and the hut cleaned out the herbs in the pot were thoroughly steeped, and as soon as the brew was sufficiently cooled I proceeded to place a dressing on my foot.

The swelling had not decreased, but the discoloration and angry appearance of the wound were less, and while the pain was still intense, I found it more bearable than when I had first bandaged it. The soggy, boiled leaves and stems of the arnica I used as a sort of poultice, binding them upon the injured parts with strips of my rags soaked in the liquor, and then, quite exhausted with my efforts in my crippled condition and the pain in my limb, I crawled into the bunk and at once fell asleep.

I awoke much refreshed and very hungry, and to my great relief I found the pain in my foot had grown less and that the inflammation had not increased or spread.

As I cooked my meal over the fire in the hearth I planned for the future, for my good fortune in finding the cabin had caused my spirits to rise and I cast aside the doleful forebodings of dying slowly from my wound, which had possessed me.

That I must remain in the hut until my leg and foot had completely recovered was certain, for to attempt to cross the lake by raft, or to undertake any hard labor, with my foot in its present condition, would merely result in more trouble and longer delay. And then a sudden light dawned upon me, and, had I been able, I would have danced for joy as I realized that the presence of the cabin proved that it would not be necessary to cross the lake, after all. Strange that it had not occurred to me before; for everything about the place spoke of its occupancy by white men, probably gum-gatherers, trappers, or lumbermen, and beyond a doubt it was on a well-marked trail to the settlements, for, had its occupants come hither by canoe across the lake, the cabin would have been built close to shore and the trail would have led to the water's edge instead of into the forest.

As I reasoned this out my mind was immensely relieved, for I felt that as soon as my foot would enable me to proceed I could follow the trail and reach the settlements in comparative ease.

But as I must remain a virtual prisoner in the cabin for the present, I determined to make the hut as comfortable as possible and then conserve all my strength and nurse my foot back to usefulness; and, having eaten, I proceeded to put my house in order.

To repair the leaks in the roof would, I felt, be a difficult matter with my disabled leg, for I would be obliged to climb upon the roof from outside, and this I did not dare attempt. It occurred to me, however, that by placing layers of evergreen branches over the roof a great deal of the rain might be kept out, and this I was able to do by means of a forked pole. The branches, I felt, would blow off with the first wind, and to secure them in position I laid long, light poles diagonally over them, digging the upper ends of the poles into the old roof and lashing the ends at the eaves to the walls of the hut.

This simple matter occupied me until dark and proved a tremendous task, for I could use but one hand when hobbling about on my crutch and my

progress back and forth from the neighboring woods was slow and painful. When the roof was at last finished I was utterly exhausted, and as soon as I had eaten and had placed a fresh poultice on my foot I crawled into my bunk. The door was still open, I had no coverings, and the night was cold, but the fire on the hearth warmed the interior of the cabin and I felt little discomfort from the chill; in fact, I slept better and felt warmer than I had for many nights, even though my shirt had been sacrificed to make bandages.

For several days thereafter I busied myself about the cabin, caught trout for my food, and nursed my wounds, and, much to my satisfaction, I found the ankle was growing no worse and that much of the pain had left it. Still, it did not recover as rapidly as I had hoped and I began to fear that it would be a long time before I would be able to use it again, or that it might heal badly and become useless forever. In rummaging about the cabin I found several articles left by the former owners, such as a couple of empty bottles, some old tin cans, and a rusty old steel file.

The bottles proved useful for storing my arnica brew and the cans I used in cooking, but the file seemed of no value. The clothes which I had seen hanging upon the walls were far too rotten to be of any use as clothing—no doubt they were cast aside as worthless by their former owners—but they were better than nothing as coverings at night, and by piecing them together with hemlock roots I managed to form a ludicrous sort of patchwork quilt which was a real comfort.

I had been at the cabin for about a week when I was aroused one night by a strange sound. Some one or something was moving across the floor. At first I was startled. I had been so long accustomed to solitude that the presence of another alarmed me. Then it flashed upon me that perhaps a former occupant of the cabin had returned and had entered, unsuspecting my presence. Overjoyed at the thought and without stopping to consider the consequences, I called out, “Hello!” and rose up in bed.

At the sound of my voice the footsteps ceased, there was a strange guttural reply from near the fireplace, and a queer rattling sound as of some one crumpling paper.

Puzzled, and now rather alarmed, I spoke again, and instantly the grunt was repeated and I heard the sound of shuffling footsteps hurriedly approaching my bed. Terrified, I shrank back, peering into the gloom of the cabin in an endeavor to discern the outlines of my mysterious visitor. The

fire had died down, but a few red embers still gleamed upon the hearth and cast a faint glow in the immediate vicinity. At the very edge of this dim light I thought I saw a moving shadow—a great, bulky, shapeless form, crouched low, and stealing toward me. My nerves were on edge, and at the sight of this mysterious, formless, threatening figure I uttered a scream of mortal terror. With a sudden sideways motion the shadowy something stepped within the glow of the fire, and then I broke into a roar of laughter.

CHAPTER VIII

A MIDNIGHT VISITOR

My visitor was a porcupine, a great, clumsy creature who was far more frightened at my presence than I at his. Realizing that here was food come to my larder of its own free will, I scrambled out of bed, closed my door, and, carefully avoiding my prickly visitor, made my way toward the fire. Shoving a resinous stick into the coals, I soon had a flaming torch, and by its light I discovered the porcupine scratching and gnawing at the door in an endeavor to escape. I ended his career by knocking him over the nose, and then, barring the door to prevent any other nocturnal visitors from disturbing my slumber, I returned to bed.

The roasted porcupine proved most toothsome, and as I gnawed at his bones I wondered why I had not thought of hunting these creatures before. That they were abundant in the forest I knew, and I also was aware that they were considered excellent eating by all the guides and woodsmen, and yet, until one of the brutes actually forced his presence upon me porcupines had never occurred to me as a source of food. Now, however, I made up my mind to hunt up a den of the creatures at the first opportunity, for I knew that where one was found there were usually numbers of them.

I had met with some difficulty in removing the skin, with its innumerable sharp quills, but I had noticed how tough the hide was and I vaguely wondered if I could not manage to tan and use it.

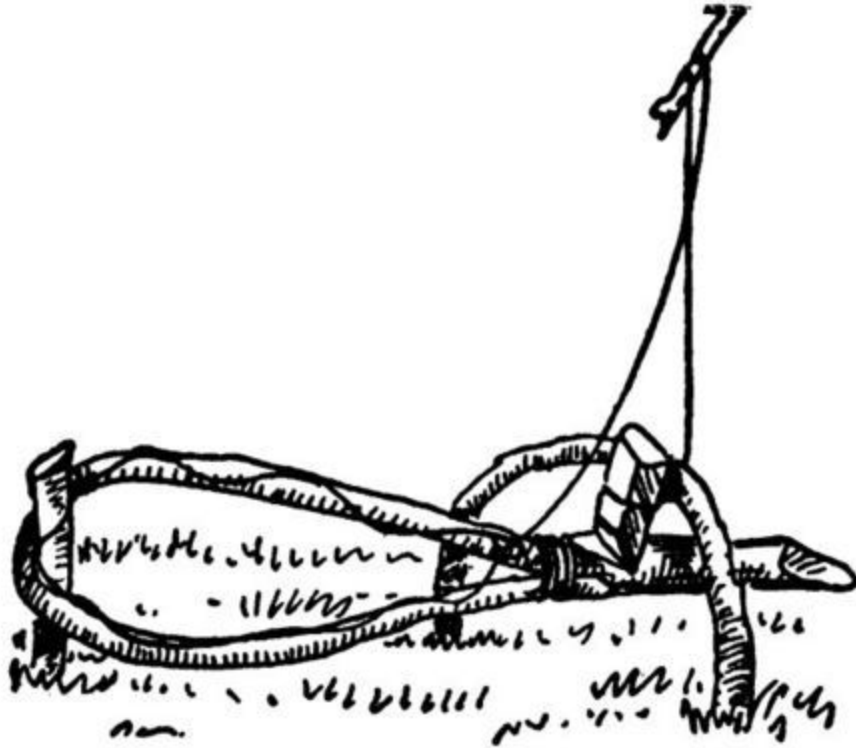
With its adhering quills it was quite useless, and I realized that if merely dried, like the skin of the beaver, this tough, thick skin would be as stiff and hard as a board. If I was to make use of the hide at all I must devise some method of removing hair and quills and must tan the skin so that it would be flexible and soft. In my youth I had often visited a tannery and I knew that oak bark and sumach were used in making leather, but, rack my memory as I would, I could not remember ever having seen the process by which the hair was removed from the hides.

I attempted to pluck out the quills by hand, but only pricked my fingers, and I found it also impossible to cut or shave the hair and quills from the skin with my knife. I had worked at this for some time and was becoming thoroughly disgusted with the matter when it flashed across my mind that I might be able to remove the hair by decomposition. Many a time I had seen the hair or feathers drop from game which had been hung too long, and I knew that long before the meat was tainted or the hide was seriously injured the hair would come away. Realizing that the skin would dry before it would rot if hung up in the air, I buried it in the soft earth in a shady spot and spent some time gathering a quantity of oak and sumach bark, which I boiled in the iron pot.

Having accomplished this, I decided to hunt for the porcupine's home. My foot was now paining me but slightly and I had become quite accustomed to the use of a crutch, so that I was able to start at once. Knowing that porcupines frequented rocky ledges, I turned my steps toward a little wooded rise. A few hundred yards from the cabin I came upon a mass of rocks with great cavities and fissures between them. This seemed a promising spot for a porcupine-den. In my crippled state it was difficult to make my way over and between the boulders, but I had plenty of time and I crawled and scrambled about, peering into every crevice and cranny. I had been thus engaged for several hours when in a deep fissure I saw numerous quills scattered about, while projecting roots in the vicinity were gnawed bare of bark. Here was the den, I felt sure, and without hesitation I scrambled down into it and peered into the darkness.

I could distinguish nothing at first, but as my eyes became accustomed to the obscurity of the cavern I saw a darker spot in the farther corner of the den and heard the rustling, rattling noise of a porcupine's quills as the creature raised them in defense. Now that I had found my porcupine, the next question was how I was to capture him. I had no mind to attempt to grasp the rascal with my bare hands, and, like a fool, I had not brought my primitive weapons with me. I thought of building a fire and smoking the creature out and then knocking him over the head with a club, but my fire-making apparatus was in the cabin and I had no mind to make the journey again if it could be avoided. I was wondering how I could solve the puzzle when it occurred to me that I might snare him exactly as I had snared the partridges. In a few moments I had secured a strong strip of moosewood bark, had formed a noose in it, and with it attached lightly to the end of a

slender pole I poked it into the cave and tried to slip it over the porcupine's head.



THE PARTRIDGE-SNARE

But stupid and clumsy as he was, the brute backed away from the noose each time I poked it toward him, and if over and over again I was compelled to withdraw it and form the slip-noose anew. I was tired and cramped and was about to give in despair when I felt a tug at the bark, and, pulling upon it, found I had at least secured a hold upon my prey. Dropping the pole, I hauled on the strip of bark, and a moment later drew the porcupine from his lair, grunting, scratching, and bristling with anger.

He was snared by one hind foot. I knocked him over the head and, slinging him to the end of a pole, returned to my hut, well pleased with the hunt.

The next day I dug up his mate's skin and, much to my delight, found the hair had started and the quills came away readily. The second skin was buried, and a few days later both were free of quills and hair, and after

washing them in the lake I placed them in the bark liquor which I had prepared. Five days later I removed them, rinsed them off, and was immensely elated to find that they had assumed a rich brown tint and had been transformed to tough leather. Now that my mind had turned to tanning, I recalled many little details. Among other matters I recollected how the tanners worked the hides upon rounded timbers or horses, and, while I did not know the reason why this was done, I assumed it was an essential part of the tanning process, and so I placed the porcupines' skins upon a fallen tree and worked at them diligently until they were thoroughly dried and all bits of adhering flesh and fat had been removed. The results of my labor were two soft, brown pieces of leather nearly two feet in length by a foot in width, and I felt immensely proud of my success at leathermaking. My only regret was that I could not find more of the prickly beasts, but, although I hunted diligently, I was forced to the conclusion that there were no more in the neighborhood.

The pain had now quite gone from my foot, but it was still impossible to use it. The slightest pressure upon it caused me excruciating torture and the ankle seemed to have no strength. I was terribly worried over it, for I feared I would never be able to use it again, and with only one good foot I dared not wander far from the cabin, for fear I might slip or fall and injure my good leg and thus be left helpless in the forest.

On one or two occasions I had hobbled off along the old trail for a mile or more, but it appeared to lead into the heart of the wilderness and I was convinced that if it led to the settlements it must be by a long and roundabout route.

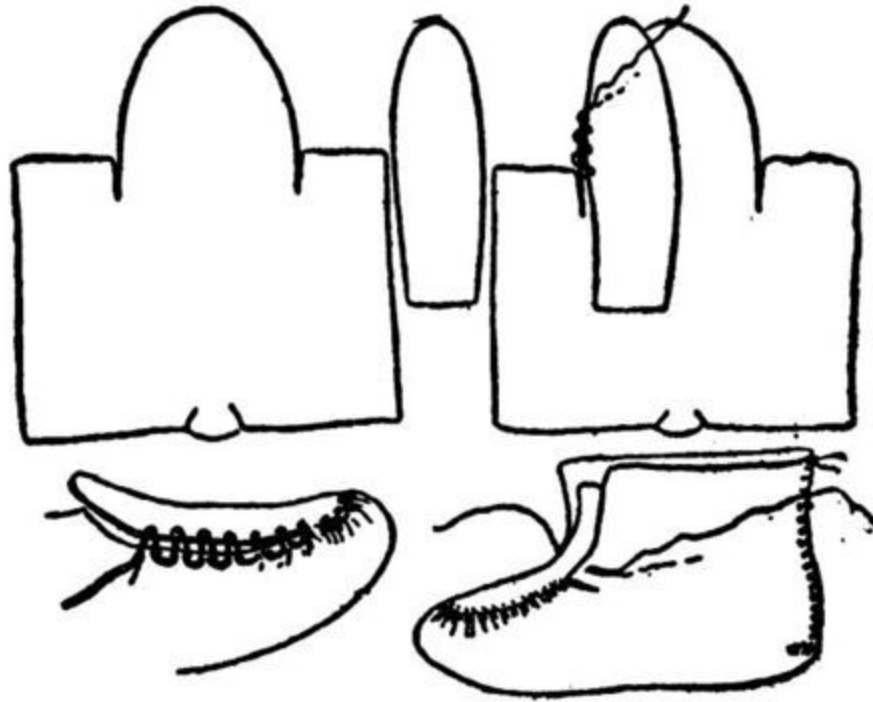
The weather was now becoming very cold, frosts were of nightly occurrence, and once or twice a thin layer of ice skimmed the water at the edges of the pools. I suffered a great deal from the cold, hardened and accustomed as I was to exposure, and I knew that unless I made my way out of the forest or managed to invent some sort of covering for my body I would soon succumb to the weather.

Had I been able to secure enough porcupine-skins, or even rabbit-skins, I might have made some sort of garment, but both rabbits and porcupines were very scarce. I had snared and shot a number of partridges, and their skins, sewed together, I made into a vest-like garment, but the delicate skins tore apart with the least strain and I used the affair only when sleeping or sitting in the cabin.

The ragged, rotten old clothes which I had used as a quilt were some protection, but these soon gave out and I was faced with the serious problem of freezing to death or securing enough hides to make a covering for my nakedness. The chances seemed all in favor of freezing.

My shoes, or rather, my shoe—for I could use but one foot—had now worn through, and I determined to attempt making moccasins from the tough porcupine-hides. Fortunately I had seen Joe make moccasins, and after one or two trials I succeeded in producing a very serviceable form of foot-gear. I remembered that Joe always marked a pattern on the leather around his foot, but I did not feel sure how he gauged the size and, fearing to waste any of my valuable leather by a mistake, I hit upon the plan of ripping open my old shoe and using this for a pattern. Placing the split shoe, sole down, upon the porcupine-skin, I marked around it with a bit of charred stick and cut out the form with my knife. Then from the trimmings I cut tongues, as I had seen Joe do, and these I sewed to the uppers so that the latter were puckered, or gathered up, over the instep. Finally I sewed up the heels, cut string from the hides, and my first moccasin was complete.

All this occupied a great deal of time and labor and was far more difficult than it sounds, for my only tool was my knife, and as I worked slowly and clumsily with this I longed for an awl or needle. With my mind on such things I remembered that I had heard of Indians using bone needles, and I decided to try and make one with which to sew the other moccasin. There were plenty of bones at hand, and I selected several and commenced rubbing one upon a flat stone to grind it to a point.



HOW I MADE MOCCASINS

I had worked diligently for some time when I suddenly remembered the old grindstone and, laughing at my own stupidity, for I had repeatedly used the stone for sharpening my knife, I proceeded to sharpen the bones in a simple and easy manner.

I had ground down one bone to form a fairly fine needle-pointed tool when my glance fell upon the old rusty file. The sharp tang of the file suggested an awl at once, and, marveling that I had not thought of it sooner, I proceeded to grind it to a fine point.

In a few minutes I had an excellent awl and with its aid I found the completion of my moccasins easy. The heavy file made the tool clumsy, however, and as soon as the moccasins were finished I set to work to break the awl-like end from the file itself. It seemed a simple matter and the reader may smile at my spending a moment's thought over breaking off one end of a file, but to me it was an affair of great importance, for I was afraid of snapping the steel in the wrong spot and thus ruining the awl which I found so useful. With only stones as hammers, I realized it would be difficult to break the metal in the desired place, and I pondered on the

matter for some time before it occurred to me to cut the piece from the file by means of the grindstone.

While I was cutting through the file it flashed upon me that the remaining portion of the tool might be transformed into a most efficient weapon. To be sure, I had no real need of a weapon, for my flimsy bow and arrows and frog-spear had served all my needs. But I had seen deer tracks on several occasions and had even caught a fleeting glimpse of the creatures more than once, and while I had made no effort to molest them, knowing the hopelessness of such an attempt, I had longed to secure them and their hides. With a powerful spear, such as I could form from the old file, I thought it might be possible to kill a deer or some other large animal. Therefore I promptly set to work to grind the steel down to sharp edges and a point. I soon found, however, that the grindstone made scarcely any impression upon the file, and that, if I was to succeed, I must soften the metal by means of fire. I hesitated to do this, for fear that I would not be able to retemper it properly, but at last I thrust the file into the coals of my fire. When it was red-hot I raked it forth and allowed it to cool slowly. I now found the stone bit into the steel rapidly, and I felt genuine pride in my accomplishment as I watched the old file slowly assume the shape of a shining spear-head. It was hard work, however, and I stopped to rest many times before the task was half finished. In fact, I spent a large part of two days at this labor, and several times I was on the point of giving up the work as not being worth the trouble and exertion. At last, however, the spear-head was completed, and again placing it in the fire, I heated it to a cherry red, pulled it out from the coals, and dumped it into the kettle full of cold water. I found it had taken an excellent temper, and I then selected a straight, light pole of ash, bound the head to it with sinews and roots, and my spear was completed. I was proud of the result of all my work, for it certainly looked like a most efficient weapon, but I was in some doubt as to my ability to throw it accurately. I hobbled forth to try my skill. I had no mind to ruin or break the weapon by using it against a tree, but I remembered a rotten old stump which I had seen near the edge of the woods. This offered an excellent target.

I approached within a dozen yards of the stump, steadied myself on my crutch, and, balancing the spear, threw it with all my strength. The weapon fell far short of the mark and descended wrong end first. Then I realized that the pole was far too heavy for the head.

Cutting off a foot or so of the haft, I tried again, and found that the spear traveled in a straighter line and it struck near the base of the stump. This was encouraging, and by trimming off the pole a little at a time and by numerous trials I at last had the satisfaction of seeing the keen point bury itself in the rotten wood.

Again and again I hurled the weapon, each time gaining greater skill and retreating farther from the stump, until I could drive the spear into the stump at twenty paces every time.

My arm was now so tired with the unaccustomed exercise that I abandoned my target practice and returned to the cabin, well pleased with the success of my spear-making and quite convinced that if I could but find a deer I would have a very good chance of killing him.

During the day, while busy with my work, and interested, I felt quite optimistic and planned most wonderful things, but when alone in my cabin at night I often grew terribly despondent and saw matters in a very different light. While throwing the spear in the bright sunshine it had not seemed at all unreasonable to think of trailing a deer and killing him like a primitive hunter, but as I thought over the matter that night it seemed a most visionary and ridiculous scheme. To trail a deer and bring him down with a javelin would be a difficult feat at any time, and now, with my crippled foot, I realized that I would be hopelessly handicapped. In vain I tried to persuade myself that, as I had seen deer before, I might see them again, and that chance might favor me; in vain I tried to think up some other method of obtaining the wherewithal for the garments I so greatly needed. I could see nothing promising, nothing hopeful ahead, and, finally, convinced that I was doomed to die here in this deserted cabin in the wilderness, I fell asleep.

I was aroused from my slumber by the sound of something scratching at the door. Instantly I was wide awake, with all my faculties on the alert. Presently the sound was repeated; there was no doubt that some creature was nosing about and endeavoring to enter the cabin. Another porcupine, I thought to myself, and, slipping cautiously from my bed, I grasped my spear and, as silently as my injured foot would permit, stole toward the door. This was a rude slab affair, without lock, bar, or latch, and kept closed by a strip of bark looped over a peg in the wall.

Slipping off this fastening, I threw the door open, and as I did so I stumbled back in terror, and only the heavy door saved me from tumbling

head over heels into the hut. I had expected to see a clumsy, harmless porcupine, and instead, clearly outlined in the bright moonlight, his grinning teeth and gleaming wicked eyes fixed on my face, stood a great, shaggy figure. My visitor was an enormous bear!

For a brief second I stood transfixed with surprise and fear, and then, without stopping to think, I hurled my spear full at the creature with all my strength, slammed the door to, and braced myself against it.

CHAPTER IX

WINTER SETS IN

Each second I expected to feel the enraged bear tearing at the door behind me, and I cursed my folly in having attacked him, for I realized that, unmolested, he would have, no doubt, retreated at sight of a human being, whereas, maddened by the sting of my weapon, he might force the door and tear me to pieces. Even if he dared not do this, he might lie in wait outside and attack me as I went forth in the morning, and I shivered with cold and fear at the consequences of my rash act as I stood there against the door in the black darkness of the cabin.

Minutes passed, and not a sound broke the silence outside, and at last, chilled to the bone, unable to stand the strain longer, and convinced that the bear would not attempt to force his way in, I crept to the fireplace, fanned the coals into a blaze, and sat huddled within the warmth of the hearth. Starting in terror at each sound outside, and not daring to go back to bed, I passed the long hours until dawn.

Daylight has a marvelous effect upon a man's courage, and my fears of the night seemed quite childish as the golden rays of the rising sun penetrated the chinks of my cabin and the scream of a blue jay reached my ears.

Probably my spear had missed the bear completely, I thought, and the poor brute, far more frightened than myself, had put many miles between himself and the cabin where he had met with such a surprising reception. Thus reasoning, and smiling at my foolish terror of a harmless bear, I rose, opened the door, and peered out. Not a sign of the bear could be seen, and I hobbled out into the crisp air with the glorious sunshine sparkling on the hoar frost. I glanced about for my spear and stopped short; almost at my feet was a great crimson splotch of blood upon the fallen leaves. My spear had found its mark, and the next instant I caught sight of it lying upon the ground a few yards away. I hurried to it and picked it up. It was covered with blood, blood was spattered upon the leaves and bushes all about.

Thrilled with the excitement of the chase, forgetting the risk I ran, and with spear in hand, I followed the crimson trail toward the woods. Peering intently into each clump of brush or tangled thicket, listening for some sound which might betray the presence of the wounded bear, I traced the blood-drops. At one spot among a clump of high brakes he had stopped to rest, for the coarse ferns were crushed and broken and stained with blood. In another spot he had forced his way blindly through a tangle of brambles, for blood spattered the glossy leaves and bits of black hair clung to the sharp thorns. At any moment I might come upon him wounded and at bay, and I proceeded with greater caution. At last I reached the shadow of the woods and on the dark moss and earth the trail became indistinct and hard to follow. A few yards ahead was an old fallen tree and, approaching this, I seated myself to rest and listen. As I did so I glanced to the other side, and there, within a yard of me, huddled in a great, shaggy heap, was the bear—stone-dead!

For a moment I could scarcely believe my eyes, but there was no doubt of it. The piggy little eyes were wide open and glassy, no breath heaved the great, furry sides, and the gleaming, white teeth and lolling tongue were thick with frothy blood.

Leaning toward him, I poked the sharp tip of my spear into his nose, but there was no response, no growl, no flicker of an eyelid, not even a twitch of the nostrils, and, convinced that no spark of life remained, I hurried around the log and, like a man bereft of his senses, danced about my victim. To an onlooker I would have been a strange, weird sight as, tattered and torn, half naked, with one foot bandaged in old rags, I jumped and hopped about on my rude crutch, yelling like an Indian, brandishing my spear, and crying taunts at the dead monster beside me. But my actions did not seem ludicrous at the time. I had been transformed to a primitive savage hunter glorying in his victory over the brute beast. Thousands of years had been swept from me and I acted as my ancestors once acted when they slew the saber-toothed tiger or the great cave-bear. In the dim, distant past of the stone age.

At last I calmed down and examined my quarry with vast satisfaction. Here was food, warmth, and clothing, and I patted the rude spear which had enabled me to secure them. It would be a hard job to skin the beast and to carry the hide and meat to my hut, I knew, but in view of the comfort it would bring and its value to me the thought of the labor troubled me little.

Although I had not breakfasted and was very hungry, I determined to lose no time, and started in to skin the bear at once. It was a far harder job than I had counted on, for the creature weighed three or four hundred pounds. After I had stripped the hide from one side I was obliged to turn the carcass over, and this, with only sticks to aid me and with the handicap of my crippled foot, proved a difficult task. Luckily, there was a slight slope where the bear had fallen and this helped me a great deal. By prying up the bear with a pole on one side, until his legs were uppermost, and then by hauling on withes attached to his feet, I at last toppled him over. I now returned to camp for breakfast and took with me a good-sized steak cut from the bear. The meat was excellent. Refreshed and strengthened by my meal, I returned to the bear's carcass.

By noon I had finished skinning the creature and, with a vast amount of labor, I dragged the hide to the cabin. Two more trips were made to secure a supply of meat, and then, completely exhausted by my hard day's work, I ate my evening meal, threw myself into my bunk, and slept soundly until the next morning.

To tan this great, shaggy skin was, I knew, quite beyond my powers, for I had no receptacle in which to place it, and I therefore decided to scrape it clean and dry it carefully, so that it would serve as a robe or covering, even though it was stiff and hard. I was determined, however, to use some of it for clothing, and for this purpose I cut off a good-sized piece and put it in the tanning liquor with the hair on. Eventually this was made into a shirt-like coat, which proved wonderfully warm and comfortable, although it was a most shapeless, ill-made affair. The bear furnished far more meat than I could eat for a long time, and I made up my mind to try to preserve a portion of it by drying. My experiments with the beaver meat and fish had been a failure, but, nevertheless, I decided to try once more, and this time I hung the strips of meat in the chimney of the cabin to smoke. I cannot say that the result of this method of preserving meat was a huge success, for the flesh was hard, dry, and smoky in flavor, but it was far better than nothing. After being soaked in water it was edible, and I knew that with a supply on hand I could not starve.

Very soon after I had killed the bear my foot became worse and I was obliged to spend most of my time in bed, for even the exertion of walking on a crutch caused me agony. I did not know, at the time, what caused the foot to become worse, but thought very likely I had struck or pressed upon

it while trailing or working at the bear, or that perhaps the hard labor of those few days was too great. At any rate, the swelling recommenced, the leg became inflamed and pained me horribly, and I was fearful that blood-poisoning or gangrene had set in. I kept it constantly poulticed with the arnica and, despite the agony it caused, rubbed it with bear's grease. But while it grew no worse, it showed no signs of getting better.

The weather had now become very cold, ice formed about the borders of the lake at night, the sky was gray and lowering, the chill north winds swept the few remaining leaves from the trees, and I realized that I could not hope to escape from the forest before winter set in.

Day after day I lay upon my bunk, only moving forth to obtain water or fire-wood. Miserable as I was, I was thankful indeed for the comfortable bear-skin which covered me, the supply of rank, smoked meat which I had provided, and, above all else, for the stout log cabin which I had so fortunately discovered.

Although I had no comforts or luxuries, yet I had all of the real necessities of life, and, had my foot been strong and well, I could have looked forward to spending a winter in the woods without great foreboding. But with my bad foot I was in a desperate situation, for the supply of bear meat was diminishing rapidly and I could not go forth to hunt for other food. At last, however, there seemed to be a change for the better in my foot and then an abscess began to form. When this at last broke and I drew forth a large sliver of wood from the wound, I felt immediate relief and realized why the injury had proved so troublesome.

As soon as the splinter was removed the ankle and foot commenced to recover rapidly, and within a week the pain had left it and, much to my joy, I found that I could move the foot and could even rest some weight upon it. But I realized that it would take time for my foot to regain its full strength and that to use it too soon would only result in further trouble, and I nursed it with every possible care. Day by day it became stronger and soon I was able to limp about the cabin, although I still used a crutch when out of doors. I began to plan for my tramp toward the settlements and to hope for an escape from the forest very soon.

Then one morning I awoke to find snow drifting through the crevices of my cabin. Opening the door, I looked forth upon a strange world of white. Already the snow was ankle-deep upon the earth, the evergreens drooped under its weight, the lake gleamed black and sullen in the midst of the vast

white landscape. Ceaselessly, silently, the flakes fell from the leaden sky, shutting off the farther shores and the interminable forests as with a dense, white curtain.

All day long the snow fell, all through that night and until noon of the next day, and when the pale, wintry sun again broke through the clouds its watery rays glittered in dazzling brilliancy from a veritable fairyland. But to my eyes there was little to admire, for the earth was buried deep under many inches of snow, which had made me a helpless prisoner in the depths of the wilderness.

How hopeless it would be to attempt to tramp to the settlements under such conditions was borne upon me as I broke my way through the snow toward the edge of the woods to secure a supply of wood, the next morning; but while I was disgusted at thus being shut off by this first snowfall just as my foot was becoming of use again, I realized that, after all, the snow had helped me.

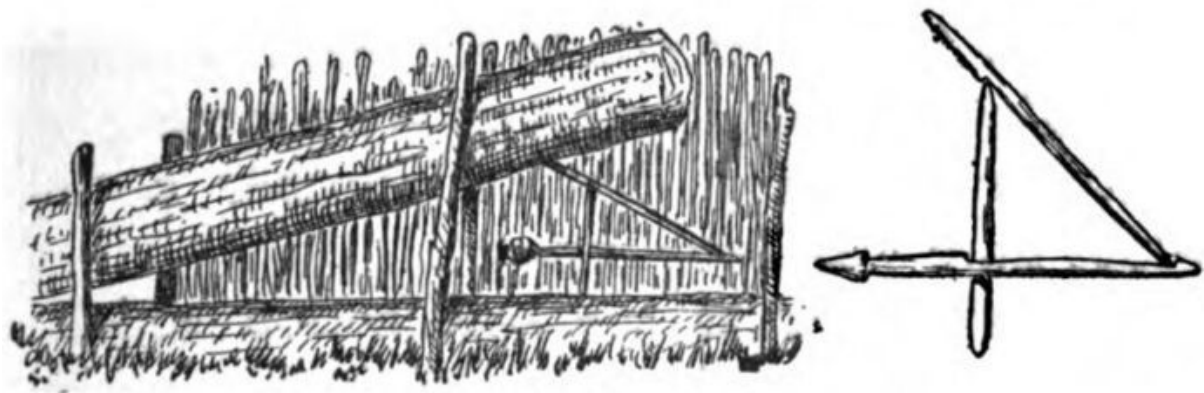
Everywhere upon the surface of the fresh snow were the footprints of birds and animals. It seemed incredible that there could be so many wild creatures dwelling close at hand, unseen and unsuspected. Here, the well-marked trail of a hare crossed the snow; back and forth across it zigzagged the snake-like track of a marten; to one side, and evidently following the others, were the imprints of a fox's feet, and I could distinguish the marks left by partridges, squirrels, and many other woodland denizens which I could not identify.

Surely, with these trails to guide me, I could set traps and secure both food and furs, and I grew greatly interested and wandered here and there, striving to read the stories the creatures had written on the snow.

Even close to the hut many creatures had passed and repassed during the night. I noticed where a hare had scampered about the cabin. Some larger animal had dug through the snow before the door to secure some old bones I had dropped, and at least a dozen smaller creatures had made merry about my dwelling while I slept, all unconscious of their presence.

As my foot was now strong enough to enable me to walk upon it with the aid of a cane, I decided to start out after breakfast and set some traps, and so, wrapping the bear-skin about me, armed with my bow and arrows, and using my spear as a staff, I set forth. I must have been a wild and savage figure as I limped through the snow that morning, wrapped in the bear-skin, with my hair falling to my shoulders, and with an unkempt, ragged beard

covering my face, and I might well have been mistaken for the original wild man. But no one was there to see me and I gave no thought to my appearance, but trudged away through the snow toward the forest, my eyes fixed upon the trail before me, and well protected from the icy wind by the thick fur bound about my body. The tracks I was following puzzled me, for they looked like those of a miniature bear, but they were easy to trail and led directly to an old hollow stub. As there seemed to be no other trail, I decided that this must be the home of the creature, whatever it was. Curious to know what sort of animal had made the tracks and had visited my cabin, I pounded upon the stub and was rewarded by a slight scratching noise from within, and an instant later a queer, quizzical face peered forth from the opening above me and stared down questioningly at the strange being who had knocked upon its home. The sharp nose and bright eyes were visible only for an instant, but the brief glimpse was enough, and I recognized the owner as a raccoon. Well, raccoons were good to eat and their fur was warm, and I made up my mind to capture the 'coon that night. My first thought was to set a trap beside the tree, but a moment's reflection changed my plans, for I realized that the 'coon would probably return to the cabin again and that it would be far easier to trap him by my own door than by his.



DEADFALL SET WITH FIGURE-FOUR TRIGGER

There were still plenty of tracks to be followed, and for several hours I busied myself locating the homes of hares and setting twitch-ups to capture them. Then, as my hands were becoming numb and I feared catching cold

in my weak ankle, I retraced my steps to the cabin. Late that afternoon I set a deadfall outside the door for the raccoon, building it like the one in which I had caught the beaver, but furnishing it with a figure-four trigger, and baiting it with scraps of meat.

The next morning I hurried to the door as soon as I awoke and, much to my joy, found that a fine, fat raccoon had fallen a victim to my trap.

I lost no time in skinning him, and dined off broiled 'coon, which I found excellent. Then I set out for the woods to inspect my snares. Several were untouched, two were sprung but contained nothing, but in another a fine, white hare was dangling.

I cannot express the satisfaction which I felt upon the success of my first day's trapping, for I realized that I would not want for food where animal life was so plentiful and so unaccustomed to man. I knew also that rabbit and 'coon-skins could be made into the warmest and most comfortable of garments.

I was so much encouraged that I went much farther into the woods, set a number of new snares, and returned by another route to the cabin. This brought me close to my old camp and the border of the lake, and as I came out from the edge of the woods to the shore I heard a strange, subdued chuckling sound from beyond a low, brushy point. Wondering what creature was there, I crawled cautiously forward, peered at the little cove beyond, and looked upon a flock of wild ducks sunning themselves at the edge of the lake where a spring kept the water free from ice.

Fitting an arrow to my bow, I rose silently, but, cautious as I was, the birds caught sight of me and instantly took wing with loud quacks of alarm. In a vague hope of bringing one down I fired, but the arrow fell short, and dropped into the lake beyond my reach, and the ducks safely winged their way toward the farther shores. But I minded the loss of my arrow far more than the loss of the ducks, for I had plenty of meat for my present needs and the weather was now cold enough to enable me to keep a supply on hand for a long time. For several days I spent my time setting traps and skinning and tanning the hides, for each morning I found hares in my twitch-ups, and I also succeeded in capturing another raccoon. The tanned rabbit-skins I sewed into rude mittens and a sort of undershirt, the two 'coon-skins provided leggings, and rabbit-skins were made into a cap. As my feet suffered a great deal from the cold, I lined them with rabbit-skin, with the fur inside, and when all this had been accomplished I found I was perfectly

protected from the weather as long as I exercised, while my bear-skin robe made a splendid blanket at night, or served as a cloak when I was sitting still during the days.

About two weeks had now elapsed since the first snowfall, my foot was strong enough so that I could walk upon it without a cane, save on rough ground, and I was well clothed in furs and was quite comfortable, and could look forward without fear to spending the winter in the woods, which now seemed inevitable.

To preserve the supply of meat I had suspended the carcasses of the hares on a pole at one side of the cabin, where they were frozen stiff and out of reach of prowling foxes or other creatures. Each evening I set the deadfall by the door to capture any 'coon or other animal which might approach my hut to pick up scraps from my meals. Just as I was snuggling down beneath my bear-skin robe one night I heard a slight thud outside, and, thinking some creature had sprung the deadfall, I threw off the robe and started to go forth to secure my prize. But ere my feet touched the floor I was riveted to the spot and a cold shiver ran down my spine as the silence of the night was pierced by a terrifying, moaning wail, ending in a blood-curdling scream.

For one long second I sat motionless on the edge of my bunk while that awful sound echoed through the night, then, like a frightened child, I ducked under my bear-skin robe, pulled it over my head, and lay huddled and quaking in superstitious terror. For a space there was silence, then some heavy body landed with a crash upon the roof above my bunk, and the awful, banshee-like wail pealed forth within a few inches of my head. A cold sweat broke from my skin, I shivered with abject fear, my hair seemed to rise upon my scalp, and as the last sobbing note of the terrible sound died out and the awful something began tearing at the roof, my nerves gave way, and I shrieked aloud.

CHAPTER X

BACK TO THE PRIMITIVE

The sound of my own voice was a great relief. Suddenly my foolish terror vanished, and with a sudden reaction of feeling I broke into peals of laughter as I realized that the blood-curdling cry was that of a lynx and not the wail of a banshee or anything supernatural.

No doubt my screams frightened the creature quite as much as his cries had terrorized me, for there was no further sound from the roof and the howling was not repeated.

Feeling confident that the lynx had left the vicinity, I again snuggled down in my bunk and slept undisturbed until morning.

But when I stepped out of my cabin the next day the effects of the lynx's visit were evident. The deadfall was sprung and the bait had been taken, the snow was covered with the big footprints of the creature, and my store of frozen meat was torn to pieces and scattered upon the snow, while a good portion of it had been devoured or carried off by the great cat.

That he would return for more food when darkness fell I was convinced, and I determined to capture him if it was possible. As long as he was at large I could not keep a store of frozen meat in safety, and I thought longingly of his thick fur coat, which would be a most welcome addition to my wardrobe.

As I went into the woods to look at my traps and snares I traced the lynx tracks for some distance, until I reached a spot where he had leaped into a partly fallen tree and the trail was lost.

Thinking this tree might be a regular runway for the creature, I spent some time setting a large and heavy deadfall upon it and then made the rounds of my traps. But I was doomed to disappointment, for a thieving fisher-cat had been before me and only fragments of torn skin and fur, a few drops of blood, and some scattered feathers remained as proofs that my snares had captured hares and partridges.

As long as this thief was about I could not expect to obtain game, for I well knew that once the rascal had discovered my traps he would visit them as regularly as myself, and that it would be necessary to capture him to insure my food-supply.

I had often heard Joe and the other woodsmen tell tales of the sagacity and cunning of the fisher-cats and I realized that I would have a hard task to capture the creature which robbed my traps. Nevertheless, I could but try, and with the greatest care I set a deadfall near each of my traps and arranged the triggers as I had that of the beaver-trap. Then behind each trap I built a little inclosure or fence of sticks, covered this with slabs of bark, and within these placed pieces of the frozen hares left by the lynx.

I then returned to the cabin and in the afternoon set two large deadfalls and baited them with the remains of the hares' carcasses. During the day I had given a great deal of thought to the capture of the lynx, and various plans had occurred to me which I cast aside as impracticable. Had I possessed any sort of serviceable weapons it would not have been such a difficult matter to kill the beast, for I knew that, driven by hunger, a lynx will become very bold, and that by lying in wait I might easily obtain a good shot at him from within the cabin. I could not throw my spear from inside the hut, however, and I knew how hopeless it would be to attempt to approach the lynx in the open, while to fire at him with my flimsy, bone-tipped arrows would be utterly useless.

This led me to consider the possibility of making a more powerful bow and better arrows. I knew that the Indians used stone-headed weapons in former times and I had often seen the stone arrow-heads and had even found many myself, and, while I was familiar with their appearance, I had no idea how the savages formed them.

But I was convinced that if a naked primitive Indian could make a stone arrow-head, a white man who had overcome as many difficulties as myself should be able to accomplish the same feat, and I decided to try my hand at making stone arrow-heads at once.

I had seen arrow-heads of white quartz, of flint, and of various other stones, and I therefore came to the conclusion that the kind of rock made little difference; and as there were more pebbles and stones around the borders of the lake than anywhere else in the vicinity, I made my way to the shore and picked up a number of stones which I thought might serve my purpose.

Again inside my hut, I proceeded to crack the pebbles with a large stone. While some of the rocks broke into small bits, or thick, squarish pieces, others split into flakes or slivers which were quite thin and bore a remote resemblance to the forms I desired.

I reasoned that the Indians must have worked with stones for tools, and my common sense told me that the simplest and easiest method of transforming a rough flake of stone to an arrow-head would be to chip or break off the edges by nicking them with a rock.

Selecting a promising flake of quartz, I attempted to work it into shape and chipped away steadily for half an hour or so. It was work to which I was not accustomed, and I scratched, bruised, and cut my hands, but the bit of stone gradually assumed a rough, spear-like shape. I was becoming elated at my success when, without warning, the quartz split in two. I was thoroughly disgusted. Tossing the two pieces petulantly into the fire, I tried another piece of stone. This was even more disappointing than the first, for with the third blow of my stone hammer the rock flew to pieces and this followed the other into the flames. Then it occurred to me that I might grind the stones on my grindstone, and I at once tried this scheme. Instead of cutting the rock the grindstone was cut by the rock, and I realized that this method was impracticable. Then I thought that possibly some softer stone might be ground into shape, and I made another trip to the edge of the lake and returned with a number of pieces of a slate-like stone which seemed fairly soft. This gave way rapidly to the grindstone and I soon succeeded in grinding out two arrow-heads which pleased me greatly. These I bound on to my arrows in place of the bone heads, and then, in order to test them, I fired one at a piece of hide. The arrow flew much straighter and harder than those with the bone heads, but when it struck the skin the slate splintered and the arrow fell to the earth without even penetrating the hide.

I was now thoroughly convinced that my attempts at making stone arrow-heads was a complete failure and I seated myself before the fire to rest and think. As I sat there, gazing idly at the flames, I noticed the bits of quartz glowing red among the coals, and, impelled by a vague curiosity, I raked them out upon the hearth. Thinking to cool the stones so I could examine them, I poured some water upon them and instantly the hot quartz flew into pieces. Surprised at this, I picked up one of the fragments and was struck by its knife-like edge and smooth surface, and like an inspiration it dawned upon me that here, perhaps, was a solution of my problem.

If water poured upon hot quartz would cause the stone to sliver off in this way, why would it not be possible to heat stones, drop water upon them in the proper places, and thus break off pieces until the desired shape was produced? It was certainly worth trying, and without more ado I set to work to try the experiment.

Placing a lump of quartz in the fire, I waited until it was red-hot, and then, pulling it out, I poured a little water upon it. With a sharp crack it burst into several pieces. Selecting the best of these, I again placed them on the coals. When they were well heated I drew one out and very carefully dropped water upon one edge. Pieces flaked from it wherever the water touched the hot stone. While my first attempt was a failure and the piece of quartz refused to assume the form of an arrow-head, yet I realized that this was due to my lack of skill and care and that my theory was correct.

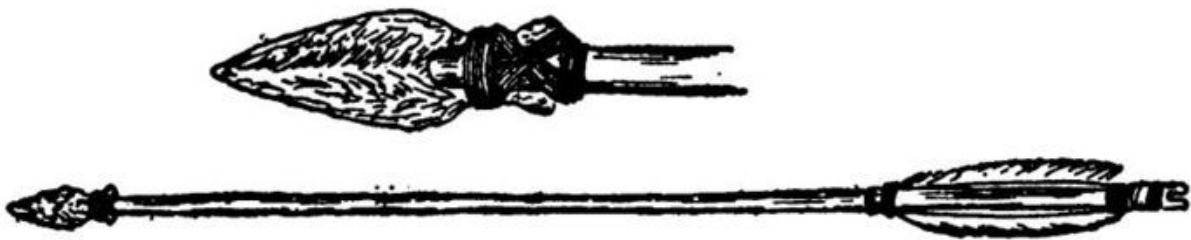
Over and over again I heated stones and flaked them into shape by means of drops of cold water, and although I did not succeed in making a single arrow-head before darkness came and I was obliged to cease, yet I felt convinced that with practice I could produce keen, well-shaped arrow-points, and I went to bed determined to resume my labors on the morrow and to persevere until I was successful.

I heard no sound from the lynx, or any other prowler, that night, and found the deadfalls undisturbed when I opened my door in the morning. As I approached the tree where I had placed the lynx-trap, I saw it was sprung, but the lynx was not in it, although a few wisps of dark-brown hair and numerous bloodstains proved that some creature had been struck by the heavy descending log. All about there were lynx tracks in the snow. Here and there I saw smaller tracks, and for a few moments I was puzzled, for the hairs upon the tree trunk did not look exactly like those of a lynx and much of the blood was trodden into the snow by the creature's feet. Then, as I examined the trap more carefully, I discovered the tip of a bushy black tail and realized what had happened. The fisher-cat had been caught in the deadfall and had been found and devoured by the lynx. I smiled as I thought how one thief had made away with another, but I regretted the loss of the fisher's warm skin. I reset the deadfall and then visited my other traps. Three hares were hanging in the twitch-ups, and the fact that they had not been disturbed proved that my surmises in regard to the fisher's fate had been correct. One of the deadfalls contained a skunk, which I did not use, although its fine coat of black-and-white fur was a great temptation despite

its odor. Another deadfall contained a marten, while the others had not been disturbed.

The rest of the day I devoted to working at my arrow-heads and before noon I had the satisfaction of producing a very creditable arrow-head of quartz. I was anxious to test this, but I hesitated for fear of breaking it and thus wasting all the time and labor I had spent. Finally I decided to take the risk and, having bound it to one of the arrow-shafts, I fired it at one of the hares I had caught. With a thud it struck the carcass, penetrated skin, muscle, and bone, and came to rest with half its length projecting beyond the farther side of the hare. I hurried forward to examine the point, expecting to find it chipped or broken, but it was absolutely uninjured. I shouted with joy as I realized that my perseverance was rewarded, that I now possessed a weapon of real penetrating power, and that, provided my aim was true, I could successfully bring down many a creature that otherwise would have been beyond my reach.

I argued if my crude, weak bow could drive this stone-headed shaft completely through a half-frozen hare, that with a stronger bow I might even kill a deer or the lynx, and I at once went to work on more arrow-heads, wisely deciding that I had best complete this work while I was in practice and leaving the making of a new bow until later.



A PRIMITIVE ARROW

By nightfall a half-dozen arrow-points were finished, for, once I had discovered the “knack” of making them—by touching the red-hot flakes of stone with a wet stick where I wished to chip off the edges—I found I could produce excellent results rapidly and easily. I had thought the tip which I tested on the hare was perfect, but by comparison with the last one I made it seemed crude and rough, and I retired that night well pleased with my

success and skill, and I fell asleep planning the bow which I promised myself I would make the following day.

That night I was again aroused by the piercing scream of the lynx, and although I knew well what it was, yet I could not avoid shivering as the weird cadence rose and fell, echoing from the forest and ending in its unearthly moan, like a soul in mortal torment. I had always thought of lynxes with a feeling of contempt, and had looked upon them as cowardly, overgrown cats—sneaking thieves and destroyers of small game—and I tried to laugh away my unreasonable fear. I told myself that I was perfectly safe here in the cabin and that even outside the creature would turn tail and bound off at sight of a human being, but despite every effort, when the cries again rang out from close at hand I felt my knees shake, while strange, crawling sensations ran up and down the back of my neck.

Listening intently, I could plainly hear the light creak of snow beneath the beast's feet as he moved about outside, and, finding inaction utterly unbearable, I slipped out of my bunk, tiptoed to the door, and peered out through a crevice.

Outside it was as bright as day, with the full moon shining upon the snow, and there, within a dozen feet, was the lynx; a huge, tawny creature whose strength showed in every movement.

He was nosing about near the deadfall, and his attitudes as he peered at the trap, cocked his tufted ears forward, and turned his head first to one side and then another—as if reasoning out a method of securing the coveted bait without injuring himself—made him appear unnaturally human.

At sight of the creature my unreasonable fears disappeared, for here was something tangible and he was so close, so plainly visible, and presented such a splendid mark, that I determined to try a shot at him with my stone-tipped arrow.

The crevice between the door and wall was very narrow, scarce an inch in width, and, had I stopped to think, I would have known that to attempt to fire an arrow through this and strike the mark was utterly foolish. Placing the arrow on the bowstring, I slipped the head through the opening and started to draw the bow. As I did so the arrow touched the side of the crevice and dislodged a tiny bit of old bark, which dropped to the floor. Slight as the noise was, the keen ears of the lynx heard it, and with a startled growl he faced toward me. Never shall I forget the fierce, malevolent gleam in those great, green eyes, the snarling lips and gleaming

teeth, and the bristling fringe of white beard. No longer did I see only a hulking lynx out there upon the snow; instead I seemed to gaze upon some supernatural, awful monster whose baleful eyes stared into mine and made me feel defenseless in spite of my shield of stout logs. His gaze filled me with a strange fear, an unnamed dread, and yet fascinated me as with hypnotic power.

Motionless I stood there, as if transfixed, my bow half drawn, my arrow poised, while for a brief space the lynx stared at the blank wall. Then, as he crouched low as if about to spring, the spell was broken, my fingers released the string, there was a sharp twang, the rustle of the speeding arrow, and the next second the cabin shook as the vicious brute hurled himself against the door with a piercing scream of rage. For an instant I feared the door would give way if the lynx sprang again, and, terror-stricken at the fury I had aroused by my action in firing at him, I grasped my spear and knife, determined to sell my life dearly if the beast gained entrance. But there was no repetition of the attack, and at last, gathering up courage, I again looked forth from my peephole, but no sign of the lynx was to be seen. Still, I was afraid that he might return, and before climbing into my bunk I piled the table, all the fire-wood, the iron pot, and the grindstone against the door.

For a long time I lay awake, listening for some noise which would betray the return of the lynx, but no sound, save the distant bark of a fox and the querulous note of a screech-owl, broke the silence of the winter forest, and finally I fell asleep.

When I again opened my eyes and, glancing about, saw the pile of things against the door, I laughed at my fears of the night, and, jumping out of bed and wrapping myself in my robe, stepped out of doors to see what signs had been left by my midnight visitor.

Evidently the brute had vented his ill-temper on my arrow, for the shaft was broken and bitten to pieces, but the head was uninjured, and as no blood-spots were visible I decided that the lynx had not been hit. The deadfall was still unsprung and the bait within it was untouched. By the distance between his footprints I knew the first blind rage of the lynx had been superseded by terror and that he had used all his speed in getting as far from the cabin as possible.

Upon the door deep, white grooves were cut into the wood where the sharp claws of the lynx had struck and I shuddered to think of the injuries

those keen talons could inflict on human flesh. I was thankful indeed that stout wood was between me and the giant cat when he launched himself forward in a frenzy at being disturbed in his prowling.

Judging by the fate of the fisher-cat, I rather expected to find that any game which I had captured had gone to feed the hungry lynx, and in this surmise I was not at fault. Bits of white fur were all that remained of at least two hares, while scattered feathers told the tale of a snared partridge which had also helped to satisfy the lynx's appetite.

Matters now began to look serious. If the lynx was to rob my traps nightly there was little use in setting them and I would soon be face to face with the problem of killing the lynx or starving. He was evidently too wise to be caught in any trap which I could make, and my only hope lay in securing a supply of game by means of my bow and arrows, or of killing the lynx by the same method.

Fortunately I had quite a supply of frozen meat on hand, which I kept inside the cabin at night after freezing it during the day, and as I had no wish to become a purveyor to the lynx I left my snares unset and returned to the hut prepared to make a new bow.

Now that I was comfortably clothed and housed and could not hope to get out of the woods for a long time, I could devote more skill and time to my bow-making. In the three months which had passed since I made my first bow I had learned a great deal about the forest's resources, the properties of various woods, and the skilful use of my knife.

Now I determined to make a weapon which would be worth while and I spent several hours searching through the young growth at the edges of the woods and near the lake for a suitable staff from which to make my bow. Finally I found what I sought, a thoroughly seasoned, but sound, pole of hornbeam which was so tough, springy, and hard that I felt confident it would work into a bow of exceptional strength and power.

As I returned with this, I noticed that the sky had grown overcast and that a few great flakes of snow were falling lazily through the cold, motionless air. A short time before I would have been greatly troubled by the indications of a heavy snowfall, but now, with my foot completely recovered, well clothed, and fully convinced that it was useless to think of making my way to the settlements before spring, a few inches or feet of snow, more or less, was of little moment.

I found the hornbeam very hard to work, and by the time I was compelled to put it aside, on account of darkness, I had accomplished little save to whittle the staff to half-round form.

CHAPTER XI

AN UNEXPECTED MEETING

When I looked from my door the next morning I found a blizzard raging, while the snow was drifted almost to the eaves of my tiny cabin.

I was indeed thankful that I had enough food on hand to satisfy my wants for a number of days, for to hunt or trap in such weather was an utter impossibility. My stock of fire-wood was none too large, however, and I decided it would be wise to gather more before the storm prevented me from leaving my hut.

Accordingly, I bundled up in my furs and started out, but before I had traveled a hundred feet I gave up in despair and retraced my way toward the cabin, for the snow was up to my waist and to plow my way through it was absolutely impossible.

I was hopelessly snow-bound, for the present at least, but I cheered myself with the thought that when the storm eased the surface of the snow might freeze and form a crust which would support my weight.

For a day or two, at any rate, I would not suffer, and as the snow, banked around the hut, prevented the chill wind from entering, I was able to keep comfortably warm with a smaller fire than usual and could thus save a great deal of fuel. I was thinking on such matters, while working at my bow, when the idea of snow-shoes occurred to me. It was strange that I had not thought of them before, for I had used snow-shoes for sport and knew that every camp and house in the northern woods had snow-shoes hanging upon the walls for winter use. To think of snow-shoes was easy, but to make them was a very different matter, as I soon discovered.

I knew full well that I would not be able to produce a really fine pair of shoes, such as were used by the guides and dwellers in the north, but I thought that I would have no trouble in making some sort of affairs which would serve my purpose and would prevent me from sinking into the snow. My idea was to make the frames of elastic green wood, and string or net them with strips of hide or bark. To secure the withes for the frames I was

obliged to face the storm and after a terrific struggle succeeded in reaching a little clump of young growth close to the cabin. Cutting a number of the slender shoots of birch, I fought my way back to the hut and, utterly exhausted by my exertions, threw myself on the bunk to recover my breath and strength.

I found little trouble in bending the birch withes in oval form. Lashing the two ends of each together, I placed braces or spreaders between the sides to hold them apart and then started to put on the netting. But here my troubles commenced. If I drew the thongs tight in one direction it loosened those which ran the other way; the strips of hide slipped on the frames; the frames sprung out of shape and twisted, and finally I discovered that to cover the frames with netting, even if I succeeded in fixing it in place, would require more thongs of hide or more strips of bark than I could furnish. Until I attempted the work I had no idea of the quantity of material which was required to cover snow-shoes with their netting, nor of the immense amount of labor necessary to make them. My respect for the makers of snow-shoes was vastly increased by the time I had labored for a few hours at my clumsy attempts, and I marveled that articles entailing so much work and such ingenuity could be sold at the prices I had heard quoted.

At last I *did* succeed in producing a nondescript, ridiculous thing which bore a remote resemblance to a snow-shoe, and, anxious to test its practicability before attempting to make its mate, I placed it on the snow outside my door and stepped cautiously upon it. For a moment it supported me upon the surface of the snow, and I was just commencing to congratulate myself upon my success when, without warning, the frame doubled up, the netting collapsed, and my foot plunged into the snow, throwing me sprawling in the drift. Floundering about, I extricated myself, fished up the wrecked thing upon which I had spent so much toil, and re-entered the cabin, heartily sick of snow-shoe making.

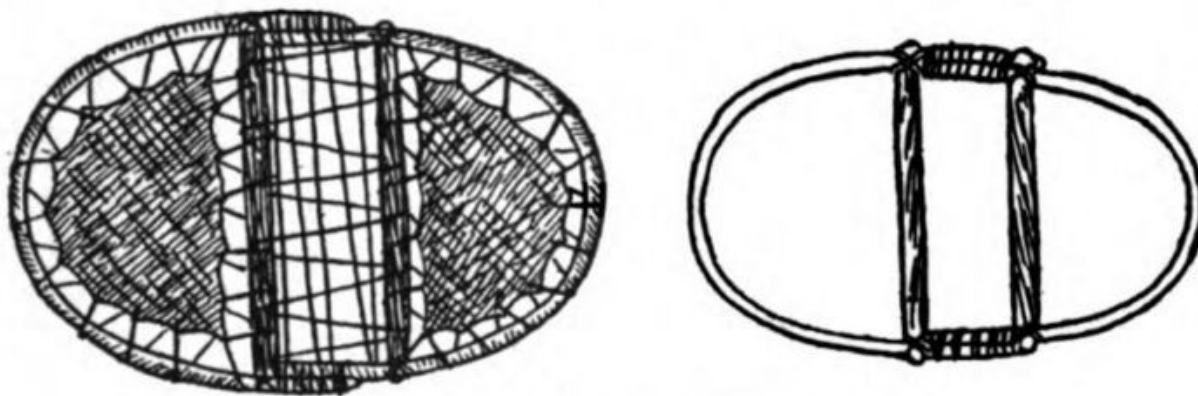
But as I busied myself preparing a meal my thoughts constantly reverted to the snow-shoes, for I realized that they were a real necessity, and I strove to reason out the causes for my failure and to devise some method of overcoming the difficulties. My experience with stone arrow-heads had taught me a lesson and had showed me how simple an apparently difficult thing may be, once you discover the secret of doing it properly, and I had

little doubt that to make a serviceable pair of snow-shoes would prove very easy, if I only knew how.

I decided that the main difficulty lay in making a network which was strong enough to support my weight and which brought an equal strain upon all sides of the frame. Then I began to wonder why a net was really necessary and why an unbroken flat center would not serve the purpose as well or even better. The Norwegian skees, I knew, were merely narrow, flat boards, and yet they supported men upon the snow, and if this was the case why shouldn't any flat object of the same area support my weight, regardless of its form? This led me to enumerate the objects which were within my reach and which would present a broad, flat surface to the snow, and I thought of birch bark. This, however, I felt sure, was too soft and pliable for the purpose, and I bent all my energies to devising some method of stiffening the bark and keeping it spread flat.

I thought of lashing sticks across it, but I realized that unless the bark was crisscrossed by a number of sticks such a method would be useless, and that so many sticks would make the shoes far too heavy. Then it occurred to me that the bark might be lashed within a frame, but this idea I also discarded, as I knew the bark would split and break under my feet, and then, like a flash of light, I thought of skins. Why was it not feasible to cut a piece of stout hide to the proper form, lash it within an oval frame to hold it taut, and thus make snow-shoes without the labor and difficulties of weaving a netting?

I was so pleased at my idea that I immediately started to put my plan into concrete form. Feeling that the frames must be far stronger than those I had used before, I selected heavier poles, and then, finding it was impossible to bend them in a complete circle or oval, I made each frame of two separate pieces, each forming half an oval, and joined the ends together by lashings. Stout braces were then placed within these frames to spread them apart and to strengthen the ends of the withes where they were joined, and these were firmly secured by additional lashings.



HOW THE SNOW-SHOES WERE MADE

The next thing was to fit the hide inside the frames, and here I once more came face to face with a problem which took some time to solve. Of rabbit or hare skins I had an abundance, but these, I knew, were too frail and delicate to serve my purpose. The 'coon-skins were about my limbs as leggings, the two porcupine-skins had been converted into moccasins, and nothing else remained save the bear-skin robe.

I disliked cutting into this, and, moreover, I wanted hide minus hair, but there was nothing else to be done and, with some misgivings, I cut two oval pieces, of the size and shape of the frames, from my bear-skin. With my knife I cut and shaved the hair from these pieces of hide, and then, by means of roots, thongs, and strips of bark, I lashed the oval pieces of skin inside the frames. The result of all this work was a pair of platter-like objects of rawhide, and, while the stiff skin was far from smooth or tight, I judged that they might hold me up. I hurried out to try them. Placing the two queer contraptions upon the snow, I clambered upon them, and to my great delight found they scarcely sank into the snow at all around the edges, but the centers sagged badly, and I found it difficult to maintain my balance upon them.

I had not yet fitted straps or fastenings for my feet, but I felt sure that to travel over the snow would be quite impossible if the centers of the snowshoes sagged to such an extent, and that, while the principle was all right, I must manage to draw the hide smooth and tight before my invention would prove a success.

It was now growing dark and I was compelled to stop work, but all through the preparation of my evening meal I was busily thinking and

trying to evolve some means for perfecting the shoes. Not until I had climbed into my bunk did the simple expedient of wetting the skin, while lashing it in place, occur to me. This I felt positive would result in stretching the skin as tight as a drumhead when it shrunk in drying. Having thus seemingly overcome this difficulty, I tried to conjure up a mental picture of the arrangement of foot-straps on the snow-shoes I had used years before. As my brain visualized the graceful, racquet-shaped shoes, with their amber-colored netting, their tiny, ornamental, worsted tassels, and their glossy hickory frames, each detail slowly came back to me. I remembered the slightly upturned toes, the broad, buckskin foot-straps, and—yes—the open spaces in the centers of the network into which one's toes sank as one's feet bend when walking. As all this came back to me I wondered that I could have forgotten it, and then, feeling that I would have a practical pair of snow-shoes before the next night, I fell asleep to the roar of the gale outside.

Morning found the storm over, but with fully four feet of snow covering the earth, while the drift before my door was so high that I could not see over it. With the memories of the night before still fresh in my mind I unlashed the pieces of skin from the frames and placed them to soak in water while I ate my breakfast. Now that I remembered that a hole *must* be left for one's toe in the center of a snow-shoe the task seemed greatly simplified. As soon as I had eaten I drew a piece of the soggy, softened hide from the water and, cutting it in two, proceeded to lace one portion of it in the frame between the bowed, semicircular rim and the nearest brace. As soon as this piece was in place I laced the other piece in the opposite end, hung the shoe to the roof of the cabin to dry, and repeated the operations with the other shoe.

By the time this was completed the first shoe was partly dry, and I was well pleased to find that, as I had foreseen, the skin was shrinking as tight and hard as a sheet of iron. I then busied myself with my bow and worked steadily until late in the afternoon, when, the snow-shoes having dried thoroughly, I hurried out to test them. They were a great success and supported my weight perfectly, scarcely sinking beneath the surface of the snow as I stood upon them. Tying the straps about my ankles, I started forward. For a few minutes I had some difficulty in walking, yet the knack soon came back to me and I found no trouble in traveling about on the primitive snow-shoes I had evolved.

My first need was fire-wood and I shuffled off across the snow to the woods. It was a strange sensation to be thus walking into the forest on a level with the "second floor," so to speak, but it enabled me to obtain a fine lot of dry, resinous branches which had previously been far beyond my reach.

Well pleased with the success of my snow-shoes, and with a great load of wood, I returned to the cabin, and then spent the time until dark digging away the drift before my door and using a snow-shoe for a shovel.

Realizing that it would be useless to attempt to trap in the deep snow, I determined to set out on a hunt the next morning, hoping to find some birds or animals which I might be able to kill with my bow and arrows. My new bow was far from completed, but for my immediate purposes the old bow would serve and within a few days the new one would be ready for use.

When I started out the following morning I found numerous tracks of small creatures, and was soon within the forest searching each snowladen evergreen for partridges or other game.

I had traveled for some distance without success when I came upon the unmistakable tracks of the lynx. His broad, padded, furry feet had served him as well as snow-shoes and he had sunk but slightly into the snow, but I could not tell whether he had passed that way within a few minutes or the day before. I felt that there was little danger of the creature attacking me here in broad daylight, and I commenced trailing his steps, glancing keenly at each tree as I proceeded. Presently I came to a spot where the lynx had crouched low, and then, for the first time, I realized that he had been trailing some other creature. Partly obliterated by the marks of the cat's feet were other tracks, small, deep, and with a furrow between them as if some animal had dragged himself slowly through the snow into which his feet had sunk. I was puzzled to know what had made these tracks, and, curious to learn the outcome of the tragedy indicated by the telltale marks in the snow, I crept cautiously forward.

I had gone scarcely a hundred yards when I rounded a projecting ledge and came within sight of a little, open glade. Near the center of the swale a dark spot stood boldly out upon the snow, and before I had time to realize what it was the lynx leaped up and, bounding off, disappeared in the thick woods beyond. But the dark object still remained in the center of the glade, and, approaching it, I discovered that it was a full-grown deer. It was evident that the lynx had overtaken and killed the buck while the latter was

floundering in the deep snow and was unable to travel rapidly enough to escape, for the deer's throat was torn open and a good portion of the shoulder had been eaten by the lynx. There was plenty of good, fresh meat left upon the carcass, however, and the skin was scarcely injured, and I congratulated myself upon having followed the lynx tracks, for my curiosity had thus led me to a supply of food which would last me for a long time. The lynx had unwittingly saved me a deal of trouble, and I forgave him all the fright and loss he had caused me hitherto.

Immediately I set to work cleaning the deer, leaving the entrails for the lynx, and then, tying the legs together, I shouldered the carcass and commenced my return journey. It was a heavy load, and the additional weight caused my snow-shoes to sink deeply into the snow. I found progress slow and difficult, and decided to try dragging the deer instead of carrying it.

As I stood resting for a few moments before resuming my journey, I had a strange, uneasy sensation of being watched and I peered nervously about, but no sign of a living thing could I see. Laughing at my own foolishness, I picked up the strip of bark which I had fastened to the deer's antlers and resumed my journey. The deer slid along over the snow quite easily and, by keeping to the more open parts of the woods and avoiding trees and branches which might catch upon the deer's legs and feet, I made fairly rapid progress. But despite every effort, I could not shake off the feeling that I was being watched and followed. Several times I wheeled about quickly and swept the surrounding forest with my eyes, but I saw nothing tangible, although once I thought I caught a glimpse of a flitting, shadowy form behind a clump of thick firs.

Perhaps, I thought, the lynx is following me in the hope of regaining his lost prey, and I began to fear that the creature might actually gather sufficient courage to attack me.

Then that demoniacal face which I had seen outside my cabin in the moonlight began to haunt me, and unconsciously I hurried faster and faster until, as I approached the edge of the woods and saw the clearing with my hut before me, I was making the best speed which my clumsy foot-gear and the deer's carcass would permit.

Once in the open, and with the forest behind me, I felt easier, and, panting for breath, stopped and wheeled about, and as I did so I knew that my vague, unreasoning fear had not been unfounded. Squatting at the edge

of the woods was the lynx, every hair bristling with fury, his green eyes gleaming with cruelty and hate, and his lips drawn back across the great, white teeth in a threatening snarl.

For an instant he sat there, as if half minded to leap forth to the attack, and then, with a mighty spring, he bounded into a near-by spruce. The branches swayed and the snow came down in avalanches as he leaped from bough to bough, and from the shelter of the woods the creature's fearful cry of defiance rang out.

Even there in the sunshine and within a few yards of my hut I shivered at the eerie sound, but I realized that, after all, the lynx was too cowardly to attack me in the open. Relieved, now that I knew my phantom-like pursuer was the lynx, I dragged the deer to the cabin, slammed the door to, and threw myself upon the bunk, thoroughly exhausted with my hard morning's work.

CHAPTER XII

STRANGE ADVENTURES

After a short rest and a hearty meal of broiled venison I felt greatly refreshed. Then I skinned the deer and hung the hide upon the wall to dry. I had no means of tanning it, unless I cut it into small pieces, and I could not remove the hair by burying it, but I had no real need of more leather at the time and this did not trouble me. Moreover, I felt sure that the skin could be softened and tanned later on, if it was required, and that meanwhile the rawhide, with its thick hair, would serve as a robe or blanket for my bunk.

I soon found that the lynx had served me far better than I had thought at first, for the deer furnished a great many useful things which would prove of the utmost value.

In the hard, pointed horns I saw material which could be used in making awls, arrow-heads, and other objects. The strong leg-tendons were just what I required for my bowstring. The great cords of the neck would serve as thongs or lashings, and I even saved the hoofs in the hope of finding use for them. By the time the skin was scraped and spread up, the cords and tendons removed and stretched to dry, and the meat had been cut up and hung up to freeze, the afternoon was spent and darkness prevented further work.

I had expected that the lynx would prowl about and disturb me, but I heard nothing of him that night. The next day I devoted all my time and energies to finishing my new bow. It promised to be a powerful and beautiful weapon. The following morning I fitted it with a string of braided deer-sinew and, stepping outside, proceeded to test its power.

I used one of my old bone-tipped shafts—for I had determined to discard them for better ones. It took all my strength to draw the arrow to the head. My target was a tree some fifty yards distant. The loud, musical twang of the string, the speed of the whizzing shaft, and the sounding thump with which it struck the tree filled me with delight, and I hurried forward to examine the arrow where it stuck, quivering, in the tree trunk.

Although it was a light, bone-headed affair, it had buried its tip for fully an inch in the frozen bark, and while the head was ruined and the shaft split and shattered from the blow, I was overjoyed at the result of the shot.

Returning to the hut, I now gave my attention to making a supply of arrows. I had already gathered a good number of straight sticks for the shafts, and I soon bound the stone heads I had made to these, using fine deer-sinews for lashings, and by this means making a far neater and better job than I had been able to accomplish before. Moreover, I had profited by my experience in shrinking hide upon the snow-shoes, and wrapped the arrows with wet sinew, which tightened until it sank into the wood when it dried. These arrows I feathered with strips cut from partridge-wing quills, binding them on with slender, thread-like sinew. Having used up all my stone arrow-points, I proceeded to make some more tips from the deer's horns. This was an easy matter compared with many things which I had accomplished, for I could cut off the tips of the antlers, grind them to sharp points, and work down the bases to fit the shafts, all by means of the grindstone. While engaged at this occupation I wondered how I would have fared had I not found the cabin, the grindstone, the file, and the other contents of the hut. In an open lean-to I would doubtless have frozen to death long ago; the file had enabled me to kill the bear and secure a warm robe; while the grindstone had provided means for making the spear. The hut had unquestionably saved my life, but I felt that even without the file and the grindstone I might have succeeded very well, for my traps and snares had provided food and clothing without the aid of the hut's contents. I could have used birch-bark utensils instead of the kettle, and nothing which I had found in the cabin had helped me in making stone arrow-heads. I was grateful for everything I had found, however, for I realized how much I had been helped by the odds and ends which the former occupants of the cabin had not thought worth taking away, and I breathed a blessing upon them for what they had left behind.

With my strong, powerful bow and quartz-tipped arrows I felt well armed and able to attack any creature which I might meet—not excepting the abominable lynx. I was so anxious to test my new weapons on real game that I decided to set out the next morning without waiting to complete my horn-tipped arrows.

As I stood outside my hut, hesitating which way to go, I thought of the trail which led away from the cabin and immediately decided to follow this.

I had given up all thoughts of making my way out of the forest until spring, but my snow-shoes were such a success that I could travel almost as readily as if no snow covered the earth, and I became quite excited as I reasoned that, after all, there was no reason why I should not explore the trail for as far as I could walk and return in a day. For all I knew the settlements might be much nearer than I suspected, and while I was far too wise to dream of cutting myself off from the comforts and safety of my cabin by making a long trip through the winter woods, I might learn something of value by following the old trail for a few miles.

In any case, I had not gone in this direction since the snow fell and I was as likely to find game there as anywhere else, and, so thinking, I trudged off along the track. In the autumn, before snow had hidden the ground, the trail had been easy to follow, for half-obliterated ax-marks on the trees, old moss-grown stumps where trees had been felled to form a rough road, and other signs made the way unmistakable. Now, however, the old stumps and logs were many feet beneath the snow, while many of the “blazes” on the trees were hidden by it, and only by using great care and by noting the more open space in the forest—which indicated where the trees had been cut—was I able to follow the trail.

I saw my first game when about a mile from my cabin—a sleek, red fox—which trotted out of the woods ahead and stopped short and gazed at me curiously as his keen nose caught my scent.

Hastily fitting an arrow to my bow, I drew it to the head and let drive at Reynard, but at the twang of the bowstring he was up and away like a flash of red light, and the whizzing, stone-tipped arrow buried itself harmlessly in the snow. I was somewhat disappointed, but I was also encouraged when I found the arrow had struck fair and square exactly where the fox had stood, for I knew then that there was no question of the accuracy of my weapons.

It took me a long time to dig the arrow out, for it had penetrated several feet of snow, but I recovered it at last and resumed my tramp. Soon after this I started a white hare from a clump of drooping evergreen boughs and marked him down where he squatted beside a fallen branch, a score of paces distant. Had I not actually seen him stop I should never have been able to distinguish him, for his white fur rendered him invisible against the snow, and only by keeping my eyes fixed upon him could I be sure of his presence. As the speeding arrow struck home there was a little commotion in the snow and, hurrying forward, I found the hare pierced through and

through. It was really an excellent shot, and I felt wonderfully proud of my skill as I slung the hare at my belt and continued on my way.

I had traveled perhaps three miles from my cabin when the woods came suddenly to an end and I found myself standing at the edge of a large clearing. I was so absolutely dumfounded that I could scarce believe my eyes, for scattered about the clearing were half a dozen log houses. I cannot describe the sensations that swept over me in the few brief minutes that I stood there, gazing speechless at the tiny settlement. Mingled with my overwhelming joy at sight of the buildings was a feeling of chagrin to think that I had lived alone for months, had suffered agonies, and had endured hardships and privations, when by an hour's walk I could have been once more among my fellow-men.

With a loud halloo I hurried forward, and then, for the first time, I noticed that no smoke rose from the chimneys of the log buildings, that the smooth expanse of snow was unbroken by human footprints or trodden paths, and that no yelping curs—invariable accessories to outlying settlements—disputed my approach.

Before I reached the first house my spirits fell, for I realized that the place was deserted, that no human being was there to welcome me, and a sort of vague fear crept over me. There was something unnatural, something “creepy,” mysterious, and weird about this lifeless village in the wilderness, and as I reached the first building I hesitated to push open the door for dread of what I might find within.

But I had no need to fear; the house was empty, although garments hung upon the walls, the bunks were filled with moldy bedding, and utensils and rude furniture were scattered about. My curiosity now overcame my groundless fears and I made the rounds of the entire settlement, but in every building it was the same—not a sign of life anywhere—and yet somehow I felt convinced that human beings had been there not long before. Where they had gone, what had caused them to leave, was a mystery, for the buildings were in good repair, there was an abundance of timber all about, and a few dry corn-stalks projecting above the snow showed that the former inhabitants had cultivated the ground. That it was a permanent settlement and not a temporary wood-cutters' camp was evident, for articles of feminine apparel were upon the walls of some of the houses and a few cheap toys were in two of the dwellings.

The buildings were so much better than my little cabin, that for a moment I considered moving my belongings and taking possession of the place, but the thought had scarcely entered my mind when I realized that I would feel far more lonely and depressed when surrounded by the silent, deserted houses than in my own cabin in the heart of the woods. But if I could not consider taking up my quarters in the village, I could at least make my life far easier by helping myself to the many useful articles the former inhabitants had left. Thus thinking, I started toward the first house, determined to make a systematic search of every building.

I was about half-way across an open space near the center of the clearing, which I supposed had been cultivated land, when the toe of my snow-shoe caught upon some object and I plunged forward into the snow. Picking myself up, I glanced around to see what had thus tripped me, and to my surprise saw a hewn timber, or plank, projecting through the snow.

Curious to learn what this meant, I dug away the snow about it and in a few moments disclosed a second timber nailed at right angles to the first. There was no doubt about it—the object over which I had stumbled was a cross! At first its import did not dawn upon me. No doubt, I thought, the inhabitants had been French Canadians and had erected the cross above a little shrine, and, rising, I passed around it and was about to continue on my way when I caught a glimpse of letters cut into the wood. Scraping away the ice and snow which half concealed them, I studied the inscription for a moment and a wave of deadly horror swept over me, for, in rudely carved, misspelled words I read the following:

Joseph and Marie Bemis His Wife
Dead of Smallpox October 25
God Have Mercy on Their Souls

Instantly I realized why the village was deserted, why no human being was there, why clothing, household goods, and even the children's toys had been left behind. The place had been swept by the plague and those who survived, if indeed any *had* escaped, had left the stricken spot to its silent dead.

And with the thought came terror; all unwittingly I had exposed myself to the awful malady. I had entered the houses reeking with pestilence, had inhaled the stale air within the buildings where men and women had breathed their last, and had handled the very clothes and bedding which had

covered their bodies during illness and death. Beyond a doubt I had already contracted the dread disease; by now the germs of smallpox might be coursing through my blood. Sick at the thought of what my fate might be and haunted by the specter of loathsome death that stalked in the silent village, I turned and dashed madly back along the trail.

From my path a flock of partridges whirled up and, perching upon a near-by tree, gazed curiously at the fur-clad being stumbling headlong through the woods. A hare leaped from the snow and scampered to one side within easy bow-shot; but I heeded neither bird nor beast, for my one thought was to leave the accursed buildings behind, to regain my cabin, and to throw myself upon my rude bunk to await the deadly sickness I felt sure would be my fate.

Breathless, exhausted, and panting I reached the hut, threw off my outer garments, and huddled in my bed to spend an awful afternoon and night, as, alternately shaking with cold chills and burning with fever, I waited for the first symptoms of the malady.

To add to my misery the lynx howled in the woods near by, and in its awful cries I seemed to hear a dire foreboding of my death; a note of triumphant malice, as if the creature knew my plight and realized that ere long I would be helpless, that, ill and dying, I could offer no resistance, and that, tearing his way through the flimsy roof, he could spring on me and glut himself upon my wasted flesh.

Like some terrible nightmare the long hours passed, for I was more terrified than ever before in my life, and no savage creature, no danger of the forest, not even the dread of losing my foot, had filled me with such mortal fear as the thought of lying alone and uncared for in my hut while dying slowly of the smallpox.

But toward morning sheer exhaustion compelled me to sleep, and the day was well advanced when I again awoke. Then, feeling strong, hungry, and as well as ever, much of my first fear left me and, pulling myself together, I looked the matter squarely in the face.

After all, I reasoned, I might not have contracted the disease. I had been vaccinated a few years before, and even if this was no longer efficacious I might be naturally immune to smallpox. And then another cheering thought came to me. Perhaps the two who had died were the only victims and the others had left before the disease had spread. Moreover, I now remembered there was no date other than October 25 upon the rude epitaph, and, for all I

knew, the cross might have been erected several years before and the village might have been occupied for long thereafter by healthy people. All these sensible thoughts comforted and cheered me and revived my spirits greatly. I decided there was no use of worrying over what might never happen and that if I *had* contracted the disease there would be time enough to fear the results when sickness came on.

Meanwhile I was busily preparing my breakfast, for I was very hungry, having quite forgotten to eat during my fright the day before. I was half through my meal when a new idea came to me.

I had found this very cabin deserted. Was it not probable that its occupants had also been afflicted with smallpox? For an instant I was quite overcome with the thought that I might have been dwelling for many weeks where people had suffered or died with the contagious disease, but the next moment I burst into a hearty laugh, for I realized that if, as I suspected, there *had* been smallpox in the cabin, the fact that I had not contracted it was proof that I had little need to fear. Thus reassured and casting all worry aside, I gathered up my weapons and went forth to search for game.

This morning I decided to take a new route and to cross the lake, which, I felt sure, was frozen hard enough to bear my weight in safety, for I still had a lingering feeling that the way to the settlements lay across the lake, and I was anxious to explore the farther side.

Traveling upon the smooth, unobstructed surface of the lake was easy, and I soon reached the opposite shores. I found the forest far heavier here, while rocky ledges, cliffs, and ravines made travel very difficult. In fact, I found it impossible to penetrate far into the woods, and in the hopes of finding some trail, pass, or opening I skirted the shore of the lake from one end to the other. But everywhere I found the same wild, impassable country, and, becoming firmly convinced that the settlements did not lie in this direction, I headed around the upper end of the lake toward home.

I had seen no game during my tramp, although tracks of foxes, hares, and other creatures were numerous, and, reaching my own side of the lake, I entered the forest and proceeded to seek carefully for game. I had walked for some distance and was well within the woods when I again felt the sensation of being followed and watched which I had experienced when dragging home the deer. At first I thought this was pure imagination, for I had seen no signs of lynx tracks in the vicinity, but nevertheless, I could not resist the desire to glance furtively about from time to time. Finally the

feeling became so strong that it got upon my nerves and almost unconsciously I began to swing around on my tracks toward my cabin. Presently I came upon my own trail, and then I knew for a certainty that instinct had not played me false, for beside the broad, oval marks of my snow-shoes were the unmistakable tracks of the lynx. The beast was trailing me! At first I felt fear, but this quickly gave way to anger at the lynx for daring to track me, and I became possessed with the determination to slay him and be rid of his presence once and for all.

To attempt to approach within reach of the lynx would, I felt, be useless, and I knew he was far too sagacious to be caught in my traps. I decided to match my own skill and cunning against his and to bring him to his death through his own persistence in following me.

Carefully stepping in my former tracks I continued on for some distance and then, turning about, retraced my steps, still stepping in the marks already made by my snow-shoes. Presently I came to a spot where a low limb projected above the tracks and, grasping this, I drew myself up, untied the snow-shoes from my feet, and very carefully worked my way along the branch to the trunk of the tree. It was leafless and bare and afforded no shelter, and I knew that the lynx would see me long before I saw him and so, dropping to the snow on the farther side of the trunk, I made my way to a near-by evergreen and, climbing up, concealed myself among its thick branches.

With an arrow fitted to my bow I waited, peering forth through the aisles of the forest in the direction whence I had come.

The time passed slowly. A great, scarlet-crested, pileated woodpecker flitted to a neighboring tree and the forest echoed to the resounding blows of his powerful beak. A flock of redpolls twittered among the branches above my head, and crossbills clambered, parrot-like, among the drooping cones, shearing off the scales with their scissors-like beaks in their search for pine seeds. A gray Canada jay alighted upon one of my snow-shoes and pecked at the dry bear's hide, eyeing me saucily meanwhile. From a treetop a squirrel chattered, and from all about came the plaintive calls of chickadees. The forest was full of busy life, each tiny creature busily gleaning its livelihood and all unmindful of my presence.

Five, ten, fifteen minutes passed, and then at last I saw him. Slouching along in a loose-jointed stride, the lynx skulked on after me in the distance, keeping as much as possible behind trees and fallen branches, sniffing at

my footprints and ever and anon stopping to peer about, while turning his head first this way, then that, as if to smell the air for a suspicious scent. Presently he reached the spot where my two trails met and instantly the creature became all alert. For a few steps he trotted back upon the old trail and then, turning, came back and went along the new trail for a few feet. Evidently he was somewhat puzzled, for his every action betrayed the fact. Then curiosity gave place to suspicion, and with a single lithe bound he leaped into a tree and, crouching close to the trunk, peered about as if striving to catch sight of me. But he could see nothing that resembled the being he was hunting and, reassured, he sprang down and, after a moment's hesitation, came trotting toward my hiding-place.

Already the wild things about had seen the great cat and had scented danger in the tawny, threatening form. With a piercing cry the woodpecker ceased his tattoo and winged his way swiftly out of sight. The squirrels in the treetops ceased their chatter and hugged the bark, motionless. The twitter of the redpolls and crossbills was silenced. Even the bold whisky-jack, deciding that discretion was the better part of valor, perched far out of reach and contented himself with taunting the lynx with raucous cries. Only the chickadees seemed undisturbed. Regardless of the approaching form, they continued to flutter about and to utter their sweet calls, as if they realized they were too tiny to attract the savage creature with his lust for blood.

Now the lynx was within a hundred yards; I could see his baleful green eyes, his half-opened mouth, and his keen, white teeth. Each instant he was approaching closer. Never had I suffered from buck-fever before, but now my hands shook, my teeth chattered, and a strange, choking sensation gripped my throat. The lynx was within easy bow-shot, but still I hesitated, striving to steady my nerves, determined to take no chances, and watching with fixed gaze as he came nearer and nearer. At last he reached the spot where I had doubled on my tracks, sniffed about, raised himself on his hind legs, and before I realized what he was about he sprang to the very branch upon which I had drawn myself. Instantly he caught my scent upon the bark, crouched low with bristling fur, and turned his fierce eyes directly upon my hiding-place. By some supernatural instinct he seemed to have divined my ruse and to have discovered me. Perhaps he did not actually see me, perhaps it was mere chance which led him to jump into the tree, but at the moment I felt convinced that he saw me as plainly as I saw him, that he

had overcome all his natural cowardice and was bent on attacking me, and that in another instant he would launch himself across the intervening space and fly at my throat with those great, hooked claws and gleaming teeth. Scarce a score of feet separated us. At any moment he might crouch and spring. Drawing back the bow with trembling fingers, I let the arrow fly. Even as I drew the bow I knew the lynx detected the motion, and as the string twanged I saw his great form shoot into space, I had a glimpse of the outstretched feet and bared talons, I heard a snarl of rage, and the next second I fell crashing to the snow as the lynx plunged through the screen of branches about me.

The force of my fall buried me under the snow; before I could rise, even before I realized fully what had happened, something landed on me with a thud that knocked the little remaining breath from my body, and with a wild, frightened yell I struck out blindly with fists and feet. What followed was the madness of nightmare. Blinded by snow, frightened half out of my wits, dazed by my fall, struggling, kicking, striking, I was whirled about like a giant teetotum, while my yells and shouts mingled with snarls, growls, and piercing screams. Glimpses of evergreen-trees and blue sky, avalanches of snow and a brown, furry form, revolved in a kaleidoscopic blur. Suddenly all motion ceased; I found myself lying, panting but unharmed, in a crater of snow, and, sitting up, I rubbed my eyes and glanced about. Instantly I realized what had happened. The lynx had missed his mark, had fallen squarely on top of me, and together we had fought and struggled in the snow, each more frightened than the other, each striving to break loose, and both scratching, striking, and screaming with all our strength.

No wonder I had been dazed and my mind turned topsy-turvy. The marvel was that I was still alive and well and not minus eyes, ears, and strips of flesh. My furs had saved me from the lynx's claws, and the snow had acted as a pad, but my clothing was in tatters, bunches of lynx fur were scattered about, and all around the snow was churned, tossed, and furrowed where we had spun hither and thither like a mad pinwheel.

For a brief instant I was filled with dread for fear the creature was waiting close at hand to resume the attack, and then, the humor of the situation dawning upon me, I roared with laughter until tears filled my eyes.

My bow was close at hand, where it had dropped from the tree, my snow-shoes were lying half buried in the snow, and my various other

possessions were scattered about, but not a sign of the lynx could I see. But as I rose and started to gather up my belongings I saw drops of scarlet upon the snow beside the creature's tracks. A little farther on I picked up the broken shaft of my arrow, covered with blood, and then I knew that my antagonist had not escaped unscathed.

For a few yards I followed the telltale blood-drops, until both the stains and tracks ended where the lynx had evidently taken to the trees. I was not sorry; I had had my fill of adventures for one day, and, turning on my tracks, I made my way back toward the cabin. How badly the lynx was wounded I could not tell. Perhaps it was merely a scratch from which he suffered little, or perchance the stone tip of my arrow buried itself in his vitals and caused his death. At any rate, he never troubled me again and his weird cry never disturbed my slumbers in the future. This being the case, I cared little whether through fright he had fled the country or whether he had slunk away to die within his den.

CHAPTER XIII

SPRING APPROACHES

For some time after my adventure with the lynx no event worth recording occurred and I led a humdrum, lonely life. I had much to be thankful for, however, for I had little difficulty in obtaining game. With the disappearance of the lynx I again resorted to traps. I was comfortably housed and I suffered little from the cold. No symptoms of smallpox had developed, my foot and ankle were as well and strong as ever, and I was in the best of health and condition.

There was little to be done save to hunt and trap, gather fire-wood, and sleep and eat. To occupy my time I made a large number of stone arrow-heads and numerous arrows, for I realized that I would lose and break many, and that when I finally started away in the spring I must be well provided with weapons.

I also tried fishing in the lake, cutting holes through the ice with my knife and spear, and setting lines which would raise a signal when there was a bite, as I had done when pickerel-fishing in my youth.

On these lines I used hooks of horn which I made by grinding down the forked portions of the deer's antlers, where the prongs sprung from the main branches. It was tedious work, cutting these on the grindstone and with my knife, but the results more than repaid me for all the labor, for I caught a number of pickerel, several bass, and a few splendid lake-trout, and the fish proved a most welcome change of diet.

There were many heavy snowfalls during the winter, and had it not been for my snow-shoes I would have fared badly. Even as it was I was compelled to remain indoors for days at a time as the storms raged without, and by the time the days began to lengthen and the worst of the winter was over my hut was completely snowed under. Only the rude chimney and the tunnel-like entrance to my door were visible, but the deeper the snow piled around the cabin the warmer and more comfortable it was within.

But if the weather kept me much indoors I made good use of my time, and during the long winter I made many discoveries and accomplished many things which added wonderfully to my happiness and comfort.

One of my greatest troubles was lack of light, for the winter days were short, and within the windowless hut it was dark by mid-afternoon, and I was thus obliged to cease work and sit idle for many hours each day. The fire gave enough light to enable me to see to cook and even to perform certain tasks which required little care, such as cutting up fire-wood, whittling sticks to form triggers for traps, and similar work, but the fitful, unsteady glare of the flames was not sufficient to enable me to accomplish any fine or delicate work.

I had often thought of making an artificial light of some sort and had tried torches of birch bark, resinous sticks, and other devices, but none of these was satisfactory. The bark burned brightly and cast a fine light, but it flared wildly, sputtered, crackled, and soon burned out, while the pine knots made the interior of the cabin unbearable with their dense smoke.

Then it occurred to me that I might make tallow candles, for I had an abundance of fine bear's grease and deer's fat stored away, and this, I judged, would serve as well as mutton tallow. For wicks I decided to use strips torn from the remnants of my civilized clothes, and, having conceived the idea, I immediately proceeded to put it into execution. I knew that I would have to melt the fat and cast it in molds to form the candles, but I had now learned to make use of the resources at my command, and birch bark at once came to mind as suitable for making the molds. By rolling strips of the bark around a smooth stick, wrapping them with sinew, and then withdrawing the cylinders from the stick I formed the molds, but at my first attempt I found the melted grease ran out between the edges of the bark and from beneath the bottom of the molds as fast as I poured it in.

For a time I was greatly puzzled to devise a method of sealing up the cylinders, and then a happy idea occurred to me, and stepping to the door I dug a quantity of snow and packed it solidly in a corner of the hut away from the fire. Slipping one of the birch-bark molds over the stick, I pushed both down into the cake of snow, and then, withdrawing the stick, left the cylinder of bark remaining in the snow. By means of a small, slender stick I pushed one end of a narrow strip of cotton into the snow at the bottom of the mold, and poured in the melted tallow. Although the hot grease melted the surrounding snow somewhat, yet little escaped from the mold, and as

soon as it commenced to thicken I moved the wick to the center of the mass of grease and held it in position until the tallow hardened. I could hardly wait until the tallow was fully hard before withdrawing the mold from the snow to examine the result of the experiment. By pushing on the lower end of the tallow with the stick about which I had formed the mold I slipped the rude candle from the bark, set it upright on the table, and touched a blazing stick to the wick. Instantly it burst into flame and burned brightly, casting a steady light about the room. I was immensely pleased, for with half a dozen such lights I could see to work despite the darkness outside. My joy was short-lived, however, for the wick burned rapidly, curled over to one side, flared and smoked, and melted the candle away.

Evidently something was wrong, some small detail had been overlooked, and, casting the remnants of my first candle into the can of melted grease, I sat down to try and reason it out. Soon it came to my mind that all the candle-wicks I had ever seen were twisted or braided. I thought that with a braided wick my candles might succeed, and, tearing a strip of cloth into narrow ribbons, I braided it tightly. But I soon found the result would be far too coarse for a wick, and again I was at a loss. Not for a long time did I hit upon the plan of unraveling cloth and braiding the yarns together to produce the compact, firm cord which I desired. When at last this was accomplished and another candle was cast, with the braided wick, it proved a great success and thereafter I never wanted for light. But it was slow work making the birch-bark molds and casting the candles in them, and I sought for some easier and more rapid means of making them, and kept my eyes open for something better than the birch-bark cylinders for molds. It was some time before I thought of anything better, but finally I remembered that as a boy I had often made willow whistles and had slipped the cylinders of bark from the branches without splitting them. I had noticed willows near the outlet to the lake and soon I gathered a number of smooth, straight branches free from knots or sprouts. At first the bark refused to budge, for the wood was frozen, and, moreover, no sap was flowing through the bark, but I discovered that by steaming the branches or by soaking them in boiling water the bark stripped easily from the twigs, and thereafter I had little difficulty in casting my candles rapidly and easily, without wasting tallow or requiring the snow about the molds.

Oddly enough, my candle-making led me to two other most important discoveries, one of which was soap, the other a method of removing the hair

or fur from hides without waiting for them to decompose.

I discovered how to make soap quite by accident. While working at my candles I upset a quantity of grease on the hearth, and this, running down into a hollow at one side, formed quite a pool, which remained liquid, owing to the warmth of the fire. In order to cool it so I could scrape it up, I poured water upon it, but when I tried to remove it I discovered that it was full of wood-ashes and was soft, slippery, and stringy. Deciding that it was not worth bothering about, I abandoned the idea of recovering it, and proceeded to wash my hands. Much to my surprise, the water bubbled and frothed and my hands became cleaner than they had been for months. Then, all at once, it dawned upon me that I had made soap—crude, greasy, imperfect soap, to be sure, but still soap which possessed wonderful cleansing properties.

Any school-boy should have known that wood-ashes contained potash and that potash or lye and water would transform fat to soap, but during all my life in the woods it had never once occurred to me.

Now, however, I lost no more time. By soaking fine wood-ashes in water and then boiling this with grease I soon succeeded in producing soft soap which served my purposes exceedingly well.

It was a great comfort to be able to wash myself thoroughly, although the free potash in the soap stung and burnt my skin.

But of far greater importance to me than the soap itself was the fact that in making it I learned how to remove the hair or fur from hides both rapidly and easily. The first batch of soap which I made was prepared with deer's tallow, but I could ill afford to use this material for such a purpose. I carefully gathered together all the fat which I could obtain from the hares, partridges, and other game which I secured. In addition I saved all the remnants of meat from my meals, as well as the bones, all of which I boiled together with the lye-water. Among the other odds and ends were small pieces of skin, such as the feet of the hares, etc., and after the mass had boiled thoroughly I strained off the clear, liquid soap. While doing this I discovered that every sign of hair had been removed from the pieces of skin and this fact led me to experiment, and I soon found that by soaking hides in a mixture of water and wood-ashes I could remove all the hair or fur without injuring the skin in the least. Moreover, the action of the potash removed the grease from the skin and left it clean and I was thus saved the tedious process of scraping the hides with my knife. Of course the first few

trials of this method were something of a failure, for the lye was too strong and the skins were weak, rotten, and worthless, but by gradually decreasing the quantity of ashes and the length of time the skins were allowed to soak I found that the hair could be loosened so it was easily rubbed off without affecting the toughness of the skin itself.

Now that I had learned how to remove the hair from the skins, I was anxious to devise a method of tanning them without cutting them up as I had been obliged to do hitherto. My only means of tanning was to soak the pieces of hide in the iron pot. But the pot was small and it was needed for many other purposes. I could not use it while the skins were soaking, and I was obliged to cleanse it most carefully after the tanning, yet there was no other receptacle which would serve and I had no means of making any. I had already thought of hollowing out a log to form a trough, but after several hours' work which resulted only in a cavity hardly large enough to tan a chipmunk's skin, I gave up in despair.

Birch bark seemed to offer the only solution, but I found that pieces of bark of sufficient size for my purpose were always full of holes and cracks. I pondered over the matter for a long time before I thought of making a vat, or tank, after the manner of a birch-bark canoe. I had often used Indian canoes and knew, in a general way, how the pieces of bark were sewed together and rendered water-tight by means of pitch and gum. While I had too little confidence in my skill to attempt building a canoe, yet I decided it was quite possible to follow out similar methods and make a good-sized tank or basin.

My first step was to construct an oval frame about three feet in length by a foot and a half wide, which I formed from light birch withes. To this I lashed other withes, in the manner of the ribs to a canoe, and then by weaving small branches in and out and lashing them to the ribs I at last produced a sort of basket-like affair without any square corners or straight lines. This form was not intentional on my part, but was the result of necessity. As this turned out, it was fortunate, for I ultimately discovered that to make a good, tight joint at a sharp corner or a right angle was exceedingly difficult. The framework being completed, I placed it upon a large sheet of bark and by dint of a great deal of work and many trials I drew the bark tightly over the frame and secured it firmly to the edges with strips of moosewood bark and hemlock roots. Other kinds of bark were then secured over the frame until the whole was covered, and then came the

most difficult work of all, which was to sew the overlapping edges of the various pieces of bark together. Despite every care, I found that holes made by my file-awl would tear out when I drew the root-threads tight, and it was not until I hit upon the plan of burning holes through the bark with the hot awl that I met with any success.

But practice makes perfect, even when one labors with primitive tools in the heart of the wilderness, and before I had half finished with my work I learned to bore holes with a deer's-horn awl and to run the roots at an angle, or crisscross, from hole to hole, so that the strain did not come with the grain of the bark. It was slow, tedious work even then, however, and several days' steady labor was required before the last seam was closed.

Rude and uncouth as was this great, bathtub-like affair, I was greatly pleased at my handiwork and I was in a great hurry to patch up the seams and holes with pitch and to try it.

There was an abundance of pitch and gum to be had with little trouble, and I gathered a great quantity of it. This I melted and smeared over the seams of the bark, and when at last I had daubed every crack, seam, and knot-hole with the sticky mess I felt that I had successfully accomplished a great undertaking.

The birch-bark tub held water, only leaking slightly in a few spots, which were readily closed up by daubing on more pitch, and I at once started in to prepare enough tanning liquor to fill it. Then I almost undid all my hard work by my own stupidity, for as soon as the first potful of liquor was ready I poured it into the bark tub and the hot liquor softened the pitch and the tan ran out over the floor.

Fortunately it opened up only one seam, and this I soon repaired, but it served to teach me a lesson, and thereafter I cooled each lot of liquor by placing it outside in the intensely cold air before pouring it into the vat.

It must not be supposed that all this was accomplished in a day or in several days. I was obliged to attend to my traps, hunt for game, cook my meals, cut and gather fire-wood, and do many other necessary things meanwhile, and, moreover, I found that a great deal of time was required to gather the oak and sumach bark. As I could only boil a small quantity of bark at one time and as I was obliged to use my pot for other purposes between times, the work progressed very slowly, and fully two weeks elapsed from the time I finished the birch-bark tub until I had filled it with sufficient liquor to tan a large skin.

Long before the tub was quarter-full I commenced to make use of it, however, and after removing the fur from the skins, by means of wood-ashes and water, I had placed several hares' hides in it. The result was all I expected, and while the thin, papery rabbit-skins had little strength, yet I was greatly elated at being able to remove the hair and make leather. My intention was to attempt tanning the deer's hide as soon as I had accumulated enough liquor to cover it. Since I really had no need of a second fur robe, the bear's skin serving all my purposes, I determined to try to remove the hair from the deer-skin.

To be sure, I had no second receptacle in which to soak the hide in wood-ashes and water, but by stringing up the hide by its four corners and filling it with water I softened it, and then, turning it over, I filled the hair side with the potash solution. I was in some doubt about the success of this experiment. I watched the skin with great care and frequently tested the hair. At last I was rewarded by finding that it came away readily. As soon as this happened I drained off the solution, scraped and rubbed off the hair, and placed the hide in the tanning liquor.

Each day I turned and soused it about and gradually it assumed the texture and appearance of leather. When at last it was drawn out, drained and worked and rubbed until dry, I possessed a fine, large piece of excellent buckskin which I knew would make splendid moccasins and strong, durable garments for use in the spring, when my furs would have to be discarded and I started on my journey to the settlements.

My success was so great that I longed to tan the bear-skin with the fur upon it, for it was a stiff, ungainly thing in its plain dried state, but I could not spare it while cold weather lasted, and I was compelled to continue using it as it was.

Now that I had the means of making leather, I was most anxious to secure hides to tan, and I made long trips into the woods, searching for larger game than hares and partridges. But either the larger creatures had migrated to some part of the country where there was less snow, or else they had seldom been in the vicinity, for, despite every effort, I found few signs of their presence and secured still less. Only twice did I see where deer had passed, and one raccoon, a fox, and two martens were the sole results of all my hunts, aside from the hares and partridges which I had little difficulty in obtaining.

At first I was greatly puzzled by this scarcity of big game, for I was in a wilderness where deer, moose, bear, and even caribou might be found, but after giving the matter due thought I came to the conclusion that it was, no doubt, due to the proximity of the deserted settlement I had found.

Ever since I had first devised my simple calendar I had kept account of the days, weeks, and months, and now I knew that the worst of the winter was past. With each day spring was approaching, and before many months were over I would be on my way toward civilization and my life in the wilderness would be a thing of the past.

With the thought I was filled with pride and happiness to think that I had overcome all obstacles which had arisen, that the long winter—which I had so greatly dreaded—was nearly over, that, after all, it had not been such a terrible experience, and that I had managed to live in comparative comfort and in plenty by my unaided efforts.

Strange as it may seem, I really felt rather sorry at the thought of leaving the cabin where I had lived so long, for I had become so accustomed to it that it seemed quite like home, and the woods about, the friendly jays, the twittering redpolls, and gray-clad chickadees, and even the dainty little wild mouse that crept forth from its hiding-place to share my meals, all seemed like old friends. Even without my calendar I should have known that the backbone of winter was broken, for each morning, when I stepped forth, the rays of sunlight were brighter, the sky held a softer, warmer hue, at midday the snow upon the cabin roof melted and trickled down in little streams that turned to great icicles at night, and the snow was covered with a crust each morning.

Slowly, imperceptibly, the height of the drifted snow about the hut decreased, the roof and walls rose above the surrounding surface of white, and stumps and bushes long hidden from view jutted upward through the snow.

Great cracks spread across the vast, white plain that marked the lake; a few days later broad, black spaces spoke of open water, and when, one morning, I heard the honk of geese, and, looking up, saw a great, V-shaped flock of black specks winging swiftly toward the lake, I knew that spring was near at hand.

CHAPTER XIV

I FIND A COMPANION

Impatient as I was to start on my journey, I had no intention of taking any risks, for I well knew the treachery of spring weather, and that a belated snow-storm might yet arrive.

I therefore determined to wait until the weather was unquestionably settled and there was no further chance of a snowfall, which I knew would not be for several months.

This enforced period of waiting seemed to pass more slowly than all the time I had spent in the wilderness, and I strove to keep my hands and brain busy. Already the spring thaws had set in, the lake was free from ice, bare patches of earth showed here and there upon exposed hillsides, and brooks and rivulets were transformed to roaring torrents.

The weather had become so warm that I was glad to discard my fur garments, and in their place I donned trouser-like leggings and a loose, shirt-like blouse which I had fashioned from deer-skin. They were roughly made, devoid of fit, but strong, warm, and comfortable. In this rude garb, with my face hidden under an untrimmed beard, and hair falling to my shoulders, I was a veritable Crusoe of the wilderness.

With the breaking of winter game became more plentiful and I found many deer tracks in the woods, and while following one of these I made a great discovery which, had I known of it sooner, would have made my life in the woods much more bearable.

The track I was following led to a portion of the forest which I had not visited before, and soon it was joined by numerous other tracks, until I found myself walking along a well-beaten path or trail.

It was evidently a much-traveled runway, and, feeling confident that I would find deer close at hand, I moved forward with the utmost caution, stealing through the thickets, taking advantage of every bit of cover, and peering from behind trees at each turn or twist in the trail. I had thus proceeded for fully half a mile when I came in view of a small, open space,

bare of vegetation, and, standing near the center, with head bent to the ground as if feeding upon the mud, was a splendid deer.

Here indeed was luck, for the creature was unaware of my presence and I was within easy bow-shot. Raising myself cautiously above the fallen tree behind which I crouched, I fitted my best arrow to my bow, drew it to the head, and aimed at the sleek, brown body. At the twang of the string the deer tossed up his head and leaped forward, but the speeding arrow was swifter than his muscles; it struck fair behind his shoulder-blade, and he plunged forward to the earth, killed as quickly as if by a rifle-ball.

Much as I needed his hide and meat, I could not help regretting that it had been necessary to destroy the life of such a splendid creature, and I marveled that I had ever been able to hunt and kill such harmless, beautiful animals and call it sport.

While bending over the deer I noticed for the first time that the spot whereon I stood was covered with little patches of white and that the deer's muzzle was coated with the same white substance. Wondering at this, I stooped, scraped up a bit of the strange material, and touched it to my tongue. It was salt.

Instantly I realized why the numerous tracks had led hither, why the deer had been so intently licking at the caked, muddy ground; the place was a "salt-lick," the first I had ever seen, but of which I had often heard.



“I AIMED AT THE SLEEK, BROWN BODY”

Throughout my life in the woods I had missed salt terribly, and while I had become accustomed to going without it, all my old longing came back to me as I tasted this muddy, bitter salt. I regretted deeply not having found it before. But I intended to revel in it now, and that night I ate meat with salt for the first time in many months, and it tasted wonderfully good.

A short time after this I again bent my steps toward the salt-lick to obtain a fresh supply of the precious material, and, while I had no immediate need of more venison or of additional hides, yet I approached as cautiously as before, for I was curious to see if more deer were at the spot.

I had almost reached the old tree behind which I had hidden on my former trip when my ears caught a most unusual sound. It was low and faint and resembled the moaning wail of a sick child, and yet there was a whining note to it which did not sound human.

At first I could not locate it, but, by turning my head and listening intently, I decided it issued from a thick clump of brush beyond the salt-lick. No living creature was in sight, and, rising, I moved toward the sound with bow and arrow ready for instant use, for I had no idea what danger might lurk within the thicket. As I drew near I noticed that the bare earth

was torn and that the salt and mud were reddened with blood, and I halted as it flashed upon me that the thicket might conceal a lynx mauling a deer which it had killed. But as I hesitated the sound again issued from the bushes, and its wailing, sobbing sound was so unmistakably the cry of some suffering being in mortal pain that I cast fear aside and pushed into the brush.

The next instant I stopped short and drew back, for lying upon a blood-stained bed of dead leaves was a gaunt, gray wolf! I had no need to fear, however; the poor creature was absolutely helpless, and at sound of my footsteps merely turned his eyes inquiringly in my direction, unable even to lift his head from off the ground. A great gash in his neck and innumerable cuts and tears upon his body showed how badly he was wounded, and my first thought was to step to him and mercifully end his misery. But as I bent above him and drew my knife his dimming eyes gazed at me with such a pleading, wistful expression that my heart revolted at the thought of killing him and I sheathed the knife, determined to do my utmost to ease his sufferings.

He was pitifully weak from loss of blood and was emaciated from lack of food, and I doubted if anything could be done for him, but he seemed to read my thoughts and, instead of snapping or growling, licked my hand in dog-like fashion as I tenderly examined the gaping wound in his throat.

It was such a strange, friendly act that my heart was won, and I vowed that if the creature *did* survive his hurts I would strive to make him my companion. I felt that at some time he must have been partially tamed. I soon found that none of his wounds were really serious in themselves, for no vital organ had been injured and only loss of blood and his hunger-weakened state had brought him so near to death. As I bathed his wounds and poured water from a near-by spring into his parched mouth I wondered what woodland tragedy had resulted in the wolf's condition, and came to the conclusion that he had attacked a deer and, being weak with hunger, had been beaten off, gored, and trampled upon by the victorious buck.

The water seemed to revive him greatly. To stop the flow of blood from his wounds I bound them up as best I could with strips of bark and salt. Although I knew this must smart and burn the raw flesh terribly, yet the wolf scarcely uttered a sound of complaint, but bore the torture stoically while watching my every motion with his tawny eyes. As I worked I smiled to myself to think what a strange sight we presented—a wild man of the

woods, clad in rude, uncouth garments, with unkempt hair and beard, ministering to a shaggy, wounded wolf in the heart of the wilderness.

And when at last the creature's wounds were dressed and his moans had ceased, I realized that he must have food, and, speaking to him as I would to a child, and cautioning him to lie still and to be patient, I left him and went into the woods in search of game with which to feed him.

Birds were abundant and the saucy Canada jays, or whisky-jacks, were very tame, and while I had always left these feathered creatures unmolested, I felt no hesitation in killing one or two of the jays for the sake of my wolf-patient.

Returning with these, I cut them up and fed them to the wolf, who devoured them ravenously and heaved an almost human sigh of satisfaction as he licked his chops over the last bit of raw flesh.

The question now arose as to what I was to do with the poor beast. He was too weak to walk, and to carry him to my hut was out of the question, for to lift or move him would again open his wounds and start the blood flowing afresh. On the other hand, if left here unprotected in the woods he might succumb to the chill night air or else some prowling creature might kill and devour him. Perhaps I was sentimentally foolish to waste time and thought over a wounded wolf, but to my lonesome mind he was a fellow-being in distress, and already I had found great pleasure in attending to his wants and talking to him, and I was as anxious for his safety and comfort as if he had been a faithful dog.

There seemed to be but one thing to do, which was to build a shelter over him, provide him with food and water, and leave him alone until the following day, and I at once set about to erect a lean-to above him. When the rude protection was at last completed I filled a birch-bark dish with water and placed it within reach of his head, cut up another jay and laid this close by, and then, covering him with soft fir boughs and carefully placing others between his body and the damp earth, I left him gazing gratefully after me.

The next morning I hurried to the wolf as soon as I had eaten breakfast, carrying a quantity of meat with me, and to my joy found the creature much stronger than on the previous afternoon. He was now able partly to raise his body, and ate the venison and drank water eagerly, and apparently recognized me as his friend, for when I approached he thumped his brushy tail upon the leaves in welcome—an act which pleased me immensely and

which also surprised me, for I had never heard that wolves expressed their feelings in this dog-like manner.

I had brought materials with me with which to dress his wounds properly, and while I was in some dread that he might snap at me or might resent my acts while doing this, I soon found I had no cause for fear. Throughout the tedious and, no doubt, painful process of washing the cuts, smearing them with bear's grease and arnica and bandaging them, he lay almost motionless, only now and then uttering a low moan or a subdued growl to indicate an unusually severe twinge of pain. He was still very weak and lame, but despite this he made a brave effort to rise and follow me when I finally left him, but, finding it impossible, he sank back upon his bed of fir boughs and howled dismally as he saw me depart.

The next day I found the wolf well on the road toward recovery and sitting on his haunches, awaiting me. He greeted me with thumping tail and short yelps of pleasure, fawned upon me as I fed him, and, much to my satisfaction, limped along behind me when I started toward my hut. Knowing how lame and weak he was, I walked very slowly, stopped often to let him rest, and when at last we reached the clearing he seemed little the worse for the trip. At sight of the cabin the wolf drew back, as if suspecting danger or treachery, but when I entered the hut and returned with a junk of venison, which I held toward him, his hunger overcame all scruples and he trotted to the door. For some time he sniffed about suspiciously, with tail drooping and the hair on his neck bristling, and then, apparently convinced that he had nothing to fear, he entered the cabin and threw himself down near the fire.

I was genuinely surprised at this behavior, and in fact had marveled at his tameness and dog-like manner from the first, for it was quite at variance with all stories I had ever heard of wolves. Long afterward I learned that a pet wolf-cub had been kept by one of the inhabitants of the plague-stricken village I had found, and I was then convinced that Lobo, as I called my wolf, was this same semi-domesticated cub who had been left to fare for himself when his master died of smallpox.

One who has never lived alone in the heart of a wilderness for many weary months cannot appreciate the comfort which Lobo was to me. Day and night he was my constant companion and I talked to him as to a fellow-man, telling him my plans, asking his opinions on every question or problem that arose, and consulting him on all matters, to all of which he

replied by wags of his tail, low growls, sharp yelps, or by licking my hands or face as if he actually understood all that was said to him.

He soon recovered completely from his injuries, grew sleek and fat and, if the truth must be confessed, rather lazy, for he had plenty to eat without the trouble of hunting, and he much preferred the easy existence he led to the wild life to which he had been accustomed.

With Lobo for a comrade the time passed rapidly, and, the snow having all disappeared and the “pussies” commencing to swell upon the willows by the lake, I began making plans for leaving the forest and setting out on my tramp to the settlements.

CHAPTER XV

THE END OF THE TRAIL

I had thought the matter out many times already and had decided to set forth well provided with food, with all my weapons, my necessary utensils, and a supply of tanned skin, for I had no idea of how far I might have to tramp or how long I might be on the journey.

Few as were my possessions, yet I found that to carry them all would be impossible and that many things which would be of great help to me would have to be left behind. Among these were the pot and the grindstone and the bear-skin robe, and as I thought with regret of abandoning these a scheme came to me of utilizing Lobo as a pack-horse. But tame and good-natured as he was, the wolf would have none of this, and snarled, snapped, and bristled threateningly when I started to strap a load upon his back by way of experiment.

After all, the objects I must leave were not essential, and without them I would be far better off and more comfortable than during the early part of my wilderness life. The bear-skin, to be sure, would be a great comfort during the chilly spring nights, but I was well clad and a hare-skin undercoat would keep me warm enough, while I could broil my food, or even boil it in birch-bark dishes as well as I could cook it in the pot. The grindstone would be utterly useless, I knew, but I had grown to depend upon it so much and by its aid had been able to solve so many problems that I dreaded to think of being without it. However, it was useless to fret over the matter; the less I carried the sooner I would be out of the woods, and I had now become so self-reliant and so accustomed to overcoming difficulties that I had little fear of suffering on the way.

When at last I was sure the weather had settled and when the soggy, water-soaked ground had dried sufficiently to make walking possible, I gathered up my weapons, shouldered my packs, closed the door of the cabin which had sheltered me so long, and, with Lobo trotting beside me, bade good-by to my wilderness home.

At a turn of the old trail I stopped for a last look at the little hut and a lump rose in my throat at thought of leaving the place forever. It had served me well, the woods and waters had been kind to me, and I was about to turn my back on the only home I had known for many months and to start into the unknown.

It was a bright spring morning; birds were twittering among the swelling buds of trees and bushes, the sun was warm, and all about were signs of nature springing into life and beauty once more.

The broad green leaves of skunk cabbage spread above the black peat by the lake, jack-in-the-pulpits reared their straight stems and unfolding leaves beside rocks and fallen trees, pale pink and lavender hepaticas peeped through the fallen leaves on sunny slopes, and trailing-arbutus vines were rich with odorous, waxen blooms. It was a day to make one happy and light-hearted, to fill man or beast with the very joy of living, and Lobo gamboled and frolicked about, chasing imaginary rabbits into the thickets, yelping at the red squirrels who mocked him from the branches overhead, and betweenwhiles trotting back and forth with keen nose to earth as if seeking for some trail to follow.

I also felt that it was good to be afield on such a glorious morning, and, laughing at Lobo's antics, talking to him as to an intelligent being, whistling gaily, and filled with happiness as I realized that each step was bringing me nearer to the end of my trail, I swung rapidly ahead through the forest.

It was not long before we came in sight of the lonely, deserted buildings I had found during the winter, and as Lobo saw them he bristled up, howled, and, placing his tail between his legs, slunk at my heels. Although I knew that I must pass this spot on my way, yet I dreaded to approach it, and Lobo's actions added to my unreasonable fear and gave me a sort of "creepy" sensation, for somehow I could not shake off the foolish idea that he saw or heard things which were invisible to me.

Perhaps it was the shadow of death which hung over it, perhaps it was the contrast of the lifeless gray buildings with the bright sunshine and joyous spring air, or perhaps it was pure imagination on my part, but somehow the place seemed uncanny and unnatural. In the little space between the buildings the gaunt, bare cross rose upward, sharply outlined against the dark background of the distant woods, and now I saw that round about it were many smaller boards projecting from the ground. For some minutes I hesitated, standing within the shelter of the woods, while Lobo

whined cringingly beside me, for I could scarce bring myself to step forth into the clearing, to pass that tiny graveyard, and to run the gauntlet of those blank, staring windows and doors. But pass them I must if I was to reach the farther side where the trail led into the forest, and finally, summoning up my courage and calling myself a fool and worse, I started forward.

I had never believed in haunted houses, I scoffed at ghosts and everything that savored of the supernatural, and I had no patience with superstitions, but, ridiculous as it seems to me now, I could not help glancing furtively to right and left as we walked through that deserted village in the wilderness and gave the pitiful little graveyard a wide berth. I tried to argue with myself, to pooh-pooh my own foolishness, and to try to make myself believe it was merely the dread of smallpox that troubled me, and in a measure I succeeded. We had passed the graveyard, had left most of the buildings behind, and were approaching the entrance to an old wood road on the farther side. I breathed an involuntary sigh of relief to think that in a moment more we would be in the woods with the village behind us. Suddenly, as we came abreast of the last house with the patch of dead corn behind it, the door of the building swung open with a bang, and the dried stalks of corn swayed and rustled as if some object was tearing through them. In my heart I knew it was but a sudden gust of wind, but as Lobo uttered a melancholy howl and dashed past me I lost all control of myself and, taking to my heels, tore after him as fast as I could run. Not until I was a hundred yards down the road and the village was out of sight did I cease my wild race, and, overcoming my silly fright, seated myself upon a stump to regain my breath.

Lobo was nowhere to be seen, but presently he came trotting back along the old road, looking heartily ashamed of himself and wagging his tail and fawning about me as if to ask forgiveness for his behavior.

It all seems very childish and unpardonably silly of me, as I look back upon it now, but at the time it was real and a most unpleasant experience. Many animals seem to have an instinctive dread of a spot where human beings have lived and died. Lobo undoubtedly possessed such a fear, and I had lived so long in the woods and had been such a close companion of the wolf that perhaps I had developed this same unreasoning terror—it is the only excuse I can offer, the only way I can account for my sensations on that spring morning.

But the place was left behind. Our spirits rose again and, shouldering my packs, I resumed my journey.

The trail was easy to follow, for it had evidently once served as a logging-road, and for hour after hour we plodded on. That such a road must inevitably lead to the settlements I felt sure, for no one would have wasted time and labor in cutting a wide road for hauling lumber unless it led out of the forest. Even if it led me to an isolated sawmill or a rude lumbermen's camp it would serve my purpose.

At noon we stopped to rest and ate our midday meal of venison, and once more started on. My back ached from carrying the unaccustomed load of my packs, and my feet were sore. I had never walked long distances since finding the log cabin, and during most of the time I had traveled on snow-shoes.

The road wound and turned about and in several places forked or branched, and I was often in doubt as to which trail to take. The only way was to investigate, and I went many miles out of my way, only to find that these side trails led merely to cleared spots where the trees had been felled and cut for timber.

But while it was disappointing to be led astray, yet the presence of the cleared spaces and the diverging roads encouraged me greatly, for I knew that I was no longer in the heart of the vast wilderness, but was in a district which had been visited by large numbers of men and it could not be very far from the outposts of civilization.

How long a time had passed since the road had been cut or the lumbermen had labored here I could only guess, but the size of the young shoots which had sprung up, the moss and lichens which had grown upon the rotting stumps and discarded branches of the felled trees, and the thickets or brush which had partially obliterated the roadway convinced me that no active work had been carried on here for several years. This rather surprised and discouraged me, for I could see no reason for abandoning work, since there was an abundance of good timber still standing, and I rather feared that only a deserted camp or a burned and abandoned mill might lie at the end of my journey. However, I argued that there were many other reasons which might account for the matter. Perhaps, I thought, the lumbermen had been stealing timber from government land and had been driven off, or possibly better transportation facilities in some other district had made lumbering less profitable here. Finally I contented myself with

the thought that even if I found no inhabited camp or village at the end of the trail there must, at least, be a large river, an old railway, or a well-marked lumber road leading out to civilization, and once I struck this my way would be easy.

About mid-afternoon I heard the sound of rushing water ahead, and a few moments later we came to the banks of a river with the road leading directly to the water's edge. Before me stretched a hundred feet of tumbling, foaming torrent, and on the farther bank, opposite to where I stood, I could plainly see the continuation of the old wood road.

I was as badly off as ever, and, thoroughly disheartened, I threw myself upon the ground. Evidently to follow the road and escape from the forest I must cross the stream, and I realized how inexcusably stupid I had been not to have thought of this contingency before. I might have known that the lumber road had been used in winter and that on the ice the loads of timber could be hauled across brooks and rivers which would be impassable now, but not until I had been brought face to face with such conditions did the idea enter my head.

No doubt, I thought, the stream might be forded later in the season, for it was now abnormally high with the spring freshets. But I was in no mood to wait for weeks, or perhaps months, for the water to fall enough to allow me to wade across, and I knew that to attempt to swim through the icy current would be suicidal and would result in the loss of my outfit, even if I reached the other side alive. There seemed to be but one thing to be done, which was to walk up the stream to some narrower, shallower spot where it could be forded, and then, after crossing, retrace my way on the farther side until I once more came to the road. Sooner or later I knew that I must find a point where I could cross in safety, and as anything was better than sitting here and cursing my luck, I rose and proceeded up the river.

Had the stream been slightly lower I could have followed its shores at the edge of the woods and traveling would have been comparatively easy, but with the river in flood the water swept to the trees along the borders and I was compelled to make my way through the forest.

For hour after hour I trudged steadily on, encouraged somewhat by finding the river was decreasing in width and that here and there ledges and boulders jutted above the surface of the waters.

The sun was sinking low in the west, the shadows were lengthening, and I was considering the advisability of making camp for the night when Lobo

uttered a sharp yelp, lifted his nose in the air, and rushed off diagonally through the woods. Feeling sure that something unusual must have attracted him and with a half-formed hope that it was a human being or an occupied habitation, I forgot my weariness and hurried after him.

The wolf was now out of sight, but I could hear him yelping a short distance away, and presently I saw him standing at the edge of the woods, wagging his tail and jumping about as if greatly excited. Beyond him was open sky and my heart beat fast with anticipation, for I knew that he was at the edge of a clearing which indicated the presence of man.

A moment later I was by his side and, looking from the woods, gasped in astonishment, for at the farther side of the open space stood my own log cabin!

For a few seconds I stood motionless, feeling as if I had taken leave of my senses, and I rubbed my eyes and stared again to see if my vision had not played me false. It seemed incredible that I should once more be gazing at the same spot from which I had departed so many hours before, but there was not the slightest doubt about it. There stood the hut I knew so well. I recognized the familiar trees and landmarks about it, and beyond, glimmering like molten gold in the rays of the setting sun, were the distant waters of the lake.

I was thoroughly discouraged, tired, and disgusted to think that I had merely traveled in a vast circle, that since early morning I had been tramping for many weary miles over the old road when by following the river down for an hour or two I could have reached the end of the road where it was barred by the stream.

Hungry, footsore, and exhausted, I reached the cabin, tossed my belongings on the floor, and threw myself on my bunk.

Then, as Lobo, whining, licked my hand and looked inquiringly into my face, I rose, started the fire, and cooked our simple evening meal. The food did much to revive my spirits, and I comforted myself by thinking that, after all, the cabin was a far better spot to spend the night than an open camp; that I was no worse off than before, and that, knowing it was useless to follow the old road, I could seek a spot to cross the river close to my hut.

The next morning I made my way to the river, which I now knew was the outlet to the lake. I examined it carefully for a spot at which I might cross, but at its narrowest point it was far too wide and turbulent. Then it

came to my mind that I might be able to ford the inlet to the lake, but as I remembered the wild, rough country which bordered the lake on its farther side I abandoned this idea. The next instant Lobo leaped up with a startled growl as I gave a shout of gladness, for I had thought of a way to solve the difficulty.

It was so simple, now I had thought of it, that I marveled it had not occurred to me sooner, for my scheme was to finish the raft I had begun the preceding autumn, cross the lake near the outlet, and thus reach the farther shore of the stream.

So firmly had I been convinced that I could *walk* out of the wilderness by following the old trail that my original idea of a raft had been completely forgotten, but now that it had come to my mind, I lost no time in getting to work.

The logs which I had cut with so much labor were still where I had left them when I had met with my accident. I shuddered when I saw the heavy timber which had pinned me down and remembered the tortures I had suffered.

I would be more careful this time and would make haste slowly. As I was far better equipped to build a raft now than when I had first attempted the undertaking, I had little fear of the outcome.

My life in the woods had added greatly to my health and strength and I found little difficulty in handling logs which had formerly resisted all my efforts.

But despite this, many hours were required to cut the logs by the slow process of burning, for I was obliged to cut several, since those which I had rolled to the lake in the autumn had been carried away by the ice.

Then came the hard, tedious work of rolling the logs to the water, but at the close of the second day I had the satisfaction of seeing eight lengths of logs floating on the lake and securely moored to the bank.

I had learned much during the months which had passed since I first reached the lake, and I went about building my raft with far greater confidence and greater skill than would have been possible the preceding autumn.

Slipping strong pieces of saplings beneath the floating logs, I lashed the projecting ends securely to corresponding crosspieces above the logs, and I also passed withes of willow around these pieces between each two logs. To

draw these lashings taut and to secure the logs firmly I slipped short sticks through the withes, twisted them tightly, and bound them in position, and as the wood and lashings swelled in the water the strips of bark and twisted withes grew as hard and tight as wire.

But even then I was not satisfied, for I found that there was considerable play in the logs, and that instead of remaining rectangular in form the raft had a tendency to assume a diamond-shape.

As I had no wish to be dropped into the icy waters of the lake on my voyage, I placed numerous saplings diagonally across the raft, from corner to corner, lashed these firmly to the logs and crosspieces, secured another row of diagonal pieces above these in the opposite direction, and then found that the raft was as stiff, strong, and solid as one could wish.

Nearly a week had passed while I labored at my rude craft when at last it was finished and I prepared to set forth. I felt as if I was about to start on a voyage of discovery. Indeed, it was in truth a voyage of discovery, for beyond the waters of the little lake was a new and unknown land which I felt held greater reward for me than Columbus sought for when he sailed westward in his caravels.

And, like Columbus, I had trouble with my crew, and mutiny threatened even before the voyage commenced, for Lobo refused to go aboard and no amount of coaxing, threats, or argument would induce him to change his mind.

I hated to leave him behind, but I could scarce afford to remain a Crusoe of the wilderness on his account, and, thinking he might overcome his fears when he saw I was actually abandoning him, I cast off the fastenings of the raft and with a long pole pushed it a few yards from shore.

Instantly the wolf set up a dismal howl and ran up and down the shore, evidently in great distress at seeing me depart, and, feeling that he had learned a lesson, I pushed the end of the raft against the bank. But no sooner did the logs grate upon the beach than the rascal retreated to a safe distance and, squatting upon his haunches, grinned at me with his great red tongue hanging from his mouth.

Evidently the wolf considered it was some sort of a game and had no intention of taking the matter seriously, and, somewhat irritated at his behavior, I again poled the raft away from shore. For a few moments Lobo looked at me curiously, evidently expecting me to return, and then, finding

the space of water was steadily widening between raft and shore, he plunged into the lake and swam after me. I at once ceased to propel the raft and waited for the wolf to come alongside and clamber aboard, but he showed no intention of doing this and kept well out of reach, evidently preferring to trust to his own powers rather than to my makeshift vessel.

The raft floated buoyantly and the crisscross layers of saplings kept my feet dry and above the water which slopped over the logs and I was soon well out upon the lake. And then I discovered that I had overlooked a most important matter and that by my carelessness and short-sightedness I was in imminent peril.

I had trusted to poling the raft across the lake, but the bottom was now beyond my reach, I had no oars or paddle, and my raft was drifting steadily in the sweep of the current toward the outlet of the lake.

Furiously I strove to paddle the clumsy craft with the pole, but my efforts only served to whirl the raft round and round, and each moment it was drawing nearer to the rushing river and its speed was increasing.

I was now filled with real terror, and not without reason, for the stream flowed from the lake in rocky rapids and I realized that in a few moments I would be drawn into the torrent, that my raft would be dashed to pieces on the jagged rocks, and that I would be left struggling hopelessly in the icy current.

Then, just as the raft shot toward the maelstrom my pole touched bottom, I threw all my weight upon it with a last, despairing shove, the raft lurched, tipped, and plunged, and the next instant I was thrown, spluttering and surprised, but safe, into a shallow backwater.

Scrambling to my feet, for the water was scarce six inches deep, I splashed ashore, and, glancing back, saw my raft, splintered and rent, bobbing among the foam and rocks of the rapids.

It was a close shave, but I had accomplished my purpose and at last was on the southern side of the lake, and a moment later Lobo came bounding to my side and shaking the water from his shaggy coat.

My pack containing provisions and hides had gone with the raft, for I had slipped it from my shoulder to more easily handle the pole, but I still retained my precious bow and arrows and I discovered my spear sticking in the shallow water where it had been cast by the final lurch of the raft.

With my weapons and fire-making appliances intact I had little cause to worry over the loss of my supplies. Thankful that I had escaped with so little damage, I set off down the banks of the stream.

To my left a high, wooded hill rose steeply, but the forest by the riverside was fairly open and level, and I had little trouble in making my way downstream, although several times I was obliged to make wide detours to avoid jutting ledges and ramparts of rock.

In a few hours I reached the spot where the old wood road ended at the river's bank and, turning to the south, I left the brawling stream behind and entered the old trail.

On either side were lofty, wooded hills, and these increased in height and steepness until, when some five miles from the river, I found myself in a deep, narrow cleft or pass. Then, rounding a precipitous, rock-strewn promontory, the road entered a wider valley with a low, rounded hill on the farther side and with a tiny brook babbling through the center.

Stepping to one side, I approached the brook to drink, when suddenly I stopped short—listening with bated breath and thumping heart, for my ears had caught a faint, far-away sound of marvelous import—the whistle of a locomotive! Then breaking into a run, I tore madly up the hillside, with Lobo at my heels.

I reached the summit, peered through the thicket, and looked upon a broad, fair valley with a silvery river winding through its fresh green meadows.

Wildly, insanely, I shouted in a very delirium of joy, for gleaming in the soft spring sunshine were the white buildings and shingled roofs of a little village.



“I LOOKED UPON A BROAD, FAIR VALLEY”

Tears of happiness welled to my eyes as I stood feasting my gaze upon the scene and then the hoarse screech of the locomotive again roared out and a train thundered past the base of the hill at my feet.

It was too much for Lobo, and with a terrified howl he turned and sped back to the wild, leaving me standing alone upon the threshold of civilization.

THE END

*** END OF THE PROJECT GUTENBERG EBOOK MAROONED IN
THE FOREST: THE STORY OF A PRIMITIVE FIGHT FOR LIFE ***

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